

SYMMETRIES FOR OVERPARTITION RANK AND CRANK FUNCTIONS

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ABSTRACT. Continuing our study of the symmetry phenomenon in rank and crank of partitions, we turn our attention to overpartitions. Building on the previous work of the two authors on the partition rank generating function and that of the second author on the partition crank generating function, we investigate the modularity, transformation and symmetry of the overpartition rank and crank generating functions. Let $\mathcal{O}_{\mathcal{R}}(z, q)$ and $\mathcal{O}_{\mathcal{C}}(z, q)$ be the two-variable generating functions of the overpartition rank and the first residual crank statistics respectively and ζ_p be a primitive p -th root of unity. By considering the action of the group $\Gamma_0(2p)$ on the elements of the p -dissection of $\mathcal{O}_{\mathcal{R}}(\zeta_p, q)$ and $\mathcal{O}_{\mathcal{C}}(\zeta_p, q)$, we discover new symmetries for the rank and first residual crank functions for overpartitions. In the process, we improve upon the results of Jennings-Shaffer and that of Bringmann and Lovejoy on the modularity and transformation of the overpartition rank function. Finally, we give new identities for explicit dissections of overpartition rank and the first residual crank generating functions modulo 11 in terms of generalized eta products.

1. INTRODUCTION

The rank statistic for partitions was discovered by Dyson in 1944. The Dyson rank of a partition is the largest part minus the number of parts. This statistic decomposes the partitions of $5n + 4$ and $7n + 5$ into 5 and 7 equinumerous classes, as conjectured by Dyson and proved by Atkin and Swinnerton-Dyer, thus resolving the mod 5 and mod 7 congruences of Ramanujan. The crank of a partition was introduced by Andrews and Garvan in 1988. It is defined as the largest part if the partition contains no ones, and otherwise as the number of parts larger than the number of ones minus the number of ones. In their paper, Andrews and Garvan show that the crank simultaneously decomposes the partitions of $5n + 4$, $7n + 5$ and $11n + 6$ into 5, 7 and 11 equinumerous classes respectively.

The significance of the rank and crank statistics lies not only in the aspect that it partially renders a combinatorial explanation to the famous congruences of Ramanujan, but also in the fact that their generating functions have a rich modular structure. Several mathematicians have worked in this direction to unearth the modular structure underlying these two statistics following the seminal work done by Zwegers in his thesis where he shows how Ramanujan's mock theta functions occur as the holomorphic part of certain real analytic modular forms.

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In 2004, Sylvie Corteel and Jeremy Lovejoy [6] introduced the concept of overpartitions. An overpartition is a partition in which the first occurrence of a number may be overlined. For example, the 14 overpartitions of 4 are

$$4, \overline{4}, 3 + 1, \overline{3} + 1, 3 + \overline{1}, \overline{3} + \overline{1}, 2 + 2, \overline{2} + 2, 2 + 1 + 1, \\ \overline{2} + 1 + 1, 2 + \overline{1} + 1, \overline{2} + \overline{1} + 1, 1 + 1 + 1 + 1, \overline{1} + 1 + 1 + 1.$$

Analogous to the crank of an ordinary partition, Bringmann, Lovejoy and Osburn [4] introduced the first and second residual crank of an overpartition. In this paper, the authors deduce congruence properties for combinatorial functions which can be expressed in terms of the second overpartition rank moment and the corresponding residual crank moment. Several authors have since considered these residual cranks and have worked on their generalizations, finding and proving inequalities between the moments of these functions in conjunction with other overpartition statistics, among other problems. The combinatorial interpretation leading to explicit definitions for these cranks was given by us recently in [12].

Let $M(m, n)$ denote the number of overpartitions of n with first residual crank equal to m . Throughout, we assume $q = e^{2\pi i\tau}$ where $\tau \in \mathcal{H}$, the upper half complex plane. We let $\overline{M}(z, q)$ denote the two-variable generating function for the first residual crank so that

$$\mathcal{O}_C(z, \tau) = \sum_{n=0}^{\infty} \sum_m \overline{M}(m, n) z^m q^n.$$

It is the transformation, modularity and symmetry of this function under special congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$, which we hereinafter call the overpartition crank function $\mathcal{O}_C(z, \tau)$, when z is a primitive p -th root of unity, that we study in the first half of this paper. This residual crank can be expressed in terms of a Klein form whose transformation is well known in literature and helps us establish our results. We summarize our two major results for the overpartition crank without delving into further details here.

Theorem 1.1. *Let $p \geq 3$ be prime. Then the function*

$$\frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathcal{O}_C(\zeta_p, \tau)$$

is a modular form of weight 1 on $\Gamma_0(p^2) \cap \Gamma_1(2p)$.

The modularity and symmetry of the elements of the p -dissection of $\frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathcal{O}_C(\zeta_p, \tau)$ is our other point of interest in the paper. These elements are defined below.

Definition 1.2. Let $p \geq 3$ be prime, $0 \leq m \leq p-1$ and $1 \leq \ell \leq p-1$. Define

$$(1.1) \quad \mathcal{OK}_{p,m}^{(C)}(\zeta_p^\ell, z) := q^{m/p} \prod_{n=1}^{\infty} \frac{(1 - q^{pn})^2}{(1 - q^{2pn})} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{p-1} \overline{M}(k, p, pn + m) \zeta_p^{k\ell} \right) q^n.$$

The result on the modularity of these elements is deferred to Section 4. Below is our other major result of the paper for the crank of overpartitions. The action of the congruence subgroup $\Gamma_0(2p)$ on these elements leads to our symmetry result, analogous to the symmetry result for crank of partitions [19, Theorem 1.6].

Theorem 1.3. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

$$(1.2) \quad \mathcal{OK}_{p,m}^{(C)}(\zeta_p^\ell, z) \mid [A]_1 = \frac{\sin(\frac{\ell\pi}{p})}{\sin(\frac{\ell d\pi}{p})} \zeta_p^{mak} \mathcal{OK}_{p,ma^2}^{(C)}(\zeta_p^{\ell d}, z)$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

In the second half of this paper, we consider the Dyson's rank of an overpartition. It is defined as the largest part minus the number of parts, in particular the rank does not depend on whether or not a part is overlined.

Let $\overline{N}(m, n)$ denote the number of overpartitions of n with rank m . Let

$$\mathcal{O}_{\mathcal{R}}(z, \tau) = \sum_{n=0}^{\infty} \sum_m \overline{N}(m, n) z^m q^n.$$

In a fashion similar to the work of Bringmann and Ono [5] for the rank of partitions, Bringmann and Lovejoy [3] observed that $\mathcal{O}_{\mathcal{R}}(z, \tau)$ is the holomorphic part of a harmonic weak Maass form of weight $\frac{1}{2}$ on when $z \neq -1$ is a root of unity. The following theorem summarizes their main result in the case where z is a p -th root of unity.

Theorem 1.4. *Let $p \geq 3$ be prime and $0 < a < p$. Define*

$$\begin{aligned} \theta(\alpha, \beta; \tau) &:= \sum_{n \equiv \alpha \pmod{\beta}} n q^{n^2/2\beta}, \\ \Theta_{a,p}(\tau) &:= \theta\left(4a + p, 2p; \frac{\tau}{4p}\right), \text{ and} \\ J\left(\frac{a}{p}; z\right) &:= \frac{\pi i \tan\left(\frac{\pi a}{p}\right)}{4p} \int_{-\bar{z}}^{i\infty} \frac{(-i\tau)^{-\frac{3}{2}} \cdot \Theta_{a,p}\left(-\frac{1}{\tau}\right) d\tau}{\sqrt{-i(\tau + z)}}. \end{aligned}$$

Then

$$M\left(\frac{a}{p}; z\right) := \mathcal{O}_{\mathcal{R}}(\zeta_p^a, \tau) - J\left(\frac{a}{p}; z\right)$$

is a weak Maass form of weight $\frac{1}{2}$ on the congruence subgroup

$$\tilde{\Gamma} = \left\{ A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL_2(\mathbb{Z}) \mid \alpha \equiv \delta \equiv 1 \pmod{4p}, \gamma \equiv 0 \pmod{16p^2} \right\}.$$

In our study of the overpartition rank function $\mathcal{O}_{\mathcal{R}}(\zeta_p, \tau)$, we build around the functions considered by Jennings-Shaffer [13]. In this paper, Jennings-Shaffer reconsiders the modular structure of this function and bypassing the approach of Bringmann and Lovejoy, he constructs a completion of $\mathcal{O}_{\mathcal{R}}(\zeta_p, \tau)$ to a harmonic Maass form which is the sum of an easily understood modular form and a harmonic Maass form of Zwegers. We record this work here.

For $u, v, z \in \mathbb{C}$, $\tau \in \mathcal{H}$, and $u, v \notin \mathbb{Z} + \tau\mathbb{Z}$ we have

$$\vartheta(z; \tau) = \sum_{n \in \mathbb{Z} + \frac{1}{2}} \exp \left(\pi i n^2 \tau + 2\pi i n \left(z + \frac{1}{2} \right) \right),$$

$$\mu(u, v; \tau) = \frac{\exp(\pi i u)}{\vartheta(v; \tau)} \sum_{n=-\infty}^{\infty} \frac{(-1)^n \exp(\pi i n(n+1)\tau + 2\pi i n v)}{1 - \exp(2\pi i n \tau + 2\pi i u)}.$$

Next for $u, z \in \mathbb{C}$, $y = \text{Im}(\tau)$, and $a = \text{Im}(u)/\text{Im}(\tau)$ we define

$$E(z) = 2 \int_0^z \exp(-\pi w^2) dw$$

$$R(u; \tau) = \sum_{n \in \mathbb{Z} + \frac{1}{2}} (\text{sgn}(n) - E((n+a)\sqrt{2y})) (-1)^{n-\frac{1}{2}} \exp(-\pi i n^2 \tau - 2\pi i n u).$$

For $a, b \in \mathbb{R}$ we set

$$g_{a,b}(\tau) = \sum_{n \in \mathbb{Z} + a} n \exp(\pi i n^2 \tau + 2\pi i n b).$$

Finally for $u, v \notin \mathbb{Z} + \tau\mathbb{Z}$ we set

$$\tilde{\mu}(u, v; \tau) = \mu(u, v; \tau) + \frac{i}{2} R(u - v; \tau).$$

In his revolutionary PhD thesis [21], Zwegers studied these functions and gave their transformation formulas.

Proposition 1.5. (*Jennings-Shaffer, Corollary 2.2, [13]*) *Suppose a and c are integers, $c > 0$ and $c \nmid 2a$. Then,*

$$\begin{aligned} \mathcal{O}_{\mathcal{R}}(\zeta_c^a; \tau) &= \frac{2\zeta_c^a(1 - \zeta_c^a)}{(1 + \zeta_c^a)} P(a, c; \tau) - \frac{i2\zeta_c^a(1 - \zeta_c^a)}{(1 + \zeta_c^a)} N(a, c; \tau) \\ &+ i\sqrt{2} \frac{(1 - \zeta_c^a)}{(1 + \zeta_c^a)} \int_{-\bar{\tau}}^{i\infty} \frac{g_{0, \frac{1}{2} - \frac{2a}{c}}(2z)}{\sqrt{-i(z + \tau)}} dz \end{aligned} \quad (1.3)$$

where

$$N(a, c; \tau) = q^{-\frac{1}{4}} \tilde{\mu} \left(\frac{2a}{c}, \tau; 2\tau \right), \quad P(a, c; \tau) = \frac{(-q; q)_{\infty} (q^2; q^2)_{\infty}^2}{(q; q)_{\infty} [\zeta_c^{2a}; q^2]_{\infty}}.$$

Jennings-Shaffer gives explicit and compact transformation formulas to determine a larger subgroup of $\text{SL}_2(\mathbb{Z})$ on which the completion of $\mathcal{O}_{\mathcal{R}}(\zeta_p^a, \tau)$ viz.

$$\begin{aligned} M(a, c; \tau) &= \mathcal{O}_{\mathcal{R}}(\zeta_c^a; \tau) - i\sqrt{2} \frac{(1 - \zeta_c^a)}{(1 + \zeta_c^a)} \int_{-\bar{\tau}}^{i\infty} \frac{g_{0, \frac{1}{2} - \frac{2a}{c}}(2z)}{\sqrt{-i(z + \tau)}} dz \\ &= \frac{2\zeta_c^a(1 - \zeta_c^a)}{(1 + \zeta_c^a)} P(a, c; \tau) - \frac{i2\zeta_c^a(1 - \zeta_c^a)}{(1 + \zeta_c^a)} N(a, c; \tau) \end{aligned} \quad (1.4)$$

is a Harmonic Maass form, and reproves the modularity result of Bringmann and Lovejoy.

Theorem 1.6. *Suppose a and c are integers, $c > 0$, and $c \nmid 2a$, then $M(a, c; \tau)$ is a harmonic weak Maass form of weight $\frac{1}{2}$ on $\Gamma_0(16c^2) \cap \Gamma_1(4c)$.*

The case $a = 1$ and $c = p$ for prime $p \geq 3$ is the special case of the result of Bringmann and Lovejoy stated previously in Theorem 1.4.

In this paper, with the introduction of a simple eta multiplier to $\mathcal{O}_{\mathcal{R}}(\zeta_p, \tau)$, we narrow down the transformation of all the functions involved in the expression for $\mathcal{O}_{\mathcal{R}}(\zeta_p, \tau)$ (Proposition 1.5) to the simpler congruence subgroup $\Gamma_0(2p^2) \cap \Gamma_1(p)$ sans the multipliers appearing in the results of Jennings-Shaffer (Corollaries 3.3 and 4.1, [13]). This problem also involves finding the overpartition analogue of the completed rank function $\mathcal{R}_p(z)$ given by Garvan [10].

$$\mathcal{R}_p(z) := q^{-\frac{1}{24}} R(\zeta_p, q) - \chi_{12}(p) \sum_{a=1}^{\frac{1}{2}(p-1)} (-1)^a \left(\zeta_p^{3a+\frac{1}{2}(p+1)} + \zeta_p^{-3a-\frac{1}{2}(p+1)} - \zeta_p^{3a+\frac{1}{2}(p-1)} - \zeta_p^{-3a-\frac{1}{2}(p-1)} \right) q^{\frac{a}{2}(p-3a)-\frac{p^2}{24}} \Phi_{p,a}(q^p),$$

where

$$\Phi_{p,a}(q) := \begin{cases} \sum_{n=0}^{\infty} \frac{q^{pn^2}}{(q^a; q^p)_{n+1} (q^{p-a}; q^p)_n}, & \text{if } 0 < 6a < p, \\ -1 + \sum_{n=0}^{\infty} \frac{q^{pn^2}}{(q^a; q^p)_{n+1} (q^{p-a}; q^p)_n}, & \text{if } p < 6a < 3p, \end{cases}$$

and

$$\chi_{12}(n) := \left(\frac{12}{n} \right) = \begin{cases} 1 & \text{if } n \equiv \pm 1 \pmod{12}, \\ -1 & \text{if } n \equiv \pm 5 \pmod{12}, \\ 0 & \text{otherwise.} \end{cases}$$

This in turn involves finding the overpartition rank analogue of the correction factor $\Phi_{p,a}(q^p)$. We find that this correction factor for the overpartition rank function includes a Lambert series which

- (1) appears in the dissection elements of the overpartition rank generating function modulo 7 viz. $\mathcal{O}(\zeta_7, \tau)$ of Jennings-Shaffer (Theorem 1.1, [13]) and modulo 3 and 5 viz. $\mathcal{O}(\zeta_3, \tau)$ and $\mathcal{O}(\zeta_5, \tau)$ of Lovejoy and Osburn (Theorem 1.1, 1.2, [16]), and
- (2) has a connection with the cancellation of the Mordell integral of the theta function which is the non-holomorphic part of $\mathcal{O}(\zeta_p, \tau)$ (Corollary 2.2, [13]).

We will eventually see in the course of this paper that this factor allows us to complete the overpartition rank function to a weakly holomorphic modular form on a larger and nicer congruence subgroup improving the results of Bringmann and Lovejoy [3] and Jennings-Shaffer [13]. To that end,

Definition 1.7. Let $p \geq 3$ be prime and $1 \leq k \leq \frac{1}{2}(p-1)$. Define

$$(1.5) \quad \Phi_{p,k}(q) := \frac{(q^{2p}; q^{2p})_{\infty}}{(q^p; q^p)_{\infty}^2} \sum_{n=-\infty}^{\infty} (-1)^n \frac{q^{pn(n+1)}}{1 - q^{pn+k}}$$

and

$$(1.6) \quad \mathfrak{D}_R(\zeta_p, \tau) := \mathcal{O}_R(\zeta_p; \tau) - \frac{2(1 - \zeta_p)}{1 + \zeta_p} \sum_{k=1}^{\frac{1}{2}(p-1)} (-1)^k (\zeta_p^{-2k} - \zeta_p^{2k}) \Phi_{p,k}(q^p).$$

Thus one of our main results

Theorem 1.8. Let $p \geq 3$ be prime. Then the function

$$\frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathfrak{D}_R(\zeta_p, \tau)$$

is a weakly holomorphic modular form of weight 1 on $\Gamma_0(2p^2) \cap \Gamma_1(p)$.

Furthermore, we also define the elements of the p -dissection of $\frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathfrak{D}_R(\zeta_p, \tau)$.

These elements feature the functions $\mathcal{OK}_{p,m}^{(R)}(\zeta_p^d; z)$ defined below into two cases for when m is a quadratic residue or non-residue modulo p , wherein fragments of the correction factor above (Equation (1.5)) appears in the quadratic residue case. We study their modularity and symmetry, akin to the case of rank of partitions (Definition 1.4, [11]).

Definition 1.9. Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Define $\mathcal{OK}_{p,m}^{(R)}(\zeta_p^d; z)$ as follows :

(i) For $m = 0$ or $\left(\frac{-m}{p}\right) = -1$ define

$$(1.7) \quad \mathcal{OK}_{p,m}^{(R)}(\zeta_p^d; z) := q^{m/p} \frac{\prod_{n=1}^{\infty} (1 - q^{pn})^2}{\prod_{n=1}^{\infty} (1 - q^{2pn})} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{p-1} \overline{N}(k, p, pn) \zeta_p^{kd} \right) q^n,$$

where $q = \exp(2\pi iz)$.

(ii) For $\left(\frac{-m}{p}\right) = 1$ define

$$(1.8) \quad \mathcal{OK}_{p,m}^{(R)}(\zeta_p^d; z) := q^{m/p} \frac{\prod_{n=1}^{\infty} (1 - q^{pn})^2}{\prod_{n=1}^{\infty} (1 - q^{2pn})} \left(\sum_{n=0}^{\infty} \left(\sum_{k=0}^{p-1} \overline{N}(k, p, pn + m) \zeta_p^{kd} \right) q^n \right. \\ \left. - q^{\frac{1}{p}(a(p-a)-m)} \frac{2(1 - \zeta_p)}{1 + \zeta_p} (-1)^a (\zeta_p^{-2a} - \zeta_p^{2a}) \Phi_{p,a}(q) \right),$$

where $1 \leq a \leq \frac{1}{2}(p-1)$ has been chosen so that

$$-m \equiv a^2 \pmod{p}.$$

The result on the modularity of these elements is deferred to Section 5. Below is our other major result of the paper for the rank of overpartitions. The action of the congruence subgroup $\Gamma_0(2p)$ on these elements leads to our symmetry result, similar to the symmetry result for crank of overpartitions Theorem 1.3 and analogous to the symmetry result for rank of partitions [11, Theorem 1.6].

Theorem 1.10. *Let $p \geq 3$ be prime and $0 \leq m \leq p - 1$. Then*

$$(1.9) \quad \mathcal{OK}_{p,m}^{(R)}(\zeta_p, z) \mid [A]_1 = \frac{1 - \zeta_p}{1 - \zeta_p^d} \frac{1 + \zeta_p^d}{1 + \zeta_p} \zeta_p^{mak} \mathcal{OK}_{p,ma^2}^{(R)}(\zeta_p^d, z)$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Remarks :

- (1) Our results strengthen the results of Bringmann and Lovejoy [3] and Jennings-Shaffer [13] for the rank of overpartitions. In particular, for the case $p \geq 3$ prime, we strengthen Theorems 1.4 and 1.6. We show that the group in these results can be enlarged to $\Gamma_0(2p^2) \cap \Gamma_1(p)$ but with a simple eta multiplier $\frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)}$, on which our completed rank function is a weakly holomorphic modular form of order 1. The idea for choosing this special congruence subgroup comes from the work of Jennings-Shaffer which we explain here. The introduction of the eta multiplier $\frac{\eta(\tau)^2}{\eta(2\tau)}$ to the functions $P(a, c, \tau)$ and $N(a, c, \tau)$ involved in the expression for $\mathcal{OK}_{p,m}(\zeta_p, \tau)$ gets rid of the multipliers appearing in the results of Jennings-Shaffer (Corollaries 3.3 and 4.1, [13]), where he is able to significantly simplify the transformation of these two functions to the much simpler congruence subgroup $\Gamma_0(2p^2) \cap \Gamma_1(p)$. However, what we realize is that considering a generalization of Jennings-Shaffer's function $N_7(k; \tau)$ [13, Page 6] extended to any prime p leads us to our improved modularity result described in the point above. The resulting function $N_p(k; \tau)$ appears in our modified expression (with the correction factor and eta multipliers) for $\mathcal{OK}_{p,m}(\zeta_p, \tau)$ in Equation (1.3). See Equation (5.3). The non-holomorphic part of this function helps annihilate the non-holomorphic part arising from the Mordell integral of the theta function in Equation (1.3) and leads us to our completion of the overpartition rank function in Equation (1.6) and eventually Theorem 1.8. Our investigation of the transformation and modularity of $N_p^*(k; \tau) = \frac{\eta(\tau)^2}{\eta(2\tau)} N_p(k; \tau)$ also generalizes and improves the transformation result of Jennings-Shaffer for $N_7(k; \tau)$ [13, Proposition 5.2] to any prime $p \geq 3$. This is documented in Corollary 5.4.
- (2) A majority of our modularity and transformation results in this paper are direct analogies of those in [11] and [19].
- (3) The reason behind considering the study of the first residual crank of overpartitions in this paper is because we observe that the respective modular and transformation properties in the case of Dyson's rank of overpartitions hold for the exact same congruence subgroups of $\text{SL}_2(\mathbb{Z})$. To be precise, the modularity results for the generating function of the two modified overpartition statistics hold under $\Gamma_0(2p^2) \cap \Gamma_1(p)$, the modularity and transformation results for their dissection elements under $\Gamma(2p)$

and $\Gamma_1(2p)$ and finally the symmetry result for the same dissection elements under $\Gamma_0(2p)$. However, the second residual crank of overpartitions also appears in the hindsight in our study of the rank. It arises in the formula for the two variable overpartition rank generating function $\mathcal{O}_{\mathcal{R}}(\zeta_c^a; \tau)$ due to Jennings-Shaffer in Equation (1.3). This is the function $P(a, c, \tau)$ whose modularity and transformation along with a simple eta multiplier, when ζ_c^a is a p -th root of unity $\zeta_p, p > 3$ is investigated in Propositions 5.1 and 5.10 and Corollary 5.2.

In the final part of this paper, we find explicit p -dissections of the overpartition rank and crank functions in terms of generalized theta functions. Identities for the dissection elements of the overpartition rank function for the case $p = 7$ were found and studied by Jennings-Shaffer [13]. However we find that for higher primes $p > 7$, the computational techniques in Maple that we employed to find dissections for the partition counterparts in [11] and [19] fall short. Consequently, the idea here is to consider the ring of modular/cusp forms for $\Gamma_1(22)$, and eventually find a simpler basis consisting of Jacobi products which we accomplish using a combination of Maple and Sage. We present and prove such identities for the case $p = 11$ in the last section.

The paper is organized as follows. In Section 2, we review and build a database of the necessary notations, definitions and fundamental transformation results which we will use in the course of establishing our main results in the subsequent sections. In Section 3, we give conditions for the modularity of the generalized theta functions analogous to the conditions developed for the rank of partitions [11, Section 3]. In Sections 4 and 5, we consider the first residual crank of overpartitions and Dyson's rank of overpartitions respectively and deduce our transformation, modularity and new symmetry results. We give another symmetry result involving the cyclotomic coefficients appearing in the $m = 0$ dissection elements of the rank and crank functions in Section 6. Section 7 is devoted to calculating lower bounds for the orders of $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$ at the cusps of $\Gamma_1(2p)$ which we utilize to prove identities for the same in the subsequent section. In Section 8, we prove generalized eta-quotient identities for $\mathcal{OK}_{11,m}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{11,m}^{(R)}(\zeta_p, z)$, where $0 \leq m \leq 10$, using techniques coming from automorphic forms and with the aid of q -series packages in MAPLE. This amounts to a complete 11-dissection of the overpartition crank and rank functions.

2. PRELIMINARIES

2.1. Theta functions and transformations. We define a class of generalized theta functions that we will use frequently in expressions for our overpartition rank and crank identities and to study their transformations in the subsequent sections.

Definition 2.1. Following Biagioli (see [2]), define

$$(2.1) \quad f_{N,\rho}(z) := (-1)^{\lfloor \rho/N \rfloor} q^{(N-2\rho)^2/(8N)} (q^\rho, q^{N-\rho}, q^N; q^N)_\infty.$$

Then, for a vector $\vec{n} = (n_0, n_1, n_2, \dots, n_p) \in \mathbb{Z}^{p+1}$, define

$$(2.2) \quad j(z) = j(2p, \vec{n}, z) = \eta(2pz)^{n_0} \prod_{k=1}^p f_{2p,k}(z)^{n_k}.$$

We note that

$$(2.3) \quad f_{N,\rho}(z) = f_{N,N+\rho}(z) = f_{N,-\rho}(z),$$

and

$$(2.4) \quad f_{N,\rho}(z) = \eta(Nz) \eta_{N,\rho}(z)$$

where

$$\eta(z) := q^{\frac{1}{24}} \prod_{n=1}^{\infty} (1 - q^n),$$

and

$$\eta_{N,k}(z) = q^{\frac{N}{2}P_2(k/N)} \prod_{\substack{m>0 \\ m \equiv \pm k \pmod{N}}} (1 - q^m)$$

where $z \in \mathfrak{h}$, $P_2(t) = \{t\}^2 - \{t\} + \frac{1}{6}$ is the second periodic Bernoulli polynomial, and $\{t\} = t - [t]$ is the fractional part of t .

The transformations for $\eta(z)$ due to Knopp [14, Theorem 2, p.51] and the theta function $f_{N,\rho}(z)$ due to Biagioli [2, Lemma 2.1, p.278] are well known and have been used previously by us to study transformations in [10] and [11].

2.2. Klein forms. Our transformation results for the first residual crank of overpartitions are realized by writing them in terms of a Klein form which we define here. These functions are also considered by Jennings-Shaffer in his study of the transformation of the function $P(a, c; \tau)$ described in the introduction section. Here we correct a minor mistake in the definition adopted by him [13, Section 4] and instead state and use the definition adopted by Eum, Koo and Shin [9] as follows.

Definition 2.2. [9, Equation 1.4] For $(a_1, a_2) \in \mathbb{Q}^2 - \mathbb{Z}^2$ and $\zeta = \exp(2\pi i(a_1\tau + a_2))$, the Klein form $t_{(a_1, a_2)}(\tau)$ is defined by the following infinite product expansion

$$(2.5) \quad t_{(a_1, a_2)}(\tau) = \exp(\pi i a_2(a_1 - 1)) q^{\frac{1}{2}(a_1(a_1-1))} (1 - \zeta) \frac{(\zeta q; q)_{\infty} (\zeta^{-1} q; q)_{\infty}}{(q; q)_{\infty}^2}.$$

The transformation and order of this Klein form necessary for our study is recorded in the proposition below.

Proposition 2.3. [9, Proposition 2.1]

(i) For $(a_1, a_2) \in \mathbb{Q}^2 - \mathbb{Z}^2$ and $(b_1, b_2) \in \mathbb{Z}^2$, we have

$$t_{(a_1+b_1, a_2+b_2)}(\tau) = (-1)^{b_1 b_2 + b_1 + b_2} \exp(-\pi i(b_1 a_2 - b_2 a_1)) t_{(a_1, a_2)}(\tau).$$

(ii) For $(a_1, a_2) \in \mathbb{Q}^2 - \mathbb{Z}^2$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, we have

$$t_{(a_1, a_2)}(\tau) | [A]_1 = t_{(a_1, a_2).A}(\tau) = t_{(a_1 a + a_2 c, a_1 b + a_2 d)}(\tau).$$

(iii) For $(a_1, a_2) \in \mathbb{Q}^2 - \mathbb{Z}^2$, we have

$$\text{ord}_q t_{(a_1, a_2)}(\tau) = \frac{1}{2} \{a_1\} (\{a_1\} - 1),$$

where $\{x\}$ is the fractional part of $x \in \mathbb{R}$.

2.3. A useful permutation.

Definition 2.4. Let $p \geq 3$ be prime. For $r = 1, 3, \dots, p-2$, we define a permutation $\pi_r : [p] \rightarrow [p]$, where $[p] = \{1, 2, \dots, p\}$ by $\pi_r(i) = i'$ where $ri' \equiv \pm i \pmod{2p}$.

π_r induces a permutation on $\mathbb{Z}^p$. For $\vec{n} = (n_0, n_1, n_2, \dots, n_p)$, $\pi_r(\vec{n})$ permutes the components to $\pi_r(\vec{n}) = (n_0, n_{\pi_r(1)}, n_{\pi_r(2)}, \dots, n_{\pi_r(p)})$.

Lemma 2.5. Let $p \geq 3$ be prime and $\vec{n}$ and $j(z)$ be defined as in Definition 2.1. Then,

$$j(2p, \pi_r(\vec{n}), z) = \eta(2pz)^{n_0} \prod_{k=1}^p f_{2p, rk}(z)^{n_k}.$$

The proof of the lemma follows as in [11, Lemma 3.5].

2.4. The Atkin operator. We define the (weight k) Atkin U_p operator by

$$(2.6) \quad F \mid [U_p]_k := \frac{1}{p} \sum_{r=0}^{p-1} F\left(\frac{z+r}{p}\right) = p^{\frac{k}{2}-1} \sum_{n=0}^{p-1} F \mid [T_r]_k,$$

where

$$T_r = \begin{pmatrix} 1 & r \\ 0 & p \end{pmatrix},$$

and the more general $U_{p,m}$ is defined as

$$(2.7) \quad F \mid [U_{p,m}]_k := \frac{1}{p} \sum_{r=0}^{p-1} \exp\left(-\frac{2\pi irm}{p}\right) F\left(\frac{z+r}{p}\right) = p^{\frac{k}{2}-1} \sum_{r=0}^{p-1} \exp\left(-\frac{2\pi irm}{p}\right) F \mid [T_r]_k.$$

We note that $U_p = U_{p,0}$. In addition, if

$$F(z) = \sum_n a(n) q^n = \sum_n a(n) \exp(2\pi i z n),$$

then

$$F \mid [U_{p,m}]_k = q^{m/p} \sum_n a(pn + m) q^n = \exp(2\pi i m z / p) \sum_n a(pn + m) \exp(2\pi i n z).$$

3. MODULARITY CONDITIONS FOR GENERALIZED THETA FUNCTIONS

We present a general criteria for a generalized theta function $j(2p, \vec{n}, z)$ in our Definition 2.1 to be a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$ in the form of a theorem. This result is analogous to the criteria for the functions that we had defined to study the partition rank [11, Theorem 3.1].

Theorem 3.1. Let $p \geq 3$ be prime and suppose $\vec{n} = (n_0, n_1, n_2, \dots, n_p) \in \mathbb{Z}^{p+1}$. Then $j(2p, \vec{n}, z)$ is a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$ satisfying the modularity condition

$$j(2p, \vec{n}, z) \mid [A]_1 = \exp\left(\frac{2\pi i b m}{p}\right) j(2p, \vec{n}, z)$$

for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p)$ provided the following conditions are met :

$$\begin{aligned} (1) \quad & n_0 + \sum_{k=1}^p n_k = 2, \\ (2) \quad & n_0 + 3 \sum_{k=1}^p n_k \equiv 0 \pmod{24}, \\ (3) \quad & \sum_{k=1}^p k^2 n_k \equiv 4m \pmod{4p}. \end{aligned}$$

Proof. The Dedekind eta function is a modular form of weight $\frac{1}{2}$. Thus, $\eta(2pz)^{n_0}$ contributes $\frac{n_0}{2}$ and each of the $f_{2p,k}(z)^{n_k}$ contributes $\frac{n_k}{2}$ to the weight and the weight of $j(2p, \vec{n}, z)$ is $\frac{n_0}{2} + \sum_{k=1}^p \frac{n_k}{2}$. Condition (1) implies that this weight is 1.

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p)$. Then, by [10, Theorem 6.14, p.243], we have

$$\eta(2pz) \mid [A]_{1/2} = \nu_\eta({}^{2p}A) \eta(2pz),$$

where $\nu_\eta({}^{2p}A)$ is the eta-multiplier,

$${}^{2p}A = \begin{pmatrix} a & 2bp \\ c/2p & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Then

$$\eta(2pAz) = \nu_\eta({}^{2p}A) \sqrt{cz+d} \eta(2pz)$$

and using the Biagioli transformation [10, Theorem 6.12, p.243] for $f_{p,k}(z)$, we have

$$\begin{aligned} f_{2p,k}(z) \mid [A]_{1/2} &= (-1)^{kb + \lfloor ka/2p \rfloor + \lfloor k/2p \rfloor} \exp\left(\frac{\pi iab}{2p} k^2\right) \nu_\eta^3({}^{2p}A) f_{2p,ka}(z) \\ &= (-1)^{kb + \lfloor ka/2p \rfloor} \exp\left(\frac{\pi iab}{2p} k^2\right) \nu_\eta^3({}^{2p}A) f_{2p,k}(z), \end{aligned}$$

assuming $1 \leq k \leq p$. Therefore

$$\begin{aligned} j(2p, \vec{n}, z) \mid [A]_1 &= (-1)^{L_1(A)} \exp\left(\frac{\pi iab}{2p} L_2(A)\right) \nu_\eta^{L_3(A)}({}^{2p}A) j(z) \\ &= \exp\left(\pi i L_1(A) + \frac{\pi iab}{2p} L_2(A)\right) \nu_\eta^{L_3(A)}({}^{2p}A) j(z), \end{aligned}$$

where

$$L_1(A) = b \sum_{k=1}^p k n_k + \sum_{k=1}^p \left\lfloor \frac{ka}{2p} \right\rfloor n_k,$$

$$L_2(A) = \sum_{k=1}^p k^2 n_k,$$

$$L_3(A) = n_0 + 3 \sum_{k=1}^p n_k.$$

Now assume conditions (1) – (3) hold, and $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p)$. Since $L_3(A) \equiv 0 \pmod{24}$, the modularity condition holds if we can show that $L_1(A) + \frac{ab}{2p}L_2(A) - \frac{2bm}{p}$ is an even integer.

Since $a \equiv 1 \pmod{2p}$, we have $abL_2(A) \equiv bL_2(A) \pmod{2p}$.

Also, using (3), we have $bL_2(A) \equiv 4bm \pmod{4p}$.

Combining the two congruences, we can conclude that $L_1(A) + \frac{ab}{2p}L_2(A) - \frac{2bm}{p}$ is an integer.

Now

$$L_1(A) + \frac{ab}{2p}L_2(A) - \frac{2bm}{p} = \frac{1}{p}(pL_1(A) + \frac{ab}{2}L_2(A) - 2bm).$$

Also, condition (3) also implies that

$$L_2(A) \equiv 0 \pmod{4}$$

$$\frac{ab}{2}L_2(A) \equiv 0 \pmod{2}.$$

Thus, to show that $L_1(A) + \frac{ab}{2p}L_2(A) - \frac{2bm}{p}$ is an even integer, it is sufficient to show that $L_1(A) \equiv 0 \pmod{2}$.

We have $a \equiv 1 \pmod{2p}$. Let $a = 2pr + 1$.

Then

$$\frac{ka}{2p} = \frac{2kpr}{2p} + \frac{k}{2p} = kr + \frac{k}{2p}.$$

So

$$\left\lfloor \frac{ka}{2p} \right\rfloor = kr.$$

Thus

$$L_1(A) = \sum_{k=1}^p (bk + \left\lfloor \frac{ka}{2p} \right\rfloor) n_k$$

$$\equiv (b + r) \sum_{k=1}^p kn_k \pmod{2}$$

$$\begin{aligned} &\equiv (b+r) \sum_{k=1}^p k^2 n_k \pmod{2} \\ &\equiv 0 \pmod{2}, \end{aligned}$$

where the last congruence follows because $\sum_{k=1}^p k^2 n_k = L_2(A) \equiv 0 \pmod{4}$. $\square$

Definition 3.2. Let $\mathfrak{F}(m, p)$ be the set of functions $j(2p, \vec{n}, z)$ that satisfy the conditions of Theorem 3.1.

Theorem 3.3. Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Suppose $j(2p, \vec{n}, z) \in \mathfrak{F}(m, p)$. Then,

$$(3.1) \quad j(2p, \vec{n}, z) \mid [A]_1 = (-1)^{L(\vec{n}, a, p)} \exp\left(\frac{2\pi i abm}{p}\right) \eta(2pz)^{n_0} \prod_{k=1}^p f_{2p, ka}(z)^{n_k},$$

where

$$L(\vec{n}, a, p) = \sum_{k=1}^p \left\lfloor \frac{ka}{2p} \right\rfloor n_k.$$

for

$$A = \begin{pmatrix} a & b \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Also

$$(3.2) \quad j(2p, \pi_r(\vec{n}), z) \in \mathfrak{F}(m', p),$$

where $r = 1, 3, \dots, p-2$ and $m' \equiv r^2 m \pmod{2p}$.

Proof. Following the proof of Theorem 3.1, we use the transformation for $\eta(z)$ and $f_{p,k}(z)$ to get

$$j(2p, \vec{n}, z) \mid [A]_1 = (-1)^{L_1(A)} \exp\left(\frac{\pi i ab}{2p} L_2(A)\right) \nu_\eta^{L_3(A)}({}^{2p}A) \eta(2pz)^{n_0} \prod_{k=1}^p f_{2p, ka}(z)^{n_k}$$

where

$$\begin{aligned} L_1(A) &= b \sum_{k=1}^p k n_k + \sum_{k=1}^p \left\lfloor \frac{ka}{2p} \right\rfloor n_k, \\ L_2(A) &= \sum_{k=1}^p k^2 n_k, \\ L_3(A) &= n_0 + 3 \sum_{k=1}^p n_k. \end{aligned}$$

Since $j(2p, \vec{n}, z) \in \mathfrak{F}(m, p)$, we have that $L_3(A) \equiv 0 \pmod{24}$ and we can also deduce from the third modularity condition that

$$\sum_{k=1}^p k n_k \pmod{2} \equiv \sum_{k=1}^p k^2 n_k \pmod{2} \equiv 0 \pmod{2}.$$

It suffices to prove that

$$\frac{ab}{2}L_2(A) - 2abm$$

is an even multiple of p , which is easy to see since $L_2(A) \equiv 4m \pmod{4p}$. Equation (3.1) follows.

Now suppose $1 \leq r \leq p$ so that for $1 \leq i \leq p$ we have $\pi_r(i) = i'$ where $1 \leq i' \leq p$ and $ri' \equiv \pm i \pmod{2p}$. We note that

$$\begin{aligned} \sum_{k=1}^p n_k &= \sum_{k=1}^p n_{\pi_r(k)}, \text{ and} \\ \sum_{k=1}^p k^2 n_{\pi_r(k)} &= \sum_{k=1}^p k^2 n_{k'} \equiv r^2 \sum_{k'=1}^p (k')^2 n_{k'} \pmod{2p}. \end{aligned}$$

Equation (3.2) follows easily. $\square$

4. TRANSFORMATION, MODULARITY AND SYMMETRY OF THE FIRST RESIDUAL CRANK OF OVERPARTITIONS

The two variable generating function for the first residual crank of an overpartition given by Bringmann, Lovejoy and Osburn [4] is

$$\mathcal{O}_C(z, \tau) = \sum_{n=0}^{\infty} \sum_m \overline{M}(m, n) z^m q^n = (-q; q)_{\infty} \frac{(q; q)_{\infty}}{(zq; q)_{\infty} (z^{-1}q; q)_{\infty}}.$$

We write this function in terms of the Klein form in Definition 2.2.

$$\begin{aligned} \mathcal{O}_C^*(\zeta_p^a, \tau) &= \frac{\eta(\tau)^2}{\eta(2\tau)} \mathcal{O}_C(\zeta_p^a, \tau) \\ &= \frac{\eta(\tau)^2}{\eta(2\tau)} \frac{(-q; q)_{\infty}}{(q; q)_{\infty}} \frac{(q; q)_{\infty}^2}{(\zeta_p^a q; q)_{\infty} (\zeta_p^{-a} q; q)_{\infty}} \\ &= \frac{(q; q)_{\infty}^2}{(\zeta_p^a q; q)_{\infty} (\zeta_p^{-a} q; q)_{\infty}} \\ &= \left(\exp\left(\frac{\pi i a}{p}\right) - \exp\left(\frac{-\pi i a}{p}\right) \right) \frac{1}{t_{(0, \frac{a}{p})}(\tau)}, \end{aligned}$$

where

$$(4.1) \quad t_{(0, \frac{a}{p})}(\tau) = -\exp\left(-\frac{\pi i a}{p}\right) (1 - \zeta_p^a) \frac{(\zeta_p^a q; q)_{\infty} (\zeta_p^{-a} q; q)_{\infty}}{(q; q)_{\infty}^2}.$$

4.1. Transformation and modularity of the crank and associated functions.

Theorem 4.1. *Let $p \geq 3$ be prime. Then $\mathcal{O}_C^*(\zeta_p^a, \tau)$ is a weakly holomorphic modular form of weight 1 on $\Gamma_0(p^2) \cap \Gamma_1(2p)$.*

Proof. Let

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \Gamma_0(p^2) \cap \Gamma_1(2p).$$

Using the transformations in Proposition 2.3, we have

$$\begin{aligned} t_{(0, \frac{a}{p})}(\tau) |[A]_1 &= t_{(\frac{a\gamma}{p}, \frac{a\delta}{p})}(\tau) \\ &= t_{(0 + \frac{a\gamma}{p}, \frac{a\delta}{p} + 0)}(\tau) \\ &= (-1)^{\frac{a\gamma}{p}} \exp \left(-\pi i \left(\frac{a^2 \gamma \delta}{p^2} - 0 \right) \right) t_{(0, \frac{a\delta}{p})}(\tau). \end{aligned}$$

Then

$$\begin{aligned} \mathcal{O}_C^*(\zeta_p^a, \tau) |[A]_1 &= \frac{\exp \left(\frac{\pi i a}{p} \right) - \exp \left(\frac{-\pi i a}{p} \right)}{\exp \left(\frac{\pi i a \delta}{p} \right) - \exp \left(\frac{-\pi i a \delta}{p} \right)} (-1)^{a\gamma} \exp \left(\frac{\pi i a^2 \gamma \delta}{p^2} \right) \mathcal{O}_C^*(\zeta_p^{a\delta}, \tau) \\ &= \frac{\sin \left(\frac{a\pi}{p} \right)}{\sin \left(\frac{a\delta\pi}{p} \right)} (-1)^{a\gamma} \exp \left(\frac{\pi i a^2 \gamma \delta}{p^2} \right) \mathcal{O}_C^*(\zeta_p^{a\delta}, \tau) \\ &= (-1)^{a\gamma} \exp \left(\frac{\pi i a^2 \gamma \delta}{p^2} \right) \mathcal{O}_C^*(\zeta_p^{a\delta}, \tau), \end{aligned}$$

since

$$\sin \left(\frac{a\pi}{p} \right) = \sin \left(\frac{a\delta\pi}{p} \right) \text{ when } \delta \equiv 1 \pmod{2p}.$$

□

Theorem 4.2. *Let*

$$X_p(\tau) = \frac{\eta(p^2\tau)^2 \eta(2\tau)}{\eta(2p^2\tau) \eta(\tau)^2}.$$

Then $X_p(\tau)$ *is a modular function on* $\Gamma_0(2p^2)$.

Proof. $f(z) = \prod_{0 < \delta | N} \eta(q^\delta)^{r_\delta}$ is a weak modular form of weight $k = \frac{1}{2} \sum_{0 < \delta | N} r_\delta$ on $\Gamma_0(N)$ if it satisfies the following :

$$\begin{aligned} \sum_{0 < \delta | N} \delta r_\delta &\equiv 0 \pmod{24}, \\ \sum_{0 < \delta | N} \frac{N}{\delta} r_\delta &\equiv 0 \pmod{24}. \end{aligned}$$

Let $\delta | 2p^2$. Then for $f(z) = X_p(\tau)$, we have

$$\sum_{0 < \delta | 2p^2} \delta r_\delta = -2 + 2 + 2p^2 - 2p^2 = 0,$$

$$\sum_{0 < \delta | N} \frac{2p^2}{\delta} r_\delta = -4p^2 + p^2 + 4 - 1 = 3(1 - p^2) \equiv 0 \pmod{24}.$$

Thus, $X_p(\tau)$ is a modular function on $\Gamma_0(2p^2)$. $\square$

The modularity of the overpartition crank function stated in the introduction section as Theorem 1.1 now deduces easily as a corollary. We restate it here.

Corollary 4.3. *Let $p \geq 3$ be prime. Then $X_p(\tau) \mathcal{O}_C^*(\zeta_p^a, \tau)$ is a modular form of weight 1 on $\Gamma_0(p^2) \cap \Gamma_1(2p)$.*

Next, we consider the elements of the p -dissection of $X_p(\tau) \mathcal{O}_C^*(\zeta_p^a, \tau)$ defined in Definition 1.2. Using the definition of the Atkin operator Equation (2.7), we can equivalently write them as

$$(4.2) \quad \mathcal{OK}_{p,m}^{(C)}(\zeta_p^\ell, z) = X_p(\tau) \mathcal{O}_C^*(\zeta_p^\ell, \tau) \mid [U_{p,m}]_1.$$

The following theorem accounts for the modularity of $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ under special congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$.

Theorem 4.4. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

- (i) $\mathcal{OK}_{p,0}^{(C)}(\zeta_p, z)$ is a weakly holomorphic modular form of weight 1 on $\Gamma_1(2p)$.
- (ii) If $1 \leq m \leq (p-1)$ then $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ is a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$. In particular,

$$\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z) \mid [A]_1 = \exp\left(\frac{2\pi i b m}{p}\right) \mathcal{OK}_{p,m}^{(C)}(\zeta_p, z),$$

$$\text{for } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p).$$

Proof. We let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p)$$

so that $a \equiv d \equiv 1 \pmod{2p}$ and $c \equiv 0 \pmod{2p}$. Let $0 \leq k \leq p-1$. We take $k' \equiv b+k \pmod{p}$ so that

$$T_k A = B_k T_{k'},$$

and

$$B_k = T_k A T_{k'}^{-1} = \begin{pmatrix} a + ck & \frac{1}{p}(-k'(a + kc) + b + kd) \\ pc & d - k'c \end{pmatrix} \in \Gamma_0(p^2) \cap \Gamma_1(2p).$$

Thus,

$$\begin{aligned} & \mathcal{OK}_{p,m}^{(C)}(\zeta_p, z) \mid [A]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \exp\left(-\frac{2\pi i k m}{p}\right) X_p(\tau) \mathcal{O}_C^*(\zeta_p, \tau) \mid [(T_k A)]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \exp\left(-\frac{2\pi i (k' - b)m}{p}\right) X_p(\tau) \mathcal{O}_C^*(\zeta_p, \tau) \mid [(B_k T_{k'})]_1 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{p}} \exp\left(\frac{2\pi ibm}{p}\right) \sum_{k'=0}^{p-1} \exp\left(-\frac{2\pi ik'm}{p}\right) X_p(\tau) \mathcal{O}_C^*(\zeta_p, \tau) \mid [T_{k'}]_1 \\
&\quad (\text{by Corollary 4.3 since } B_k \in \Gamma_0(p^2) \cap \Gamma_1(2p)) \\
&= \exp\left(\frac{2\pi ibm}{p}\right) \mathcal{OK}_{p,m}^{(C)}(\zeta_p, z),
\end{aligned}$$

as required. Thus each function $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ has the desired transformation property. It is clear that each $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ is holomorphic on $\mathcal{H}$. The cusp conditions follow by a standard argument. We examine orders at each cusp in more detail in a later section. $\square$

4.2. Overpartition crank symmetry. We finally present a proof of our result on the observation of symmetry among the elements of the p -dissection of the overpartition crank function, which was stated in the introduction section in Theorem 1.3. We restate it here.

Theorem 4.5. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

$$(4.3) \quad \mathcal{OK}_{p,m}^{(C)}(\zeta_p^\ell, z) \mid [A]_1 = \frac{\sin\left(\frac{\ell\pi}{p}\right)}{\sin\left(\frac{\ell d\pi}{p}\right)} \zeta_p^{mak} \mathcal{OK}_{p,ma^2}^{(C)}(\zeta_p^{\ell d}, z)$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. We undergo the same matrix transformations as we did in the case of $\mathcal{K}_{p,m}(\zeta_p, z)$

([11], Proposition 4.7), and for $0 \leq r \leq p-1$ let $T_r = \begin{pmatrix} 1 & r \\ 0 & p \end{pmatrix}$, and

$$B_r = \begin{pmatrix} a + 2pr & \frac{1}{p}(k + rd - r'(a + 2pr)) \\ 2p^2 & d - 2r'p \end{pmatrix},$$

where $0 \leq r' \leq p-1$ is chosen so that $r' \equiv rd^2 + dk \pmod{p}$. Then

$$T_r A = B_r T_{r'}, \quad r \equiv r'a^2 - ak \pmod{p}, \quad \text{and} \quad B_r \in \Gamma_0(2p^2).$$

$$\begin{aligned}
\mathcal{OK}_{p,m}^{(C)}(\zeta_p^\ell, z) \mid [A]_1 &= X_p(\tau) \mathcal{O}_C^*(\zeta_p^\ell, \tau) \mid [U_{p,m}]_1 \mid [A]_1 \\
&= \frac{1}{\sqrt{p}} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) \mathcal{O}_C^*(\zeta_p^\ell, \tau) \mid [T_r]_1 \mid [A]_1 \\
&= \frac{1}{\sqrt{p}} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) \mathcal{O}_C^*(\zeta_p^\ell, \tau) \mid [B_r]_1 \mid [T_{r'}]_1 \\
&= \frac{1}{\sqrt{p}} \frac{\sin\left(\frac{\ell\pi}{p}\right)}{\sin\left(\frac{\ell(d-2r'p)\pi}{p}\right)} (-1)^{\ell(2p^2)} \exp\left(\frac{\pi i \ell^2 2p^2 (d - 2r'p)}{p^2}\right)
\end{aligned}$$

$$\begin{aligned}
& \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) \mathcal{O}_C^*(\zeta_p^{\ell(d-2r'p)}, \tau) \mid [T_{r'}]_1 \\
& \text{(using the second step of the transformation of } \mathcal{O}_C^*(\zeta_p^a, \tau) \mid [A]_1 \\
& \text{in the proof of Theorem 4.1)} \\
& = \frac{1}{\sqrt{p}} \frac{\sin\left(\frac{\ell\pi}{p}\right)}{\sin\left(\frac{\ell d\pi}{p}\right)} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) \mathcal{O}_C^*(\zeta_p^{\ell d}, \tau) \mid [T_{r'}]_1 \\
& = \frac{\sin\left(\frac{\ell\pi}{p}\right)}{\sin\left(\frac{\ell d\pi}{p}\right)} \zeta_p^{mak} \frac{1}{\sqrt{p}} \sum_{r'=0}^{p-1} \zeta_p^{-r'ma^2} X_p(\tau) \mathcal{O}_C^*(\zeta_p^{\ell d}, \tau) \mid [T_{r'}]_1,
\end{aligned}$$

since

$$\zeta_p^{-rm} = \zeta_p^{m(-r'a^2+ak)} = \zeta_p^{mak} \zeta_p^{-mr'a^2},$$

and as r runs through a complete residue system mod p so does r' . The result follows. $\square$

5. TRANSFORMATION, MODULARITY AND SYMMETRY OF DYSON'S RANK OF OVERPARTITIONS

Similar to the transformation, modularity and symmetry of the rank generating function for partitions, in this section, we establish analogous results for the overpartition rank generating function.

5.1. Transformation and modularity of the rank and associated functions.

Proposition 5.1. *Let $p \geq 3$ be prime and $1 \leq \ell \leq (p-1)$. Define*

$$N^*(\ell, p; \tau) := \frac{\eta(\tau)^2}{\eta(2\tau)} N(\ell, p; \tau),$$

$$P^*(\ell, p; \tau) := \frac{\eta(\tau)^2}{\eta(2\tau)} P(\ell, p; \tau).$$

Then

$$N^*(\ell, p; \tau) \mid [A]_1 = \mu(A, \ell) N^*(d\ell, p; \tau),$$

$$P^*(\ell, p; \tau) \mid [A]_1 = \mu(A, \ell) P^*(d\ell, p; \tau),$$

where

$$\mu(A, \ell) = \exp\left(-\pi i \left(\frac{2\ell(1-d)}{p} - \frac{2cd\ell^2}{p^2}\right)\right)$$

and

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. Knopp's formula for the eta multiplier gives

$$\frac{\eta(\tau)^2}{\eta(2\tau)}|[A]_1 = \frac{\nu_\eta(A)^2}{\nu_\eta(2A)} \frac{\eta(\tau)^2}{\eta(2\tau)} = \nu_\eta(2A)^3 (-1)^{-(b+\frac{a-1}{2})} i^{ab}.$$

Using [13, Corollary 3.2] we get

$$N(\ell, p; \tau) |[A]_{\frac{1}{2}} = \nu_\eta(2A)^{-3} (-1)^{b+\frac{a-1}{2}+\frac{ac}{p}} \exp \left(-\pi i \left(\frac{-2\ell^2 cd}{p^2} + \frac{2\ell(1-d)}{p} + \frac{ab}{2} \right) \right) N(d\ell, p; \tau).$$

Therefore,

$$N^*(\ell, p; \tau) |[A]_{\frac{1}{2}} = \exp \left(-\pi i \left(\frac{2\ell(1-d)}{p} - \frac{2cd\ell^2}{p^2} \right) \right) N^*(d\ell, p; \tau).$$

And, writing $P(\ell, p, \tau)$ in terms of the Klein form (Equation (2.5)), we get

$$P(\ell, p, \tau) = -\zeta_p^{-\ell} \frac{\eta(2\tau)}{\eta(\tau)^2} \frac{1}{t_{(0, \frac{2\ell}{p})}(2\tau)}.$$

Therefore, using the transformations in Proposition 2.3, we get

$$\begin{aligned} P^*(\ell, p; \tau) |[A]_1 &= -\zeta_p^\ell \frac{1}{t_{(0, \frac{2\ell}{p})}(2\tau)} |[A]_1 \\ &= -\zeta_p^\ell \frac{1}{(-1)^{\frac{\ell c}{p}} \exp \left(-\pi i \left(\frac{2\ell^2 c(d-1)}{p^2} \right) \right) t_{(\frac{\ell c}{p}, \frac{2\ell d}{p})}(2\tau)} \\ &= -\zeta_p^\ell (-1)^{\frac{\ell c}{p}} \exp \left(\pi i \left(\frac{2\ell^2 cd}{p^2} \right) \right) \frac{1}{t_{(0, \frac{2\ell d}{p})}(2\tau)} \\ &= \exp \left(-\pi i \left(\frac{2\ell(1-d)}{p} - \frac{2cd\ell^2}{p^2} \right) \right) P^*(d\ell, p; \tau). \end{aligned}$$

□

The following corollary now follows easily.

Corollary 5.2. *Let $p \geq 3$ be prime and $1 \leq \ell \leq (p-1)$. Then*

$$N^*(\ell, p; \tau) |[A]_1 = N^*(\ell, p; \tau),$$

$$P^*(\ell, p; \tau) |[A]_1 = P^*(\ell, p; \tau),$$

for

$$A \in \Gamma_0(2p^2) \cap \Gamma_1(p).$$

Now, using Theorem 1.5 and Proposition 3.5 [13], we have

$$\begin{aligned} N(\ell, p; \tau) &= \frac{-(1+\zeta_p^\ell)}{2i\zeta_p^\ell(1-\zeta_p^\ell)} \mathcal{O}_{\mathcal{R}}(\zeta_p^\ell; \tau) + \frac{1}{i} P(\ell, p; \tau) + \frac{\zeta_p^{-\ell}}{\sqrt{2}} \int_{-\bar{\tau}}^{i\infty} \frac{g_{0, \frac{1}{2}-\frac{2\ell}{p}}(2z)}{\sqrt{-i(z+\tau)}} dz \\ &= \frac{-(1+\zeta_p^\ell)}{2i\zeta_p^\ell(1-\zeta_p^\ell)} \mathcal{O}_{\mathcal{R}}(\zeta_p^\ell; \tau) + \frac{1}{i} P(\ell, p; \tau) + \frac{i\zeta_p^{-\ell}}{2\sqrt{\pi}} \sum_{n=1}^{\infty} (-1)^n (\zeta_p^{-2n\ell} - \zeta_p^{2n\ell}) \Gamma\left(\frac{1}{2}; 4\pi y n^2\right) q^{-n^2} \end{aligned}$$

$$\begin{aligned}
&= \frac{-(1 + \zeta_p^\ell)}{2i\zeta_p^\ell(1 - \zeta_p^\ell)} \mathcal{O}_{\mathcal{R}}(\zeta_p^\ell; \tau) + \frac{1}{i} P(\ell, p; \tau) + \frac{i\zeta_p^{-\ell}}{2\sqrt{\pi}} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) \\
&\quad \left(\sum_{n=0}^{\infty} (-1)^n q^{-(pn+k)^2} \Gamma\left(\frac{1}{2}; 4\pi y(pn+k)^2\right) - \sum_{n=1}^{\infty} (-1)^n q^{-(pn-k)^2} \Gamma\left(\frac{1}{2}; 4\pi y(pn-k)^2\right) \right).
\end{aligned}$$

In Section 2 [13], the author defines

$$N_7(k; \tau) = q^{-k^2+7k-\frac{49}{4}} \tilde{\mu}(14k\tau, 49\tau; 98\tau).$$

For an arbitrary prime $p \geq 3$, we define the generalized function

$$N_p(k; \tau) = q^{-(k-p/2)^2} \tilde{\mu}(2pk\tau, p^2\tau; 2p^2\tau).$$

Then

$$\begin{aligned}
N_p(k; \tau) &= q^{-2p^2\frac{1}{2}(\frac{k}{p}-\frac{1}{2})^2} \tilde{\mu}(2pk\tau, p^2\tau; 2p^2\tau) \\
(5.1) \quad &= M(k, p; 2p^2\tau),
\end{aligned}$$

where $M(a, c; \tau) = q^{-\frac{1}{2}(\frac{a}{c}-\frac{1}{2})^2} \tilde{\mu}(\frac{a\tau}{c}, \frac{\tau}{2}; \tau)$ is defined by Jennings-Shaffer in Section 5 [13]. This helps in determining the transformation of $N_p(k; \tau)$ under matrices in $SL_2(\mathbb{Z})$ using Proposition 5.1 [13].

Also, using Proposition 1.3 and 1.4 [21], we have

$$\mu(2pk\tau, p^2\tau; 2p^2\tau) = iq^{pk+\frac{p^2}{4}} \frac{(q^{2p^2}; q^{2p^2})_{\infty}}{(q^{p^2}; q^{p^2})_{\infty}^2} \sum_{n=-\infty}^{\infty} (-1)^n \frac{q^{p^2n(n+2)}}{1 - q^{2p(pn+k)}}.$$

Further, we define

$$N_p^*(k; \tau) = \frac{\eta(\tau)^2}{\eta(2\tau)} N_p(k; \tau).$$

We deduce the transformation of this function under $\Gamma_0(2p^2) \cap \Gamma_1(p)$ like the previous two functions in Corollary 5.2.

Proposition 5.3. *Let $p \geq 3$ be prime. Then*

$$N_p^*(\ell; \tau) | [A]_1 = N_p^*(a\ell; \tau)$$

for

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(2p^2).$$

Proof. We have

$$N_p^*(k; \tau) = \frac{\eta(\tau)^2}{\eta(2\tau)} N_p(k; \tau) = \frac{\eta(\tau)^2}{\eta(2\tau)} M(k, p; 2p^2\tau) = \frac{\eta(\tau)^2}{\eta(2\tau)} q^{-(k-\frac{p}{2})^2} \tilde{\mu}(2pk\tau, p^2\tau; 2p^2\tau).$$

Knopp's formula for the eta multiplier gives

$$\frac{\eta(\tau)^2}{\eta(2\tau)} | [A]_{\frac{1}{2}} = \frac{\nu_{\eta}(A)^2}{\nu_{\eta}(2A)} \frac{\eta(\tau)^2}{\eta(2\tau)} = \nu_{\eta}(2A)^3 (-1)^{-(b+\frac{a-1}{2})} i^{ab}.$$

Using the transformation for $\mu(u, v; \tau)$ [21, Theorem 1.1 (1)] due to Zwegers, we get

$$\begin{aligned}
N_p(k; \tau) | [A]_{\frac{1}{2}} &= M(k, p; 2p^2\tau) | [A]_{\frac{1}{2}} \\
&= M\left(k, p; 2p^2 \frac{a\tau + b}{c\tau + d}\right) \\
&= M\left(k, p; \tilde{A}(2p^2\tau)\right) \text{ where } \tilde{A} = \begin{pmatrix} a & 2p^2b \\ c/2p^2 & d \end{pmatrix} \\
&= \nu_\eta(\tilde{A})^{-3} \exp\left(-\pi ia \cdot 2p^2b \left(\frac{k}{p} - \frac{1}{2}\right)^2 - \pi ia^2 \cdot 2p^2\tau \left(\frac{k}{p} - \frac{1}{2}\right)^2\right) \\
&\quad \tilde{\mu}\left(\frac{ka \cdot 2p^2\tau + k \cdot 2p^2b}{p}, \frac{a \cdot 2p^2\tau + 2p^2b}{2}; 2p^2\tau\right) \\
&= \nu_\eta(\tilde{A})^{-3} \exp\left(-2\pi iabk^2 - \pi i \frac{p^2ab}{2} + 2\pi ipabk - \pi ia^2 \cdot 2p^2\tau \left(\frac{k}{p} - \frac{1}{2}\right)^2\right) \\
&\quad \tilde{\mu}(2kap\tau + 2apb, ap^2\tau + p^2b; 2p^2\tau) \\
&= \nu_\eta(\tilde{A})^{-3} (-i)^{p^2ab} \exp\left(\pi ia^2 \cdot 2p^2\tau \left(\frac{k}{p} - \frac{1}{2}\right)^2\right) (-1)^{p^2b} \tilde{\mu}(2kap\tau, ap^2\tau; 2p^2\tau) \\
&= \nu_\eta(\tilde{A})^{-3} (-i)^{p^2ab} \exp\left(-\pi ia^2 \cdot 2p^2\tau \left(\frac{k}{p} - \frac{1}{2}\right)^2\right) (-1)^{p^2b} \\
&\quad \tilde{\mu}\left(2kap\tau, p^2\tau + \frac{a-1}{2} \cdot 2p^2\tau; 2p^2\tau\right) \\
&= \nu_\eta(\tilde{A})^{-3} (-i)^{p^2ab} \exp\left(-\pi ia^2 \cdot 2p^2\tau \left(\frac{k}{p} - \frac{1}{2}\right)^2\right) (-1)^{p^2b} \\
&\quad (-1)^{\frac{a-1}{2}} \exp\left(\pi i \left(\frac{a-1}{2}\right)^2 \cdot 2p^2\tau - 2\pi i \left(\frac{a-1}{2}\right) (2kap\tau - p^2\tau)\right) \\
&\quad \tilde{\mu}(2kap\tau, p^2\tau; 2p^2\tau) \\
&= \nu_\eta(\tilde{A})^{-3} (-1)^{b + \frac{a-1}{2} + ab} i^{p^2ab} q^{-p^2\left(\frac{ak}{p} - \frac{1}{2}\right)^2} \tilde{\mu}(2pka\tau, p^2\tau; 2p^2\tau).
\end{aligned}$$

Combining the two transformation results, we have the statement of the proposition. $\square$

Corollary 5.4. *Let $p \geq 3$ be prime. Then*

$$N_p^*(\ell; \tau) | [A]_1 = N_p^*(\ell; \tau).$$

for $A \in \Gamma_0(2p^2) \cap \Gamma_1(p)$.

Next, following the proof of Proposition 5.4 [13], we can deduce that

$$N_p(k; \tau) = q^{-(k-p/2)^2} \mu(2pk\tau, p^2\tau; 2p^2\tau) + \frac{i}{2\sqrt{\pi}}.$$

$$\begin{aligned}
& \left(\sum_{n=0}^{\infty} (-1)^n q^{-(pn+k)^2} \Gamma \left(\frac{1}{2}; 4\pi y(pn+k)^2 \right) - \sum_{n=1}^{\infty} (-1)^n q^{-(pn-k)^2} \Gamma \left(\frac{1}{2}; 4\pi y(pn-k)^2 \right) \right) \\
&= i q^{2pk-k^2} \frac{(q^{2p^2}; q^{2p^2})_{\infty}}{(q^{p^2}; q^{p^2})_{\infty}^2} \sum_{n=-\infty}^{\infty} (-1)^n \frac{q^{p^2 n(n+2)}}{1 - q^{2p(pn+k)}} + \\
& \frac{i}{2\sqrt{\pi}} \left(\sum_{n=0}^{\infty} (-1)^n q^{-(pn+k)^2} \Gamma \left(\frac{1}{2}; 4\pi y(pn+k)^2 \right) - \sum_{n=1}^{\infty} (-1)^n q^{-(pn-k)^2} \Gamma \left(\frac{1}{2}; 4\pi y(pn-k)^2 \right) \right).
\end{aligned}$$

Subtracting the expressions for $N(\ell, p; \tau)$ and $N_p(k; \tau)$ with k summed over 1 to $\frac{p-1}{2}$, the non-holomorphic parts cancel out and we get

$$\begin{aligned}
N(\ell, p; \tau) - \zeta_p^{-\ell} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) N_p(k; \tau) &= \frac{-(1 + \zeta_p^{\ell})}{2i \zeta_p^{\ell} (1 - \zeta_p^{\ell})} \mathcal{O}_{\mathcal{R}}(\zeta_p^{\ell}; \tau) + \frac{1}{i} P(\ell, p; \tau) + \\
& i \zeta_p^{-\ell} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) q^{pk-k^2} \frac{(q^{2p^2}; q^{2p^2})_{\infty}}{(q^{p^2}; q^{p^2})_{\infty}^2} \sum_{n=-\infty}^{\infty} (-1)^n \frac{q^{p^2 n(n+1)}}{1 - q^{p^2 n+k}},
\end{aligned}$$

which is weakly holomorphic. Rearranging the terms we get

$$\begin{aligned}
\mathcal{O}_{\mathcal{R}}(\zeta_p^{\ell}; \tau) - \frac{2(1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) q^{pk-k^2} \frac{(q^{2p^2}; q^{2p^2})_{\infty}}{(q^{p^2}; q^{p^2})_{\infty}^2} \sum_{n=-\infty}^{\infty} (-1)^n \frac{q^{p^2 n(n+1)}}{1 - q^{p^2 n+k}} \\
= \frac{-2i \zeta_p^{\ell} (1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} N(\ell, p; \tau) + \frac{2i(1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) N_p(k; \tau) + \frac{2\zeta_p^{\ell} (1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} P(\ell, p; \tau).
\end{aligned}$$

Let

$$\begin{aligned}
\mathcal{J} \left(\frac{\ell}{p}; \tau \right) &:= \frac{-2i \zeta_p^{\ell} (1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} N(\ell, p; \tau) + \frac{2i(1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) N_p(k; \tau) \\
(5.2) \quad &+ \frac{2\zeta_p^{\ell} (1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} P(\ell, p; \tau).
\end{aligned}$$

Definition 5.5. For $p \geq 3$ prime and $1 \leq \ell \leq p-1$ define

$$(5.3) \quad \mathcal{J}^* \left(\frac{\ell}{p}; \tau \right) := X_p(\tau) \mathfrak{D}_R^*(\zeta_p^{\ell}, \tau)$$

where recall $X_p(\tau) = \frac{\eta(p^2\tau)^2 \eta(2\tau)}{\eta(2p^2\tau) \eta(\tau)^2}$, and

$$\mathfrak{D}_R^*(\zeta_p^{\ell}, \tau) = \frac{-2i \zeta_p^{\ell} (1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} N^*(\ell, p; \tau) + \frac{2i(1 - \zeta_p^{\ell})}{(1 + \zeta_p^{\ell})} \sum_{k=1}^{\frac{p-1}{2}} (-1)^k (\zeta_p^{-2\ell k} - \zeta_p^{2\ell k}) N_p^*(k; \tau)$$

(5.4)

$$+ \frac{2\zeta_p^\ell(1-\zeta_p^\ell)}{(1+\zeta_p^\ell)} P^*(\ell, p; \tau).$$

Now, with the correction factor $\Phi_{p,k}(q)$ and the corrected overpartition rank function $\mathfrak{D}_R(\zeta_p, \tau)$ defined as in Definitions (1.5) and (1.6) respectively, we can easily see that

$$(5.5) \quad \frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathfrak{D}_R(\zeta_p^\ell, \tau) = \frac{\eta(p^2\tau)^2}{\eta(2p^2\tau)} \mathcal{J}\left(\frac{\ell}{p}; \tau\right) = X_p(\tau) \mathfrak{D}_R^*(\zeta_p^\ell, \tau) = \mathcal{J}^*\left(\frac{\ell}{p}; \tau\right).$$

Thus we rewrite one of our main results, Theorem 1.8, in the equivalent form :

Theorem 5.6. *Let $p \geq 3$ be prime. Then the function $\mathcal{J}^*\left(\frac{1}{p}; \tau\right)$ is a weakly holomorphic modular form of weight 1 on $\Gamma_0(2p^2) \cap \Gamma_1(p)$.*

Proof. This follows easily from combining Theorem 4.2, Corollary 5.2 and Corollary 5.4. $\square$

Again, the functions $\mathcal{OK}_{p,m}^{(R)}(\zeta_p^\ell; \tau)$ as in Definition 1.9 can be equivalently written as

Proposition 5.7. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

$$(5.6) \quad \mathcal{OK}_{p,m}^{(R)}(\zeta_p^\ell; \tau) = \mathcal{J}^*\left(\frac{\ell}{p}; \tau\right) \mid [U_{p,m}]_1.$$

Next, the following theorem accounts for the modularity of $\mathcal{OK}_{p,m}^{(R)}(\zeta_p^\ell, z)$ under special congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$.

Theorem 5.8. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

- (i) $\mathcal{OK}_{p,0}^{(R)}(z)$ is a weakly holomorphic modular form of weight 1 on $\Gamma_1(2p)$.
- (ii) If $1 \leq m \leq (p-1)$ then $\mathcal{OK}_{p,m}^{(R)}(z)$ is a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$. In particular,

$$\mathcal{OK}_{p,m}^{(R)}(z) \mid [A]_1 = \exp\left(\frac{2\pi ibm}{p}\right) \mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$$

$$\text{for } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(2p).$$

Proof. We consider the matrix in $\Gamma_1(2p)$ and go through the same matrix transformations as in the proof of Theorem 4.4 in the previous section to arrive at our result using Theorem 5.6 and Proposition 5.7. $\square$

5.2. Overpartition rank symmetry result. We now investigate how the Atkin operator and a matrix in $\Gamma_0(2p)$ acts on each function of (5.3) to arrive at our symmetry result for the overpartition rank.

Proposition 5.9. *Let $p \geq 3$ be prime. Then*

$$X_p(\tau) N^*(1, p; \tau) \mid [U_{p,m}]_1 \mid [A]_1 = \zeta_p^{mak+d-1} X_p(\tau) N^*(d, p; \tau) \mid \left[U_{p, \overline{ma^2}} \right]_1$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. For $0 \leq r \leq p-1$ let $T_r = \begin{pmatrix} 1 & r \\ 0 & p \end{pmatrix}$, and $B_r = \begin{pmatrix} a + 2pr & \frac{1}{p}(k + rd - r'(a + 2pr)) \\ 2p^2 & d - 2r'p \end{pmatrix}$, where $0 \leq r' \leq p-1$ is chosen so that $r' \equiv rd^2 + dk \pmod{p}$. Then

$$T_r A = B_r T_{r'}, \quad r \equiv r'a^2 - ak \pmod{p}, \quad \text{and} \quad B_r \in \Gamma_0(2p^2).$$

We apply Proposition 5.1 and Theorem 4.2. We have

$$\begin{aligned} X_p(\tau) N^*(1, p; \tau) \mid [U_{p,m}]_1 \mid [A]_1 &= \frac{1}{\sqrt{p}} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) N^*(1, p; \tau) \mid [T_r]_1 \mid [A]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) N^*(1, p; \tau) \mid [B_r]_1 \mid [T_{r'}]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p(\tau) \mu(B_r, 1) N^*(d, p; \tau) \mid [T_{r'}]_1 \\ &= \frac{1}{\sqrt{p}} \zeta_p^{mak+d-1} \sum_{r'=0}^{p-1} \zeta_p^{-r'ma^2} X_p(\tau) N^*(d, p; \tau) \mid [T_{r'}]_1, \end{aligned}$$

$$\text{since } \mu(B_r, 1) = \exp \left(-\pi i \left(\frac{2(1-d+2r'p)}{p} - \frac{4p^2(d-2r'p)}{p^2} \right) \right),$$

$$\zeta_p^{-rm} = \zeta_p^{m(-r'a^2+ak)} = \zeta_p^{mak} \zeta_p^{-mr'a^2},$$

and as r runs through a complete residue system mod p so does r' . The result follows. $\square$

Proposition 5.10. *Let $p \geq 3$ be prime. Then*

$$X_p(\tau) P^*(1, p; \tau) \mid [U_{p,m}]_1 \mid [A]_1 = \zeta_p^{mak+d-1} X_p(\tau) P^*(d, p; \tau) \mid \left[U_{p, \overline{ma^2}} \right]_1$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. The proof follows similarly using Proposition 5.1. $\square$

Proposition 5.11. *Let $p \geq 3$ be prime. Then*

$$X_p(\tau) N_p^*(\ell; \tau) \mid [U_{p,m}]_1 \mid [A]_1 = \zeta_p^{mak} X_p(\tau) N_p^*(a\ell; \tau) \mid \left[U_{p, \overline{ma^2}} \right]_1$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. The proof follows using Proposition 5.3 and by going through the same series of matrix transformations and steps as it did for the function $N^*(1, p; \tau)$ in the proof of Proposition 5.9, except for the appearance of the root of unity multiplier $\mu(B_r, 1)$ in that proof which does not appear here due to the exact transformation of $N_p^*(\ell; \tau)$ as seen in Proposition 5.3. $\square$

Proposition 5.12. *Let $p \geq 3$ be prime, $1 \leq \ell, \ell' \leq p-1$, $\ell^2 \equiv -m \pmod{p}$, $\ell'^2 \equiv -ma^2 \pmod{p}$. Then*

$$\begin{aligned} & (-1)^\ell (\zeta_p^{-2\ell} - \zeta_p^{2\ell}) X_p(\tau) N_p^*(\ell; \tau) \mid [U_{p,m}]_1 \mid [A]_1 \\ &= (-1)^{\ell'} (\zeta_p^{-2\ell'd} - \zeta_p^{2\ell'd}) \zeta_p^{mak} X_p(\tau) N_p^*(\ell'; \tau) \mid [U_{p,ma^2}]_1 \end{aligned}$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. The proof follows from the previous proposition and the fact that $\ell' \equiv \pm a\ell \pmod{p}$, a is odd and $ad - 2pk = 1$. $\square$

Proposition 5.13. *Let $1 \leq k \leq \frac{1}{2}(p-1)$. Then*

$$X_p(\tau) N_p^*(k, \tau) \mid [U_{p,m}]_1 = \begin{cases} \frac{\eta(p\tau)^2}{\eta(2p\tau)} N_p(k, \frac{\tau}{p}) & \text{if } k^2 \equiv -m \pmod{p} \\ 0 & \text{otherwise.} \end{cases}$$

Proof.

$$\begin{aligned} X_p(\tau) N_p^*(k, \tau) \mid [U_{p,m}]_1 &= \frac{1}{p} \sum_{r=0}^{p-1} \zeta_p^{-rm} X_p\left(\frac{\tau+r}{p}\right) N_p^*\left(k, \frac{\tau+r}{p}\right) \\ &= \frac{1}{p} \sum_{r=0}^{p-1} \zeta_p^{-rm} \frac{\eta(p\tau + pr)^2}{\eta(2p\tau + 2pr)} \exp\left(-2\pi i \frac{\tau+r}{p} \left(k - \frac{p}{2}\right)^2\right) \\ &\quad \tilde{\mu}\left(2pk \frac{\tau+r}{p}, p^2 \frac{\tau+r}{p}; 2p^2 \frac{\tau+r}{p}\right) \\ &= \frac{1}{p} \sum_{r=0}^{p-1} \zeta_p^{-rm} \frac{\eta(p\tau)^2}{\eta(2p\tau)} \exp\left(-2\pi i \frac{\tau+r}{p} \left(k - \frac{p}{2}\right)^2\right) \\ &\quad \exp\left(\frac{-\pi i}{4} \cdot 2pr + \pi i pr + \pi i\right) \tilde{\mu}(2k\tau, p\tau; 2p\tau) \\ &= \frac{1}{p} \sum_{r=0}^{p-1} \zeta_p^{-rm-rk^2+2rkp} q^{-\frac{(k-\frac{p}{2})^2}{p}} \frac{\eta(p\tau)^2}{\eta(2p\tau)} \tilde{\mu}(2k\tau, p\tau; 2p\tau) \\ &= \frac{1}{p} \sum_{r=0}^{p-1} \zeta_p^{-r(m+k^2)} \frac{\eta(p\tau)^2}{\eta(2p\tau)} N_p\left(k, \frac{\tau}{p}\right). \end{aligned}$$

The result follows since $-r(m+k^2) \equiv 0 \pmod{p}$ if and only if $k^2 \equiv -m \pmod{p}$. $\square$

We finally present a proof of our result on the observation of symmetry among the elements of the p -dissection of the overpartition rank function, which was stated in the introduction section Theorem 1.10. We restate it here.

Theorem 5.14. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

$$(5.7) \quad \mathcal{OK}_{p,m}^{(R)}(\zeta_p, \tau) \mid [A]_1 = \frac{1-\zeta_p}{1-\zeta_p^d} \frac{1+\zeta_p^d}{1+\zeta_p} \zeta_p^{mak} \mathcal{OK}_{p,ma^2}^{(R)}(\zeta_p^d, \tau)$$

for

$$A = \begin{pmatrix} a & k \\ 2p & d \end{pmatrix} \in \Gamma_0(2p).$$

Proof. Proposition 5.7 gives

$$\mathcal{OK}_{p,m}^{(R)}(\zeta_p; \tau) = \mathcal{J}\left(\frac{1}{p}; \tau\right) \mid [U_{p,m}]_1.$$

We consider the following two cases.

CASE 1. $m = 0$ or $\left(\frac{-m}{p}\right) = -1$. In this case

$$\ell^2 \not\equiv -m \pmod{p}, \quad \text{and} \quad \ell'^2 \not\equiv -ma^2 \pmod{p},$$

for $1 \leq \ell, \ell' \leq \frac{1}{2}(p-1)$. The result then follows from Proposition 5.9, 5.10 and 5.13.

CASE 2. $\left(\frac{-m}{p}\right) = 1$. In this case choose $1 \leq \ell, \ell' \leq \frac{1}{2}(p-1)$ such that

$$\ell^2 \equiv -m \pmod{p}, \quad \text{and} \quad \ell'^2 \equiv -ma^2 \pmod{p}.$$

We have

$$\begin{aligned} & \mathcal{J}^*\left(\frac{1}{p}; \tau\right) \mid [U_{p,m}]_1 \mid [A]_1 \\ &= X_p(\tau) \left(\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} N^*(1, p; \tau) + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} \sum_{\ell=1}^{\frac{p-1}{2}} (-1)^\ell (\zeta_p^{-2\ell} - \zeta_p^{2\ell}) N_p^*(\ell; \tau) \right. \\ & \quad \left. + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} P^*(1, p; \tau) \right) \mid [U_{p,m}]_1 \mid [A]_1 \\ &= \left(\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) N^*(d, p; \tau) + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) P^*(d, p; \tau) \right) \mid \left[U_{p,ma^2} \right]_1 \\ & \quad + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} (-1)^\ell (\zeta_p^{-2\ell} - \zeta_p^{2\ell}) N_p^*(\ell; \tau) \mid [U_{p,m}]_1 \mid [A]_1 \text{ (by Propositions 5.9, 5.10 and 5.13)} \\ &= \left(\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) N^*(d, p; \tau) + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) P^*(d, p; \tau) \right) \mid \left[U_{p,ma^2} \right]_1 \\ & \quad + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} (-1)^{\ell'} (\zeta_p^{-2\ell'd} - \zeta_p^{2\ell'd}) \zeta_p^{mak} X_p(\tau) N_p^*(\ell'; \tau) \mid \left[U_{p,ma^2} \right]_1 \text{ (by Proposition 5.12)} \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) N^*(d, p; \tau) + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} \zeta_p^{mak+d-1} X_p(\tau) P^*(d, p; \tau) \right) \Big| \left[U_{p, ma^2} \right]_1 \\
&\quad + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} \sum_{j=1}^{\frac{p-1}{2}} (-1)^j (\zeta_p^{-2jd} - \zeta_p^{2jd}) \zeta_p^{mak} X_p(\tau) N_p^*(j; \tau) \Big| \left[U_{p, ma^2} \right]_1 \\
&= \frac{(1-\zeta_p)}{(1+\zeta_p)} \frac{(1+\zeta_p^d)}{(1-\zeta_p^d)} \zeta_p^{mak} \mathcal{J}^* \left(\frac{d}{p}; \tau \right) \Big| [U_{p, ma^2}]_1.
\end{aligned}$$

This completes the proof. $\square$

6. SYMMETRY OF COEFFICIENTS FOR CRANK AND RANK DISSECTION ELEMENTS CORRESPONDING TO $m = 0$

Our other symmetry results of the paper concern with the symmetries of the zeta-coefficients in identities for $\mathcal{OK}_{p,0}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{p,0}^{(R)}(\zeta_p, z)$ expressed in terms of generalized eta-functions. The symmetry for the latter is analogous to the partition rank case [11, Theorem 5.1]. We state the results for both the dissection elements, prove the one for the rank case and omit the proof for the crank element since it follows similarly.

Theorem 6.1. *Let $p \geq 3$ be prime. Suppose there are t vectors $\vec{n}_1, \vec{n}_2, \dots, \vec{n}_t \in \mathbb{Z}^{p+1}$ such that the set of functions $j(2p, \pi_r(\vec{n}_k), z), 1 \leq k \leq t, r = 1, 3, \dots, p-2$ are linearly independent (over $\mathbb{Q}$) and*

$$\mathcal{OK}_{p,0}^{(C)}(\zeta_p, z) = \sum_{k=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,r,k}(\zeta_p) j(2p, \pi_r(\vec{n}_k), z),$$

where $c_{p,r,k}(\zeta_p) \in \mathbb{Q}[\zeta_p]$. Then for $1 \leq d \leq p-1$, and $\vec{n} = (n_0, n_1, n_2, \dots, n_p)$, we have

$$c_{p,d,k}(\zeta_p) = \frac{\sin(\pi/p)}{\sin(d\pi/p)} (-1)^{L(\vec{n}, d)} c_{p,1,k}(\zeta_p^d)$$

where

$$L(\vec{n}, d) = L(\vec{n}, a, b, d, p) = bd \sum_{k=1}^p kn_k + \sum_{k=1}^p \left(\left\lfloor \frac{dka}{p} \right\rfloor + \left\lfloor \frac{dk}{p} \right\rfloor \right) n_k,$$

and a, b are chosen so that $A = \begin{pmatrix} a & b \\ 2p & d \end{pmatrix} \in \Gamma_0(2p)$.

Theorem 6.2. *Let $p \geq 3$ be prime. Suppose there are t vectors $\vec{n}_1, \vec{n}_2, \dots, \vec{n}_t \in \mathbb{Z}^{p+1}$ such that the set of functions $j(2p, \pi_r(\vec{n}_k), z), 1 \leq k \leq t, r = 1, 3, \dots, p-2$ are linearly independent (over $\mathbb{Q}$) and*

$$\mathcal{OK}_{p,0}^{(R)}(\zeta_p, z) = \sum_{k=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,r,k}(\zeta_p) j(2p, \pi_r(\vec{n}_k), z),$$

where $c_{p,r,k}(\zeta_p) \in \mathbb{Q}[\zeta_p]$. Then for $1 \leq d \leq p-1$, and $\vec{n} = (n_0, n_1, n_2, \dots, n_p)$, we have

$$c_{p,d,k}(\zeta_p) = \frac{1-\zeta_p}{1-\zeta_p^d} \frac{1+\zeta_p^d}{1+\zeta_p} (-1)^{L(\vec{n},d)} c_{p,1,k}(\zeta_p^d)$$

where

$$L(\vec{n}, d) = L(\vec{n}, a, b, d, p) = bd \sum_{k=1}^p k n_k + \sum_{k=1}^p \left(\left\lfloor \frac{dka}{p} \right\rfloor + \left\lfloor \frac{dk}{p} \right\rfloor \right) n_k,$$

and a, b are chosen so that $A = \begin{pmatrix} a & b \\ 2p & d \end{pmatrix} \in \Gamma_0(2p)$.

Proof. By Theorem 5.14, for $m = 0$, we have

$$\mathcal{OK}_{p,0}^{(R)}(\zeta_p, z) \mid [A]_1 = \frac{1-\zeta_p}{1-\zeta_p^d} \frac{1+\zeta_p^d}{1+\zeta_p} \mathcal{OK}_{p,0}^{(R)}(\zeta_p^d, z)$$

where

$$A = \begin{pmatrix} a & b \\ 2p & d \end{pmatrix} \in \Gamma_0(2p), 1 \leq a, d \leq p-1.$$

Now as in the proof of Theorem 3.1, we deduce that for an arbitrary $\vec{n}_\ell \in \{\vec{n}_1, \vec{n}_2, \dots, \vec{n}_t\}$, say $\vec{n}_\ell = (n_0, n_1, n_2, \dots, n_p)$, we have

$$j(2p, \pi_r(\vec{n}_\ell), z) \mid [A]_1 = (-1)^{L_1(A) + \frac{ab}{2p} L_2(A)} \nu_\eta^{L_3(A)} ({}^{2p}A) j(2p, \pi_{ra}(\vec{n}_\ell), z),$$

where

$$\begin{aligned} L_1(A) &= br \sum_{k=1}^p k n_k + \sum_{k=1}^p \left\lfloor \frac{rka}{p} \right\rfloor n_k + \sum_{k=1}^p \left\lfloor \frac{rk}{p} \right\rfloor n_k, \\ L_2(A) &= r^2 \sum_{k=1}^p k^2 n_k, \\ L_3(A) &= n_0 + 3 \sum_{k=1}^p n_k. \end{aligned}$$

By Theorem 3.1, we have $L_3(A) \equiv 0 \pmod{24}$ and $L_2(A) \equiv 0 \pmod{4p}$ since $m = 0$. Then

$$L_1(A) + \frac{ab}{2p} L_2(A) \equiv L_1(A) \pmod{2}.$$

Therefore

$$j(2p, \pi_r(\vec{n}_\ell), z) \mid [A]_1 = (-1)^{L_{r,\ell}(A)} j(2p, \pi_{ra}(\vec{n}_\ell), z), \text{ where } L_{r,\ell}(A) = br \sum_{k=1}^p k n_k + \sum_{k=1}^p \left(\left\lfloor \frac{rka}{p} \right\rfloor + \left\lfloor \frac{rk}{p} \right\rfloor \right) n_k.$$

Using the transformation above, we have

$$\begin{aligned} \mathcal{OK}_{p,0}^{(R)}(\zeta_p, z) \mid [A]_1 &= \sum_{\ell=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,r,\ell}(\zeta_p) j(2p, \pi_r(\vec{n}_\ell), z) \mid [A]_1 \\ &= \sum_{\ell=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,r,\ell}(\zeta_p) (-1)^{L_{r,\ell}(A)} j(2p, \pi_{ra}(\vec{n}_\ell), z). \end{aligned}$$

Since $ad \equiv 1 \pmod{2p}$, taking $r \rightarrow dr$ we have

$$\sum_{\ell=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,dr,\ell}(\zeta_p) (-1)^{L_{dr,\ell}(A)} j(2p, \pi_{dra}(\vec{n}_\ell), z) = \sum_{\ell=1}^t \sum_{\substack{r=1 \\ r \text{ odd}}}^{p-2} c_{p,dr,\ell}(\zeta_p) (-1)^{L_{dr,\ell}(A)} j(2p, \pi_r(\vec{n}_\ell), z).$$

Thus, comparing the coefficients with $\mathcal{OK}_{p,0}(\zeta_p^d, z)$, we have

$$\begin{aligned} c_{p,dr,\ell}(\zeta_p) (-1)^{L_{dr,\ell}(A)} &= \frac{1 - \zeta_p}{1 - \zeta_p^d} \frac{1 + \zeta_p^d}{1 + \zeta_p} c_{p,r,\ell}(\zeta_p^d) \\ \text{or } c_{p,d,\ell}(\zeta_p) &= (-1)^{L_{d,\ell}(A)} \frac{1 - \zeta_p}{1 - \zeta_p^d} \frac{1 + \zeta_p^d}{1 + \zeta_p} c_{p,1,\ell}(\zeta_p^d). \end{aligned}$$

□

7. LOWER BOUNDS FOR ORDER OF AT CUSPS

In this section, we calculate lower bounds for the orders of $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$ at the cusps of $\Gamma_1(2p)$, which we use in proving the identities for the same in the subsequent section.

For any cusp $\frac{a}{c}$ with $(a, c) = 1$ we define

$$\text{ord}\left(\mathcal{F}; \frac{a}{c}\right) := \text{ord}(\mathcal{F} \mid [A]_1; \infty),$$

where $A \in \text{SL}_2(\mathbb{Z})$ and $A\infty = \frac{a}{c}$.

The following result is necessary to calculate $\text{ord}\left(X_p(z); \frac{a}{c}\right)$, $\text{ord}\left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{a}{c}\right)$, $\text{ord}\left(\frac{\eta(pz)^2}{\eta(2pz)}; \frac{a}{c}\right)$ appearing in our transformations.

Proposition 7.1. [15, Corollary 2.2]. *Let $N \geq 1$ and*

$$F(z) = \prod_{m|N} \eta(mz)^{r_m},$$

where each $r_m \in \mathbb{Z}$. Then for $(a, c) = 1$,

$$\text{ord} \left(F(z); \frac{a}{c} \right) = \sum_{m|N} \frac{(m, c)^2 r_m}{24m}.$$

We also need the following result in the context of calculating the orders for $\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$ at the cusps of $\Gamma_1(2p)$. Using Equation (5.1), we can write $N_p(j; \frac{z}{p}) = M(j, p; 2pz)$. Jennings-Shaffer gives lower bounds for orders of $N(a, c; z)$, $M(a, c; mz)$ and $P(a, c; z)$ at cusps $\frac{\alpha}{\gamma}$ (see [13, Propositions 6.3, 6.4, 6.5]). We state these orders below for our functions $N(1, p; z)$, $N_p(j; z)$ and $P(1, p; z)$.

Proposition 7.2. *For $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x and $\{x\}$ the fractional part of x . For non-negative integers α and γ with $(\alpha, \gamma) = 1$, we have*

$$(7.1) \quad \text{ord}_{\text{holo}} \left(N(1, p; z); \frac{\alpha}{\gamma} \right) \geq \begin{cases} -\frac{\gamma^2}{p^2} + \frac{\gamma}{p} - \frac{1}{4} + 2\tilde{\nu} \left(\frac{\gamma}{p}, \frac{1}{2} \right) & \text{if } \gamma \text{ is even,} \\ -\frac{\gamma^2}{p^2} + \frac{\alpha\gamma}{p} - \frac{\alpha^2}{4} + \frac{1}{2}\tilde{\nu} \left(\frac{2\gamma}{p}, \alpha \right) & \text{if } \gamma \text{ is odd,} \end{cases}$$

where $\tilde{\nu}(u, w) = \frac{1}{2}(\lfloor u \rfloor - \lfloor w \rfloor)^2 + (\lfloor u \rfloor - \lfloor w \rfloor)(\{u\} - \{w\}) + k(u, w)$,

$$k(u, w) = \begin{cases} \nu(\{u\}, \{w\}) & \text{if } \{u\} - \{w\} \neq \pm \frac{1}{2}, \\ \min(\frac{1}{8}, \nu(\{u\}, \{w\})) & \text{if } \{u\} - \{w\} = \pm \frac{1}{2}, \end{cases}$$

$$\nu(u, w) = \begin{cases} \frac{u+w}{2} - \frac{1}{8} & \text{if } u+w \leq 1, \\ \frac{7}{8} - \frac{u+w}{2} & \text{if } u+w > 1. \end{cases}$$

ii)

$$(7.2) \quad \text{ord}_{\text{holo}} \left(N_p(j; \frac{z}{p}); \frac{\alpha}{\gamma} \right) \geq -\frac{g^2 x^2}{4p} \left(\frac{j}{p} - \frac{1}{2} \right)^2 + \frac{g^2}{2p} \tilde{\nu} \left(\frac{jx}{p}, \frac{x}{2} \right) \quad \text{where } g = (2p, \gamma) \text{ and } x = \frac{2p\alpha}{g}.$$

iii)

$$(7.3) \quad \text{ord}_{\text{holo}} \left(P(1, p; z); \frac{\alpha}{\gamma} \right) \geq \begin{cases} \left\{ \frac{\gamma}{p} \right\} - \left\{ \frac{\gamma}{p} \right\}^2 & \text{if } \gamma \text{ is even,} \\ \frac{1}{4} \left\{ \frac{2\gamma}{p} \right\} - \frac{1}{4} \left\{ \frac{2\gamma}{p} \right\}^2 - \frac{1}{16} & \text{if } \gamma \text{ is odd.} \end{cases}$$

Proposition 7.3. [7, Corollary 4, p.930] *Let $p \geq 3$ be prime. Then a set of inequivalent cusps $\mathcal{S}_p$ for $\Gamma_1(2p)$ is given by*

$$i\infty, 0, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{p-1}, \frac{1}{p}, \frac{2}{p}, \frac{3}{p}, \dots, \frac{\frac{1}{2}(p-1)}{p}, \frac{3}{2p}, \frac{5}{2p}, \dots, \frac{p-2}{2p}.$$

Their corresponding fan widths are

$$n(\Gamma_1(2p); i\infty) = 1, \quad n(\Gamma_1(2p); 0) = 2p, \quad n \left(\Gamma_1(2p); \frac{1}{n} \right) = \begin{cases} p & \text{if } n \text{ is even} \\ 2p & \text{if } n \text{ is odd,} \end{cases}$$

$$n \left(\Gamma_1(2p); \frac{n}{p} \right) = 2, n \left(\Gamma_1(2p); \frac{n}{2p} \right) = 1.$$

We now calculate lower bounds of the invariant order of $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$ at each cusp of $\Gamma_1(2p)$.

Theorem 7.4. *Let $p \geq 3$ be prime and $0 \leq m \leq p-1$. Then*

(i)

$$\text{ord} \left(\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z); 0 \right) \begin{cases} \geq 0 & \text{if } p = 3, 5, 7, \\ = -\frac{(p-1)(p-7)}{16p} & \text{otherwise;} \end{cases}$$

(ii)

$$\text{ord} \left(\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z); \frac{1}{n} \right) \geq \begin{cases} 0 & \text{if } n \text{ is even, } 2 \leq n \leq p-1 \\ -\frac{p^2-1}{16p} & \text{if } n \text{ is odd, } 3 \leq n \leq p-2; \end{cases}$$

(iii)

$$\text{ord} \left(\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z); \frac{n}{p} \right) \geq \frac{p^2-1}{16p}, \quad 1 \leq n \leq \frac{p-1}{2};$$

(iv)

$$\text{ord} \left(\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z); \frac{n}{2p} \right) \geq 0 \text{ where } n \text{ is odd, } 3 \leq n \leq p-2.$$

Proof. We derive lower bounds for $\text{ord} \left(\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z); \zeta \right)$ for each cusp ζ of $\Gamma_1(2p)$ not equivalent to $i\infty$. From Equation (4.2), we have

$$\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z) = X_p(z) \mathcal{O}_C^*(\zeta_p, z) \mid [U_{p,m}]_1 = \frac{1}{\sqrt{p}} (\zeta_p^{\frac{1}{2}} - \zeta_p^{\frac{-1}{2}}) \sum_{k=0}^{p-1} \zeta_p^{-rm} X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [T_k]_1.$$

We calculate

$$X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [T_k A]_1$$

for each $0 \leq k \leq p-1$ and each $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ and then pick A suitably to cover all the cusps in Proposition 7.3.

Then

$$\text{ord} \left(X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)}; \frac{a}{c} \right) = \text{ord} \left(X_p(z); \frac{a}{c} \right) + \text{ord} \left(\frac{1}{t_{(0, \frac{1}{p})}(z)}; \frac{a}{c} \right).$$

Following the proof of Theorem 4.1, we have

$$t_{(0, \frac{1}{p})}(z) \mid [A]_1 = t_{(\frac{c}{p}, \frac{d}{p})}(z)$$

for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$.

Case 1. $a + kc \not\equiv 0 \pmod{p}$. Choose $0 \leq k' \leq p-1$ such that

$$(a + kc)k' \equiv (b + kd) \pmod{p}.$$

Then

$$T_k A = C_k T_{k'},$$

where

$$C_k = T_k A T_{k'}^{-1} = \begin{pmatrix} a + ck & \frac{1}{p}(-k'(a + kc) + b + kd) \\ pc & d - k'c \end{pmatrix} \in \Gamma_0(p).$$

Then

$$\begin{aligned} (7.4) \quad X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [T_k A]_1 &= X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [C_k T_{k'}]_1 \\ &= (X_p(z) \mid [C_k T_{k'}]_0) \left(\frac{1}{t_{(c, \frac{d-k'c}{p})}(z)} \mid [T_{k'}]_1 \right). \end{aligned}$$

Case 2. $a + kc \equiv 0 \pmod{p}$. In this case we consider

$$T_k A = \begin{pmatrix} a + kc & b + kd \\ pc & pd \end{pmatrix}.$$

In this case we find that

$$T_k A = D_k P,$$

where

$$P = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix},$$

$$D_k = \begin{pmatrix} \frac{1}{p}(a + kc) & b + kd \\ c & pd \end{pmatrix} \in \mathbf{SL}_2(\mathbb{Z}).$$

Then

$$\begin{aligned} (7.5) \quad X_p(z) \frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [D_k P]_1 &= (X_p(z) \mid [D_k P]_0) \left(\frac{1}{t_{(0, \frac{1}{p})}(z)} \mid [D_k P]_1 \right) \\ &= (X_p(z) \mid [D_k P]_0) \left(\frac{1}{t_{(\frac{c}{p}, d)}(z)} \mid [P]_1 \right). \end{aligned}$$

Now we are ready to examine each cusp ζ of $\Gamma_1(2p)$. We choose

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbf{SL}_2(\mathbb{Z}), \quad \text{so that } A(\infty) = \frac{a}{c} = \zeta.$$

(i) $\zeta = 0$. Let $A = \begin{pmatrix} 0 & -1 \\ 1 & d \end{pmatrix}$ so that $A(\infty) = 0$. We assume $0 \leq k \leq p-1$.

If $k \neq 0$ then applying (7.4) with $C_k = \begin{pmatrix} k & * \\ p & d - k' \end{pmatrix}$ we have

$$\begin{aligned} \text{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z+k}{p} \right)}; 0 \right) &= \frac{1}{p} \text{ord} \left(X_p(z); \frac{k}{p} \right) + \frac{1}{p} \text{ord} \left(\frac{1}{t_{(1, \frac{d-k'c}{p})}(z)}; i\infty \right) \\ &= 0 + 0 \\ &= 0, \end{aligned}$$

where the two orders are calculated using Propositions 7.1 and 2.3 (iii) respectively.

Next applying (7.5) with $k = 0$ we have

$$\begin{aligned} \text{ord} \left(X_p \left(\frac{z}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z}{p} \right)}; 0 \right) &= p \text{ord} (X_p(z); 0) + p \text{ord} \left(\frac{1}{t_{(\frac{1}{p}, d)}(z)}; i\infty \right) \\ &= \frac{p}{24} \left(-2 + \frac{1}{2} + \frac{2}{p^2} + \frac{-1}{2p^2} \right) + \frac{p}{2} \cdot \frac{1}{p} \left(1 - \frac{1}{p} \right) \\ &= -\frac{1}{16p}(p-1)(p-7) \end{aligned}$$

again by using Propositions 7.1 and 2.3 (iii). The result (i) follows.

(ii) $\zeta = \frac{1}{n}$, where $1 \leq n \leq p-1$. Let $A = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ so that $A(\infty) = \frac{1}{n}$.

If $kn \not\equiv -1 \pmod{p}$ then applying (7.4) with $C_k = \begin{pmatrix} 1+kn & * \\ pn & 1-k'n \end{pmatrix}$ we have

$$\text{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z+k}{p} \right)}; \frac{1}{n} \right) = \frac{1}{p} \text{ord} \left(X_p(z); \frac{1+kn}{pn} \right) + \frac{1}{p} \text{ord} \left(\frac{1}{t_{(n, \frac{1-k'n}{p})}(z)}; i\infty \right)$$

From Proposition 2.3, we have

$$\text{ord} \left(\frac{1}{t_{(n, \frac{1-k'n}{p})}(z)}; i\infty \right) = 0.$$

And to calculate $\text{ord} \left(X_p(z); \frac{1+kn}{pn} \right)$ using Proposition 7.1, we split the cusps into two cases :

Case I : When n is odd, we have

$$\text{ord} \left(X_p(z); \frac{1+kn}{pn} \right) = \frac{1}{24} \left(-2 + \frac{1}{2} + 2 - \frac{1}{2} \right) = 0.$$

Case II : When n is even, we have

$$\text{ord} \left(X_p(z); \frac{1+kn}{pn} \right) = \frac{1}{24}(-2+2+2-2) = 0.$$

Next assuming $kn \equiv -1 \pmod{p}$ and applying (7.5), we have

$$\text{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z+k}{p} \right)}; \frac{1}{n} \right) = p \text{ord} \left(X_p(z); \frac{(1+kn)/p}{n} \right) + p \text{ord} \left(\frac{1}{t_{(\frac{n}{p}, 1)}(z)}; i\infty \right)$$

From Proposition 2.3, we have

$$\text{ord} \left(\frac{1}{t_{(\frac{n}{p}, 1)}(z)}; i\infty \right) = \frac{1}{2} \cdot \frac{n}{p} \left(1 - \frac{n}{p} \right).$$

And to calculate $\text{ord} \left(X_p(z); \frac{(1+kn)/p}{n} \right)$ using Proposition 7.1, we split the cusps into two cases :

Case I : When n is odd, we have

$$\text{ord} \left(X_p(z); \frac{(1+kn)/p}{n} \right) = \frac{1}{24} \left(-2 + \frac{1}{2} + \frac{2}{p^2} + \frac{-1}{2p^2} \right) = \frac{-1}{16} \left(1 - \frac{1}{p^2} \right).$$

Case II : When n is even, we have

$$\text{ord} \left(X_p(z); \frac{(1+kn)/p}{n} \right) = \frac{1}{24} \left(-2 + 2 + \frac{2}{p^2} + \frac{-2}{p^2} \right) = 0.$$

Combining these results, we have

$$\begin{aligned} \text{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z+k}{p} \right)}; \frac{1}{n} \right) &= \begin{cases} \frac{n}{2} \left(1 - \frac{n}{p} \right) + \frac{1-p^2}{16p} & \text{if } n \text{ is odd,} \\ \frac{n}{2} \left(1 - \frac{n}{p} \right) & \text{if } n \text{ is even} \end{cases} \\ &\geq \begin{cases} \frac{1-p^2}{16p} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

The result (ii) follows.

(iii) $\zeta = \frac{n}{p}$, where $1 \leq n \leq \frac{1}{2}(p-1)$. Choose b, d so that $A = \begin{pmatrix} n & b \\ p & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ and $A(\infty) = n/p$.

Since $n \not\equiv 0 \pmod{p}$ we apply (7.4) for each k with $C_k = \begin{pmatrix} n+kp & * \\ p^2 & d-k'p \end{pmatrix}$. We have

$$\begin{aligned} \text{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})} \left(\frac{z+k}{p} \right)}; \frac{n}{p} \right) &= \frac{1}{p} \text{ord} \left(X_p(z); \frac{n+kp}{p^2} \right) + \frac{1}{p} \text{ord} \left(\frac{1}{t_{(p, \frac{d-k'p}{p})}(z)}; i\infty \right) \\ &= \frac{1}{24p} \left(-2 + \frac{1}{2} + 2p^2 + \frac{-1}{2}p^2 \right) + 0 \end{aligned}$$

$$= \frac{p^2 - 1}{16p}.$$

The result (iii) follows.

(iv) $\zeta = \frac{n}{2p}$, where n is odd and $3 \leq n \leq p-2$. Let $A = \begin{pmatrix} n & b \\ 2p & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ so that $A(\infty) = n/2p$.

Since $n \not\equiv 0 \pmod{p}$ we apply (7.4) for each k with $C_k = \begin{pmatrix} n+2kp & * \\ 2p^2 & d-2k'p \end{pmatrix}$. We have

$$\begin{aligned} \mathrm{ord} \left(X_p \left(\frac{z+k}{p} \right) \frac{1}{t_{(0, \frac{1}{p})}(\frac{z+k}{p})}; \frac{n}{2p} \right) &= \frac{1}{p} \mathrm{ord} \left(X_p(z); \frac{n+2kp}{2p^2} \right) + \frac{1}{p} \mathrm{ord} \left(\frac{1}{t_{(2p, \frac{d-2k'p}{p})}(z)}; i\infty \right) \\ &= \frac{1}{24} (-2 + 2 + 2p^2 - 2p^2) + 0 \\ &= 0. \end{aligned}$$

The result (iv) follows. $\square$

Theorem 7.5. *Let $p \geq 3$ be prime, $0 \leq m \leq p-1$. Then*

(i)

$$\mathrm{ord} \left(\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z); 0 \right) \geq -\frac{(p-3)(p-5)}{16p};$$

(ii)

$$\mathrm{ord} \left(\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z); \frac{1}{n} \right) \geq \begin{cases} 0 & \text{if } n \text{ is even, } 2 \leq n \leq p-1, \\ -\frac{(4n-p+1)(4n-p-1)}{16p} & \text{if } n \text{ is odd and } 1 \leq n \leq \frac{p-1}{2}, \\ -\frac{(4n-3p+1)(4n-3p-1)}{16p} & \text{if } n \text{ is odd and } \frac{p-1}{2} < n \leq p-1, \end{cases}$$

(iii)

$$\mathrm{ord} \left(\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z); \frac{n}{p} \right) \geq \frac{p-1}{4p}, \quad 1 \leq n \leq \frac{p-1}{2};$$

(iv)

$$\mathrm{ord} \left(\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z); \frac{n}{2p} \right) \geq 0, \quad n \text{ is odd and } 3 \leq n \leq p-2.$$

Proof. We derive lower bounds for $\mathrm{ord} \left(\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z); \zeta \right)$ for each cusp ζ of $\Gamma_1(2p)$ not equivalent to $i\infty$. From Equation (5.6), we have

$$\begin{aligned} \mathcal{OK}_{p,m}^{(R)}(\zeta_p; z) &= \mathcal{J}^* \left(\frac{1}{p}; z \right) \mid [U_{p,m}]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \zeta_p^{-km} \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \mathcal{J} \left(\frac{1}{p}; z \right) \mid [T_k]_1 \\ &= \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \zeta_p^{-km} \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \left[\frac{-2i \zeta_p (1 - \zeta_p)}{(1 + \zeta_p)} N(1, p; z) \right] \end{aligned}$$

$$\begin{aligned}
& + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} \sum_{j=1}^{\frac{p-1}{2}} (-1)^j (\zeta_p^{-2j} - \zeta_p^{2j}) N_p(j; z) \\
& + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} P(1, p; z) \Big] \mid [T_k]_1 \text{ (using Equation (5.2))} \\
= & \begin{cases} \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \zeta_p^{-km} \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \left[\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} N(1, p; z) + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} P(1, p; z) \right] \mid [T_k]_1 \\ \quad \text{if } m = 0 \text{ or } \left(\frac{-m}{p}\right) = -1, \\ \frac{1}{\sqrt{p}} \sum_{k=0}^{p-1} \zeta_p^{-km} \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \left[\frac{-2i\zeta_p(1-\zeta_p)}{(1+\zeta_p)} N(1, p; z) + \frac{2\zeta_p(1-\zeta_p)}{(1+\zeta_p)} P(1, p; z) \right] \mid [T_k]_1 \\ \quad + \frac{2i(1-\zeta_p)}{(1+\zeta_p)} (-1)^\ell (\zeta_p^{-2\ell} - \zeta_p^{2\ell}) \frac{\eta(pz)^2}{\eta(2pz)} N_p(\ell; \frac{z}{p}), \text{ (using Proposition 5.13)} \\ \quad \text{where } 1 \leq \ell \leq \frac{1}{2}(p-1), \ell^2 \equiv -m \pmod{p}, \text{ and } \left(\frac{-m}{p}\right) = 1. \end{cases}
\end{aligned}$$

We calculate

$$\begin{aligned}
& \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p; z) \mid [T_k A]_1, \\
& \frac{\eta(pz)^2}{\eta(2pz)} N_p(\ell; \frac{z}{p}) \mid [A]_1 \text{ and} \\
& \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p; z) \mid [T_k A]_1
\end{aligned}$$

for each $0 \leq k \leq p-1$ and each $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ and then pick A suitably to cover all the cusps in Proposition 7.3.

Case 1. $a + kc \not\equiv 0 \pmod{p}$. Choose $0 \leq k' \leq p-1$ such that

$$(a + kc)k' \equiv (b + kd) \pmod{p}.$$

Then

$$T_k A = C_k T_{k'},$$

where

$$C_k = T_k A T_{k'}^{-1} = \begin{pmatrix} a + ck & \frac{1}{p}(-k'(a + kc) + b + kd) \\ pc & d - k'c \end{pmatrix} \in \Gamma_0(p).$$

Then

$$\begin{aligned}
(7.6) \quad \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} Y(1, p; z) \mid [T_k A]_1 &= \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} Y(1, p; z) \mid [C_k T_{k'}]_1 \\
&= \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \mid [C_k T_{k'}]_{\frac{1}{2}} \right) \left(Y(1, p; z) \mid [C_k T_{k'}]_{\frac{1}{2}} \right).
\end{aligned}$$

where $Y = N$ or P .

Case 2. $a + kc \equiv 0 \pmod{p}$. In this case we consider

$$T_k A = \begin{pmatrix} a + kc & b + kd \\ pc & pd \end{pmatrix}.$$

In this case we find that

$$T_k A = D_k P,$$

where

$$P = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix},$$

$$D_k = \begin{pmatrix} \frac{1}{p}(a + kc) & b + kd \\ c & pd \end{pmatrix} \in \mathbf{SL}_2(\mathbb{Z}).$$

Then

$$(7.7) \quad \frac{\eta(p^2 z)^2}{\eta(2p^2 z)} Y(1, p; z) \mid [T_k A]_1 = \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} \mid [D_k P]_0 \right) (Y(1, p; z) \mid [D_k P]_1)$$

where $Y = N$ or P .

Now we are ready to examine each cusp ζ of $\Gamma_1(2p)$. We choose

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbf{SL}_2(\mathbb{Z}), \quad \text{so that } A(\infty) = \frac{a}{c} = \zeta.$$

(i) $\zeta = 0$. Let $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ so that $A(\infty) = 0$. We assume $0 \leq k \leq p-1$.

If $k \neq 0$ then applying (7.6) with $C_k = \begin{pmatrix} k & * \\ p & d - k' \end{pmatrix}$ we have

$$\begin{aligned} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) \mid [T_k]_1; 0 \right) &= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{k}{p} \right) + \frac{1}{p} \text{ord} \left(N(1, p, z); \frac{k}{p} \right) \\ &\geq \frac{1}{24p} \left(2 - \frac{1}{2} \right) + \frac{1}{p} \left(-1 + k - \frac{k^2}{4} + \frac{1}{2} \tilde{\nu}(2, k) \right) \\ &= \frac{1}{16p} + \frac{1}{p} \left(-1 + k - \frac{k^2}{4} + \frac{1}{2} \tilde{\nu}(2, k) \right) \\ &= 0. \end{aligned}$$

And,

$$\begin{aligned} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p, z) \mid [T_k]_1; 0 \right) &= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{k}{p} \right) + \frac{1}{p} \text{ord} \left(P(1, p, z); \frac{k}{p} \right) \\ &\geq \frac{1}{16p} + \frac{1}{p} \left(0 - 0 - \frac{1}{16} \right) \\ &= 0, \end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

If $k = 0$ then applying (7.7) with $D_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ we have

$$\begin{aligned}
 \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) | [T_k]_1; 0 \right) &= p \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; 0 \right) + p \text{ord} (N(1, p, z); 0) \\
 &\geq \frac{p}{24} \left(\frac{2}{p^2} - \frac{1}{2p^2} \right) + p \left(\frac{-1}{p^2} + \frac{1}{2} \tilde{\nu} \left(\frac{2}{p}, 0 \right) \right) \\
 &= \frac{1}{16p} + p \left(\frac{-1}{p^2} + \frac{1}{2} \nu \left(\frac{2}{p}, 0 \right) \right) \\
 &= \frac{1}{16p} + p \left(\frac{-1}{p^2} + \frac{1}{2} \left(\frac{1}{p} - \frac{1}{8} \right) \right) \\
 &= -\frac{(p-3)(p-5)}{16p}.
 \end{aligned}$$

And,

$$\begin{aligned}
 \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p, z) | [T_k]_1; 0 \right) &= p \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; 0 \right) + p \text{ord} (P(1, p, z); 0) \\
 &= \frac{1}{16p} + p \left(\frac{-1}{4} \left(\frac{2}{p} \right)^2 + \frac{1}{4} \left(\frac{2}{p} \right) - \frac{1}{16} \right) \\
 &= -\frac{(p-3)(p-5)}{16p},
 \end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

Finally

$$\begin{aligned}
 \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); 0 \right) &= \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)}; 0 \right) + \text{ord} \left(N_p \left(\ell, \frac{z}{p} \right); 0 \right) \\
 &\geq \frac{1}{24} \left(\frac{2}{p} - \frac{1}{2p} \right) + (0 + \frac{1}{2p} \tilde{\nu}(0, 0)) \\
 &= \frac{1}{16p} - \frac{1}{8.2p} \\
 &= 0,
 \end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equation (7.2) respectively.

(ii) $\zeta = \frac{1}{n}$, where $1 \leq n \leq p-1$. Let $A = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ so that $A(\infty) = \frac{1}{n}$.

If $kn \not\equiv -1 \pmod{p}$ then applying (7.6) with $C_k = \begin{pmatrix} 1+kn & * \\ pn & 1-k'n \end{pmatrix}$ we have

$$\text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) | [T_k]_1; \frac{1}{n} \right)$$

$$\begin{aligned}
&= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{1+kn}{pn} \right) + \frac{1}{p} \text{ord} \left(N(1, p, z); \frac{1+kn}{pn} \right) \\
&\geq \begin{cases} \frac{1}{24p}(-2 + \frac{1}{2} + 2 - \frac{1}{2}) + \frac{1}{p}(-n^2 + n - \frac{1}{4} + 2\tilde{\nu}(n, \frac{1}{2})) & \text{if } n \text{ is even,} \\ \frac{1}{16p} + \frac{1}{p}(-n^2 + (1+kn)n - \frac{(1+kn)^2}{4} + \frac{1}{2}\tilde{\nu}(2n, 1+kn)) & \text{if } n \text{ is odd,} \end{cases} \\
&= \begin{cases} \frac{1}{p}(-n^2 + n - \frac{1}{4} + 2\tilde{\nu}(n, \frac{1}{2})) & \text{if } n \text{ is even,} \\ \frac{1}{16p} + \frac{1}{p}(-n^2 + (1+kn)n - \frac{(1+kn)^2}{4} + \frac{1}{2}\tilde{\nu}(2n, 1+kn)) & \text{if } n \text{ is odd,} \end{cases} \\
&= 0.
\end{aligned}$$

And,

$$\begin{aligned}
&\text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p, z) | [T_k]_1; \frac{1}{n} \right) \\
&= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{1+kn}{p} \right) + \frac{1}{p} \text{ord} \left(P(1, p, z); \frac{1+kn}{pn} \right) \\
&\geq \begin{cases} 0 + \frac{1}{p} \cdot 0 & \text{if } n \text{ is even,} \\ \frac{1}{16p} + \frac{1}{p} \cdot \frac{-1}{16} & \text{if } n \text{ is odd,} \end{cases} \\
&= 0,
\end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

If $kn \equiv -1 \pmod{p}$ then applying (7.7) with $D_k = \begin{pmatrix} \frac{1+kn}{p} & k \\ n & p \end{pmatrix}$ we have

$$\begin{aligned}
&\text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) | [D_p]_1; \frac{1}{n} \right) \\
&= p \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{1+kn}{n} \right) + p \text{ord} \left(N(1, p, z); \frac{1+kn}{n} \right) \\
&\geq \begin{cases} \frac{p}{24} \left(\frac{2}{p^2} - \frac{2}{p^2} \right) + p \left(\frac{-n^2}{p^2} + \frac{n}{p} - \frac{1}{4} + 2\tilde{\nu}\left(\frac{n}{p}, \frac{1}{2}\right) \right) & \text{if } n \text{ is even,} \\ \frac{p}{24} \left(\frac{2}{p^2} - \frac{1}{2p^2} \right) + p \left(\frac{-n^2}{p^2} + \frac{1+kn}{p} \cdot n - \frac{(1+kn)^2}{4} + \frac{1}{2}\tilde{\nu}\left(\frac{2n}{p}, \frac{1+kn}{p}\right) \right) & \text{if } n \text{ is odd,} \end{cases} \\
&= \begin{cases} p \left(\frac{-n^2}{p^2} + \frac{n}{p} - \frac{1}{4} + 2\tilde{\nu}\left(\frac{n}{p}, \frac{1}{2}\right) \right) & \text{if } n \text{ is even,} \\ \frac{1}{16p} + p \left(\frac{-n^2}{p^2} + \frac{1+kn}{p} \cdot n - \frac{(1+kn)^2}{4} + \frac{1}{2}\tilde{\nu}\left(\frac{2n}{p}, \frac{1+kn}{p}\right) \right) & \text{if } n \text{ is odd,} \end{cases} \\
&= \begin{cases} \frac{-n^2}{p} + 2n & \text{if } n \text{ is even and } 1 \leq n \leq \frac{p-1}{2}, \\ \frac{-n^2}{p} + p & \text{if } n \text{ is even and } \frac{p-1}{2} < n \leq p-1, \\ -\frac{(4n-p+1)(4n-p-1)}{16p} & \text{if } n \text{ is odd and } 1 \leq n \leq \frac{p-1}{2}, \\ -\frac{(4n-3p+1)(4n-3p-1)}{16p} & \text{if } n \text{ is odd and } \frac{p-1}{2} < n \leq p-1. \end{cases}
\end{aligned}$$

And,

$$\begin{aligned}
& \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p, z) | [D_p]_1; \frac{1}{n} \right) \\
&= p \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{1+kn}{n} \right) + p \text{ord} \left(P(1, p, z); \frac{p}{n} \right) \\
&\geq \begin{cases} \frac{p}{24} \left(\frac{2}{p^2} - \frac{2}{p^2} \right) + p \left(\frac{n}{p} - \frac{n^2}{p^2} \right) & \text{if } n \text{ is even,} \\ \frac{p}{24} \left(\frac{2}{p^2} - \frac{1}{2p^2} \right) + p \left(\frac{1}{4} \left(\frac{2n}{p} \right) - \frac{1}{4} \left(\frac{2n}{p} \right)^2 - \frac{1}{16} \right) & \text{if } n \text{ is odd and } 1 \leq n \leq \frac{p-1}{2}, \\ \frac{p}{24} \left(\frac{2}{p^2} - \frac{1}{2p^2} \right) + p \left(\frac{1}{4} \left(\frac{2n}{p} - 1 \right) - \frac{1}{4} \left(\frac{2n}{p} - 1 \right)^2 - \frac{1}{16} \right) & \text{if } n \text{ is odd and } \frac{p-1}{2} < n \leq p-1, \end{cases} \\
&= \begin{cases} n - \frac{n^2}{p} & \text{if } n \text{ is even,} \\ -\frac{(4n-p+1)(4n-p-1)}{16p} & \text{if } n \text{ is odd and } 1 \leq n \leq \frac{p-1}{2}, \\ -\frac{(4n-3p+1)(4n-3p-1)}{16p} & \text{if } n \text{ is odd and } \frac{p-1}{2} < n \leq p-1, \end{cases}
\end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

Finally

$$\begin{aligned}
& \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); \frac{1}{n} \right) = \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)}; \frac{1}{n} \right) + \text{ord} \left(N_p \left(\ell, \frac{z}{p} \right); \frac{1}{n} \right) \\
&\geq \begin{cases} \frac{2.1.2-2}{48p} - p \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{2}{p} \tilde{\nu}(\ell, \frac{p}{2}) & \text{if } n \text{ is even,} \\ \frac{2.1.2-1}{48p} - p \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{1}{2p} \tilde{\nu}(2\ell, p) & \text{if } n \text{ is odd,} \end{cases} \\
&= \begin{cases} \frac{1}{24p} - p \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{2}{p} \tilde{\nu}(\ell, \frac{p}{2}) & \text{if } n \text{ is even,} \\ \frac{1}{16p} - p \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{1}{2p} \tilde{\nu}(2\ell, p) & \text{if } n \text{ is odd,} \end{cases} \\
&= \begin{cases} \frac{1}{24p} & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd,} \end{cases}
\end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equation (7.2) respectively.

(iii) $\zeta = \frac{n}{p}$, where $1 \leq n \leq \frac{1}{2}(p-1)$. Choose b, d so that $A = \begin{pmatrix} n & b \\ p & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ and $A(\infty) = \frac{n}{p}$.

Since $n \not\equiv 0 \pmod{p}$ we apply (7.6) for each k with $C_k = \begin{pmatrix} n+kp & * \\ p^2 & d-k'p \end{pmatrix}$. We have

$$\begin{aligned}
& \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) | [T_k]_1; \frac{n}{p} \right) \\
&= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{n+kp}{p^2} \right) + \frac{1}{p} \text{ord} \left(N(1, p, z); \frac{n+kp}{p^2} \right)
\end{aligned}$$

$$\begin{aligned}
&\geq \frac{1}{24p}(2p^2 - \frac{p^2}{2}) + \frac{1}{p} \left(-p^2 + (n + kp)p - \frac{(n + kp)^2}{4} + \frac{1}{2}\tilde{\nu}(2p, n + kp) \right) \\
&= \frac{p}{16} + \frac{1}{p} \left(-p^2 + (n + kp)p - \frac{(n + kp)^2}{4} + \frac{1}{2}\tilde{\nu}(2p, n + kp) \right) \\
&= \frac{p}{16} - \frac{1}{16p}.
\end{aligned}$$

And,

$$\begin{aligned}
\text{ord} \left(\frac{\eta(p^2z)^2}{\eta(2p^2z)} P(1, p, z) | [T_k]_1; \frac{n}{p} \right) &\geq \frac{1}{24p}(2p^2 - \frac{p^2}{2}) + \frac{1}{p} \cdot \left(\frac{1}{4} \left\{ \frac{2p^2}{p} \right\} - \frac{1}{4} \left\{ \frac{2p^2}{p} \right\}^2 - \frac{1}{16} \right) \\
&= \frac{p}{16} + \frac{1}{p} \cdot \frac{-1}{16} \\
&= \frac{p}{16} - \frac{1}{16p},
\end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

Finally

$$\begin{aligned}
&\text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); \frac{n}{p} \right) \\
&= \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)}; \frac{n}{p} \right) + \text{ord} \left(N_p \left(\ell, \frac{z}{p} \right); \frac{n}{p} \right) \\
&\geq \frac{(p, p)^2 \cdot 2}{24p} + \frac{(2p, p)^2 \cdot (-1)}{48p} + \left(-pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{p}{2} \tilde{\nu} \left(\frac{2n\ell}{p}, n \right) \right) \\
&= \frac{p}{16} - pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{p}{2} \tilde{\nu} \left(\frac{2n\ell}{p}, n \right)
\end{aligned}$$

Here, we let $m = \overline{2n\ell}$. Then $2n\ell = jp + m$, where $1 \leq m \leq p - 1$, $j = \lfloor \frac{2n\ell}{p} \rfloor = \frac{2n\ell - m}{p}$ and $\frac{m}{p} = \{ \frac{2n\ell}{p} \}$. Then,

$$\begin{aligned}
&\text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); \frac{n}{p} \right) \\
&\geq \frac{p}{16} - pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{p}{2} \left(\frac{1}{2} \left(\frac{2n\ell - m}{p} - n \right)^2 + \left(\frac{2n\ell - m}{p} - n \right) \left(\frac{m}{p} \right) + \nu \left(\frac{m}{p}, 0 \right) \right) \\
&= \frac{p}{16} - pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + \frac{p}{2} \left(\frac{1}{2} \left(\frac{2n\ell - m}{p} - n \right)^2 + \left(\frac{2n\ell - m}{p} - n \right) \left(\frac{m}{p} \right) + \frac{m}{2p} - \frac{1}{8} \right) \\
&= -\frac{m^2}{4p} + \frac{m}{4} \\
&= \frac{m(p - m)}{4p}
\end{aligned}$$

$$\geq \frac{p-1}{4p} \text{ (since minimum of } f(x) = px - x^2 \text{ in } [1, p-1] \text{ is } p-1),$$

where the orders are calculated using Proposition 7.1 and Equation (7.2) respectively.

(iv) $\zeta = \frac{n}{2p}$, where n is odd and $3 \leq n \leq p-2$. Let $A = \begin{pmatrix} n & b \\ 2p & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ so that $A(\infty) = \frac{n}{2p}$.

Since $n \not\equiv 0 \pmod{p}$ we apply (7.6) for each k with $C_k = \begin{pmatrix} n+2kp & * \\ 2p^2 & d-2k'p \end{pmatrix}$. We have

$$\begin{aligned} & \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} N(1, p, z) | [T_k]_1; \frac{n}{2p} \right) \\ &= \frac{1}{p} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)}; \frac{n+2kp}{2p^2} \right) + \frac{1}{p} \text{ord} \left(N(1, p, z); \frac{n+2kp}{2p^2} \right) \\ &\geq \frac{1}{12}(p^2 - p^2) + \frac{1}{p} \left(-4p^2 + 2p - \frac{1}{4} + 2\tilde{\nu} \left(2p, \frac{1}{2} \right) \right) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

And,

$$\begin{aligned} \text{ord} \left(\frac{\eta(p^2 z)^2}{\eta(2p^2 z)} P(1, p, z) | [T_k]_1; \frac{n}{2p} \right) &= \frac{1}{12}(p^2 - p^2) + \frac{1}{p} \cdot \left(\left\{ \frac{2p^2}{p} \right\} - \left\{ \frac{2p^2}{p} \right\}^2 \right) \\ &= 0, \end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equations (7.1) and (7.3).

Finally

$$\begin{aligned} & \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); \frac{n}{2p} \right) \\ &= \text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)}; \frac{n}{2p} \right) + \text{ord} \left(N_p \left(\ell, \frac{z}{p} \right); \frac{n}{2p} \right) \\ &\geq \frac{(p, 2p)^2 \cdot 2}{24p} + \frac{(2p, 2p)^2 \cdot (-1)}{48p} + \left(-pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + 2p\tilde{\nu} \left(\frac{n\ell}{p}, \frac{n}{2} \right) \right) \\ &= -pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + 2p\tilde{\nu} \left(\frac{n\ell}{p}, \frac{n}{2} \right) \end{aligned}$$

Here, we let $m = \overline{n\ell}$. Then $n\ell = jp + m$, where $1 \leq m \leq p-1$, $j = \lfloor \frac{n\ell}{p} \rfloor = \frac{n\ell-m}{p}$ and $\frac{m}{p} = \{ \frac{n\ell}{p} \}$. Also, $\frac{n}{2} = \frac{n-1}{2} + \frac{1}{2}$. Then $\lfloor \frac{n}{2} \rfloor = \frac{n-1}{2}$ and $\{ \frac{n}{2} \} = \frac{1}{2}$. Then,

$$\text{ord} \left(\frac{\eta(pz)^2}{\eta(2pz)} N_p \left(\ell, \frac{z}{p} \right); \frac{n}{2p} \right)$$

$$\begin{aligned}
&\geq -pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + 2p \left(\frac{1}{2} \left(\frac{n\ell - m}{p} - \frac{n-1}{2} \right)^2 + \left(\frac{n\ell - m}{p} - \frac{n-1}{2} \right) \left(\frac{m}{p} - \frac{1}{2} \right) + \nu \left(\frac{m}{p}, \frac{1}{2} \right) \right) \\
&= -pn^2 \left(\frac{\ell}{p} - \frac{1}{2} \right)^2 + 2p \left(\frac{1}{2} \left(\frac{n\ell - m}{p} - \frac{n-1}{2} \right)^2 + \left(\frac{n\ell - m}{p} - \frac{n-1}{2} \right) \left(\frac{m}{p} - \frac{1}{2} \right) + \frac{\frac{m}{p} + \frac{1}{2}}{2} - \frac{1}{8} \right) \\
&= -\frac{m^2}{p} + 2m \\
&= \frac{m(2p - m)}{p} \\
&\geq \frac{2p-1}{p} \text{ (since minimum of } f(x) = 2px - x^2 \text{ in } [1, p-1] \text{ is } 2p-1) \\
&= 2 - \frac{1}{p},
\end{aligned}$$

where the orders are calculated using Proposition 7.1 and Equation (7.2) respectively. $\square$

8. OVERPARTITION RANK AND CRANK MOD 11 IDENTITIES

In this final section, we derive the 11-dissection of the overpartition rank and crank functions involving the functions $\mathcal{OK}_{11,m}^{(C)}(\zeta_{11}, z)$ and $\mathcal{OK}_{11,m}^{(R)}(\zeta_{11}, z)$ defined and studied previously. We find and prove identities expressing the above elements in terms of our generalized theta functions $j(z) = j(2p, \overrightarrow{n}, z)$ in Definition 2.1 (which in turn comprise of quotients of the Biagioli theta functions $f_{N,p}(z)$) and eta quotients with cyclotomic coefficients in $\mathbb{Q}[\zeta_{11}]$. These identities are found using a combination of automorphic-algorithmic experimental techniques in Maple and Sage. Formulas for the rank differences modulo 3 and 5 in terms of infinite products and generalized Lambert series was given by Lovejoy and Osburn [16]. As remarked by Jennings-Shaffer in [13], determining these difference formulas is equivalent to determining the 3 and 5-dissections of the overpartition rank function $\mathcal{OR}(z, \tau)$ discussed in the introduction section where we replace the variable z by the 3-rd and 5-th primitive roots of unity. In the same paper, Jennings-Shaffer determines the 7-dissection of $\mathcal{OR}(\zeta_7, \tau)$ by revisiting $\mathcal{OR}(z, \tau)$ as a harmonic weak Maass form.

We first state all of our identities and prove them in a subsequent subsection. We present the following :

- (1) For each of the rank and crank functions, we present one identity each for when m is quadratic residue and non residue modulo 11 and for the $m = 0$ case. The rest of the identities can be easily obtained using our symmetry results Theorems 4.5 and 5.14. We find that each of the dissection elements is a linear combination of 40 functions in the crank case and 45 functions in the rank case with coefficients in $\mathbb{Q}[\zeta_{11}]$.
- (2) Subsequently, we also present two nicer identities for the $m = 0$ case of the dissection with an adjustment, that is a correction factor involving an eta quotient, but this

time the coefficients are in $\mathbb{Z}[\zeta_{11}]$.

Our identities are symmetric in nature and each of them feature the following five vectors permuted using the permutation π_r defined in Definition 2.4.

$$\begin{aligned}\vec{n}_1 &= (3, 0, 0, -1, -1, 1, 0, 1, -2, 1, -2, 2), \\ \vec{n}_2 &= (39, -4, -3, -4, -5, -4, -4, -4, -2, -3, -3, -1), \\ \vec{n}_3 &= (15, -2, -1, 0, -3, 0, -2, -2, 0, -1, -3, 1), \\ \vec{n}_4 &= (39, -5, -3, -3, -4, -4, -2, -4, -4, -3, -4, -1), \\ \vec{n}_5 &= (15, -4, 3, -4, -1, -2, 0, 0, -3, 1, -4, 1).\end{aligned}$$

8.1. Statements of the identities.

8.1.1. Identity for $\mathcal{OK}_{11,0}^{(C)}(\zeta_{11}, z)$.

$$\begin{aligned}(8.1) \quad \mathcal{OK}_{11,0}^{(C)}(\zeta_{11}, z) &= (q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{M}(k, 11, 11n) \zeta_{11}^k \right) q^n \\ &= \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\ &\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\ &\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\ &\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),\end{aligned}$$

where the coefficients are :

$$\begin{aligned}a_{11,1} &= 111\zeta_{11}^9 - \frac{508}{9}\zeta_{11}^8 + \frac{517}{9}\zeta_{11}^7 - \frac{2}{9}\zeta_{11}^6 - \frac{2}{9}\zeta_{11}^5 + \frac{517}{9}\zeta_{11}^4 - \frac{508}{9}\zeta_{11}^3 + 111\zeta_{11}^2 - \frac{167}{9}, \\ a_{11,3} &= -76\zeta_{11}^9 + \frac{319}{9}\zeta_{11}^8 - \frac{508}{9}\zeta_{11}^7 - \frac{682}{9}\zeta_{11}^6 - \frac{682}{9}\zeta_{11}^5 - \frac{508}{9}\zeta_{11}^4 + \frac{319}{9}\zeta_{11}^3 - 76\zeta_{11}^2 - \frac{706}{9}, \\ a_{11,5} &= -110\zeta_{11}^9 + \frac{176}{9}\zeta_{11}^8 - \frac{482}{9}\zeta_{11}^7 - \frac{506}{9}\zeta_{11}^6 - \frac{506}{9}\zeta_{11}^5 - \frac{482}{9}\zeta_{11}^4 + \frac{176}{9}\zeta_{11}^3 - 110\zeta_{11}^2 + \frac{13}{9}, \\ a_{11,7} &= 55\zeta_{11}^9 + \frac{517}{9}\zeta_{11}^8 + \frac{176}{9}\zeta_{11}^7 + \frac{1001}{9}\zeta_{11}^6 + \frac{1001}{9}\zeta_{11}^5 + \frac{176}{9}\zeta_{11}^4 + \frac{517}{9}\zeta_{11}^3 + 55\zeta_{11}^2 + \frac{1175}{9}, \\ a_{11,9} &= 22\zeta_{11}^9 - \frac{482}{9}\zeta_{11}^8 - \frac{319}{9}\zeta_{11}^7 - \frac{484}{9}\zeta_{11}^6 - \frac{484}{9}\zeta_{11}^5 - \frac{319}{9}\zeta_{11}^4 - \frac{482}{9}\zeta_{11}^3 + 22\zeta_{11}^2 - \frac{1309}{9}, \\ b_{11,1} &= \frac{7202}{9} + 346\zeta_{11}^9 + \frac{3562}{9}\zeta_{11}^8 + \frac{578}{9}\zeta_{11}^7 + \frac{3698}{9}\zeta_{11}^6 + \frac{3698}{9}\zeta_{11}^5 + \frac{578}{9}\zeta_{11}^4 + \frac{3562}{9}\zeta_{11}^3 + 346\zeta_{11}^2, \\ b_{11,3} &= \frac{5326}{9} + 736\zeta_{11}^9 + \frac{2342}{9}\zeta_{11}^8 + \frac{3562}{9}\zeta_{11}^7 + \frac{2926}{9}\zeta_{11}^6 + \frac{2926}{9}\zeta_{11}^5 + \frac{3562}{9}\zeta_{11}^4 + \frac{2342}{9}\zeta_{11}^3 + 736\zeta_{11}^2, \\ b_{11,5} &= -\frac{3256}{9} + 114\zeta_{11}^9 - \frac{3062}{9}\zeta_{11}^8 - \frac{2536}{9}\zeta_{11}^7 - \frac{136}{9}\zeta_{11}^6 - \frac{136}{9}\zeta_{11}^5 - \frac{2536}{9}\zeta_{11}^4 - \frac{3062}{9}\zeta_{11}^3 + 114\zeta_{11}^2,\end{aligned}$$

$$\begin{aligned}
b_{11,7} &= -\frac{52}{9} + 80\zeta_{11}^9 - \frac{578}{9}\zeta_{11}^8 + \frac{3062}{9}\zeta_{11}^7 + \frac{584}{9}\zeta_{11}^6 + \frac{584}{9}\zeta_{11}^5 + \frac{3062}{9}\zeta_{11}^4 - \frac{578}{9}\zeta_{11}^3 + 80\zeta_{11}^2, \\
b_{11,9} &= \frac{1316}{9} + 196\zeta_{11}^9 + \frac{2536}{9}\zeta_{11}^8 + \frac{2342}{9}\zeta_{11}^7 - \frac{1162}{9}\zeta_{11}^6 - \frac{1162}{9}\zeta_{11}^5 + \frac{2342}{9}\zeta_{11}^4 + \frac{2536}{9}\zeta_{11}^3 + 196\zeta_{11}^2, \\
c_{11,1} &= -812\zeta_{11}^9 + \frac{1073}{3}\zeta_{11}^8 - \frac{1343}{3}\zeta_{11}^7 - \frac{638}{3}\zeta_{11}^6 - \frac{638}{3}\zeta_{11}^5 - \frac{1343}{3}\zeta_{11}^4 + \frac{1073}{3}\zeta_{11}^3 - 812\zeta_{11}^2 + \frac{55}{3}, \\
c_{11,3} &= 466\zeta_{11}^9 + \frac{238}{3}\zeta_{11}^8 + \frac{1073}{3}\zeta_{11}^7 + \frac{2036}{3}\zeta_{11}^6 + \frac{2036}{3}\zeta_{11}^5 + \frac{1073}{3}\zeta_{11}^4 + \frac{238}{3}\zeta_{11}^3 + 466\zeta_{11}^2 + \frac{2654}{3}, \\
c_{11,5} &= 722\zeta_{11}^9 - \frac{325}{3}\zeta_{11}^8 + \frac{1093}{3}\zeta_{11}^7 + \frac{1711}{3}\zeta_{11}^6 + \frac{1711}{3}\zeta_{11}^5 + \frac{1093}{3}\zeta_{11}^4 - \frac{325}{3}\zeta_{11}^3 + 722\zeta_{11}^2 + \frac{1006}{3}, \\
c_{11,7} &= 29\zeta_{11}^9 + \frac{1343}{3}\zeta_{11}^8 + \frac{325}{3}\zeta_{11}^7 + \frac{1798}{3}\zeta_{11}^6 + \frac{1798}{3}\zeta_{11}^5 + \frac{325}{3}\zeta_{11}^4 + \frac{1343}{3}\zeta_{11}^3 + 29\zeta_{11}^2 + \frac{2761}{3}, \\
c_{11,9} &= 527\zeta_{11}^9 - \frac{1093}{3}\zeta_{11}^8 + \frac{238}{3}\zeta_{11}^7 - \frac{455}{3}\zeta_{11}^6 - \frac{455}{3}\zeta_{11}^5 + \frac{238}{3}\zeta_{11}^4 - \frac{1093}{3}\zeta_{11}^3 + 527\zeta_{11}^2 - \frac{1928}{3}, \\
d_{11,1} &= d_{11,3} = d_{11,5} = d_{11,7} = d_{11,9} = 0, \\
e_{11,1} &= \frac{2398}{9} - 692\zeta_{11}^9 + \frac{1814}{9}\zeta_{11}^8 - \frac{4946}{9}\zeta_{11}^7 - \frac{650}{9}\zeta_{11}^6 - \frac{650}{9}\zeta_{11}^5 - \frac{4946}{9}\zeta_{11}^4 + \frac{1814}{9}\zeta_{11}^3 - 692\zeta_{11}^2, \\
e_{11,3} &= -\frac{9176}{9} - 816\zeta_{11}^9 - \frac{2416}{9}\zeta_{11}^8 - \frac{1814}{9}\zeta_{11}^7 - \frac{7994}{9}\zeta_{11}^6 - \frac{7994}{9}\zeta_{11}^5 - \frac{1814}{9}\zeta_{11}^4 - \frac{2416}{9}\zeta_{11}^3 - 816\zeta_{11}^2, \\
e_{11,5} &= -\frac{1832}{9} + 344\zeta_{11}^9 - \frac{5530}{9}\zeta_{11}^8 + \frac{1282}{9}\zeta_{11}^7 + \frac{2464}{9}\zeta_{11}^6 + \frac{2464}{9}\zeta_{11}^5 + \frac{1282}{9}\zeta_{11}^4 - \frac{5530}{9}\zeta_{11}^3 + 344\zeta_{11}^2, \\
e_{11,7} &= \frac{11758}{9} + 346\zeta_{11}^9 + \frac{4946}{9}\zeta_{11}^8 + \frac{5530}{9}\zeta_{11}^7 + \frac{5578}{9}\zeta_{11}^6 + \frac{5578}{9}\zeta_{11}^5 + \frac{5530}{9}\zeta_{11}^4 + \frac{4946}{9}\zeta_{11}^3 + 346\zeta_{11}^2, \\
e_{11,9} &= -\frac{680}{9} + 818\zeta_{11}^9 - \frac{1282}{9}\zeta_{11}^8 + \frac{2416}{9}\zeta_{11}^7 - \frac{632}{9}\zeta_{11}^6 - \frac{632}{9}\zeta_{11}^5 + \frac{2416}{9}\zeta_{11}^4 - \frac{1282}{9}\zeta_{11}^3 + 818\zeta_{11}^2, \\
g_{11,1} &= \frac{9328}{9} + \frac{4928}{9}\zeta_{11}^8 - \frac{3344}{9}\zeta_{11}^7 + \frac{2464}{9}\zeta_{11}^6 + \frac{2464}{9}\zeta_{11}^5 - \frac{3344}{9}\zeta_{11}^4 + \frac{4928}{9}\zeta_{11}^3, \\
g_{11,3} &= -\frac{16016}{9} - 1408\zeta_{11}^9 - \frac{7744}{9}\zeta_{11}^8 - \frac{4928}{9}\zeta_{11}^7 - \frac{10208}{9}\zeta_{11}^6 - \frac{10208}{9}\zeta_{11}^5 - \frac{4928}{9}\zeta_{11}^4 - \frac{7744}{9}\zeta_{11}^3 - 1408\zeta_{11}^2, \\
g_{11,5} &= -\frac{3344}{9} + 176\zeta_{11}^9 - \frac{7744}{9}\zeta_{11}^8 - \frac{3344}{9}\zeta_{11}^7 + \frac{2464}{9}\zeta_{11}^6 + \frac{2464}{9}\zeta_{11}^5 - \frac{3344}{9}\zeta_{11}^4 - \frac{7744}{9}\zeta_{11}^3 + 176\zeta_{11}^2, \\
g_{11,7} &= \frac{7744}{9} + \frac{3344}{9}\zeta_{11}^8 + \frac{7744}{9}\zeta_{11}^7 + \frac{2464}{9}\zeta_{11}^6 + \frac{2464}{9}\zeta_{11}^5 + \frac{7744}{9}\zeta_{11}^4 + \frac{3344}{9}\zeta_{11}^3, \\
g_{11,9} &= \frac{6160}{9} + 1232\zeta_{11}^9 + \frac{3344}{9}\zeta_{11}^8 + \frac{7744}{9}\zeta_{11}^7 + \frac{880}{9}\zeta_{11}^6 + \frac{880}{9}\zeta_{11}^5 + \frac{7744}{9}\zeta_{11}^4 + \frac{3344}{9}\zeta_{11}^3 + 1232\zeta_{11}^2, \\
h_{11,1} &= \frac{1408}{3} + \frac{704}{3}\zeta_{11}^8 - \frac{704}{3}\zeta_{11}^7 + \frac{352}{3}\zeta_{11}^6 + \frac{352}{3}\zeta_{11}^5 - \frac{704}{3}\zeta_{11}^4 + \frac{704}{3}\zeta_{11}^3, \\
h_{11,3} &= -704\zeta_{11}^9 - \frac{1408}{3}\zeta_{11}^8 - \frac{704}{3}\zeta_{11}^7 - \frac{1760}{3}\zeta_{11}^6 - \frac{1760}{3}\zeta_{11}^5 - \frac{704}{3}\zeta_{11}^4 - \frac{1408}{3}\zeta_{11}^3 - 704\zeta_{11}^2 - \frac{2816}{3}, \\
h_{11,5} &= -\frac{704}{3} - \frac{1408}{3}\zeta_{11}^8 - \frac{704}{3}\zeta_{11}^7 + \frac{352}{3}\zeta_{11}^6 + \frac{352}{3}\zeta_{11}^5 - \frac{704}{3}\zeta_{11}^4 - \frac{1408}{3}\zeta_{11}^3, \\
h_{11,7} &= \frac{1408}{3} + \frac{704}{3}\zeta_{11}^8 + \frac{1408}{3}\zeta_{11}^7 + \frac{352}{3}\zeta_{11}^6 + \frac{352}{3}\zeta_{11}^5 + \frac{1408}{3}\zeta_{11}^4 + \frac{704}{3}\zeta_{11}^3, \\
h_{11,9} &= 704\zeta_{11}^9 + \frac{704}{3}\zeta_{11}^8 + \frac{1408}{3}\zeta_{11}^7 + \frac{352}{3}\zeta_{11}^6 + \frac{352}{3}\zeta_{11}^5 + \frac{1408}{3}\zeta_{11}^4 + \frac{704}{3}\zeta_{11}^3 + 704\zeta_{11}^2 + \frac{1408}{3}, \\
m_{11,1} &= -165\zeta_{11}^9 + \frac{11}{9}\zeta_{11}^8 - \frac{1001}{9}\zeta_{11}^7 - \frac{1331}{9}\zeta_{11}^6 - \frac{1331}{9}\zeta_{11}^5 - \frac{1001}{9}\zeta_{11}^4 + \frac{11}{9}\zeta_{11}^3 - 165\zeta_{11}^2 - \frac{1496}{9}, \\
m_{11,3} &= -55\zeta_{11}^9 + \frac{682}{9}\zeta_{11}^8 + \frac{11}{9}\zeta_{11}^7 + \frac{836}{9}\zeta_{11}^6 + \frac{836}{9}\zeta_{11}^5 + \frac{11}{9}\zeta_{11}^4 + \frac{682}{9}\zeta_{11}^3 - 55\zeta_{11}^2 + \frac{1694}{9}, \\
m_{11,5} &= -55\zeta_{11}^9 - \frac{506}{9}\zeta_{11}^8 - \frac{484}{9}\zeta_{11}^7 - \frac{1342}{9}\zeta_{11}^6 - \frac{1342}{9}\zeta_{11}^5 - \frac{484}{9}\zeta_{11}^4 - \frac{506}{9}\zeta_{11}^3 - 55\zeta_{11}^2 - \frac{1672}{9}, \\
m_{11,7} &= 132\zeta_{11}^9 - \frac{1001}{9}\zeta_{11}^8 + \frac{506}{9}\zeta_{11}^7 - \frac{154}{9}\zeta_{11}^6 - \frac{154}{9}\zeta_{11}^5 + \frac{506}{9}\zeta_{11}^4 - \frac{1001}{9}\zeta_{11}^3 + 132\zeta_{11}^2 - \frac{979}{9}, \\
m_{11,9} &= -187\zeta_{11}^9 + \frac{484}{9}\zeta_{11}^8 - \frac{682}{9}\zeta_{11}^7 - \frac{847}{9}\zeta_{11}^6 - \frac{847}{9}\zeta_{11}^5 - \frac{682}{9}\zeta_{11}^4 + \frac{484}{9}\zeta_{11}^3 - 187\zeta_{11}^2 - \frac{187}{9},
\end{aligned}$$

$$\begin{aligned}
n_{11,1} &= -88\zeta_{11}^9 + \frac{44}{3}\zeta_{11}^8 - \frac{176}{3}\zeta_{11}^7 - \frac{176}{3}\zeta_{11}^6 - \frac{176}{3}\zeta_{11}^5 - \frac{176}{3}\zeta_{11}^4 + \frac{44}{3}\zeta_{11}^3 - 88\zeta_{11}^2 - \frac{176}{3}, \\
n_{11,3} &= \frac{88}{3}\zeta_{11}^8 + \frac{44}{3}\zeta_{11}^7 + \frac{176}{3}\zeta_{11}^6 + \frac{176}{3}\zeta_{11}^5 + \frac{44}{3}\zeta_{11}^4 + \frac{88}{3}\zeta_{11}^3 + \frac{308}{3}, \\
n_{11,5} &= -44\zeta_{11}^9 - \frac{44}{3}\zeta_{11}^8 - \frac{88}{3}\zeta_{11}^7 - \frac{220}{3}\zeta_{11}^6 - \frac{220}{3}\zeta_{11}^5 - \frac{88}{3}\zeta_{11}^4 - \frac{44}{3}\zeta_{11}^3 - 44\zeta_{11}^2 - \frac{220}{3}, \\
n_{11,7} &= 44\zeta_{11}^9 - \frac{176}{3}\zeta_{11}^8 + \frac{44}{3}\zeta_{11}^7 - \frac{88}{3}\zeta_{11}^6 - \frac{88}{3}\zeta_{11}^5 + \frac{44}{3}\zeta_{11}^4 - \frac{176}{3}\zeta_{11}^3 + 44\zeta_{11}^2 - \frac{220}{3}, \\
n_{11,9} &= -88\zeta_{11}^9 + \frac{88}{3}\zeta_{11}^8 - \frac{88}{3}\zeta_{11}^7 - \frac{88}{3}\zeta_{11}^6 - \frac{88}{3}\zeta_{11}^5 - \frac{88}{3}\zeta_{11}^4 + \frac{88}{3}\zeta_{11}^3 - 88\zeta_{11}^2 + \frac{44}{3}.
\end{aligned}$$

8.1.2. **Identity for $\mathcal{OK}_{11,0}^{(R)}(\zeta_{11}, z)$.**

$$\begin{aligned}
(8.2) \quad \mathcal{OK}_{11,0}^{(R)}(\zeta_{11}, z) &= (q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \bar{N}(k, 11, 11n) \zeta_{11}^k \right) q^n \\
&= \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
&\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
&\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
&\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
\end{aligned}$$

where the coefficients are :

$$\begin{aligned}
a_{11,1} &= \frac{35446}{9}\zeta_{11}^9 + \frac{10190}{9}\zeta_{11}^8 + \frac{6292}{3}\zeta_{11}^7 + \frac{21080}{9}\zeta_{11}^6 + \frac{21080}{9}\zeta_{11}^5 + \frac{6292}{3}\zeta_{11}^4 + \frac{10190}{9}\zeta_{11}^3 + \frac{35446}{9}\zeta_{11}^2 + \frac{38935}{9}, \\
a_{11,3} &= -\frac{31432}{9}\zeta_{11}^9 - \frac{7814}{9}\zeta_{11}^8 - \frac{7528}{3}\zeta_{11}^7 - \frac{14528}{9}\zeta_{11}^6 - \frac{14528}{9}\zeta_{11}^5 - \frac{7528}{3}\zeta_{11}^4 - \frac{7814}{9}\zeta_{11}^3 - \frac{31432}{9}\zeta_{11}^2 - \frac{29377}{9}, \\
a_{11,5} &= -\frac{35780}{9}\zeta_{11}^9 - \frac{7414}{9}\zeta_{11}^8 - \frac{7562}{3}\zeta_{11}^7 - \frac{19474}{9}\zeta_{11}^6 - \frac{19474}{9}\zeta_{11}^5 - \frac{7562}{3}\zeta_{11}^4 - \frac{7414}{9}\zeta_{11}^3 - \frac{35780}{9}\zeta_{11}^2 - \frac{35459}{9}, \\
a_{11,7} &= \frac{10274}{9}\zeta_{11}^9 - \frac{3476}{9}\zeta_{11}^8 + \frac{3872}{3}\zeta_{11}^7 - \frac{308}{9}\zeta_{11}^6 - \frac{308}{9}\zeta_{11}^5 + \frac{3872}{3}\zeta_{11}^4 - \frac{3476}{9}\zeta_{11}^3 + \frac{10274}{9}\zeta_{11}^2 + \frac{1439}{9}, \\
a_{11,9} &= \frac{1184}{9}\zeta_{11}^9 + \frac{4246}{9}\zeta_{11}^8 - \frac{2524}{3}\zeta_{11}^7 + \frac{6622}{9}\zeta_{11}^6 + \frac{6622}{9}\zeta_{11}^5 - \frac{2524}{3}\zeta_{11}^4 + \frac{4246}{9}\zeta_{11}^3 + \frac{1184}{9}\zeta_{11}^2 + \frac{4409}{9}, \\
b_{11,1} &= \frac{94304}{9}\zeta_{11}^9 - \frac{36332}{9}\zeta_{11}^8 + \frac{32660}{3}\zeta_{11}^7 + \frac{5608}{9}\zeta_{11}^6 + \frac{5608}{9}\zeta_{11}^5 + \frac{32660}{3}\zeta_{11}^4 - \frac{36332}{9}\zeta_{11}^3 + \frac{94304}{9}\zeta_{11}^2 + \frac{43664}{9}, \\
b_{11,3} &= \frac{242944}{9}\zeta_{11}^9 + \frac{40436}{9}\zeta_{11}^8 + \frac{55012}{3}\zeta_{11}^7 + \frac{123524}{9}\zeta_{11}^6 + \frac{123524}{9}\zeta_{11}^5 + \frac{55012}{3}\zeta_{11}^4 + \frac{40436}{9}\zeta_{11}^3 + \frac{242944}{9}\zeta_{11}^2 + \frac{265396}{9}, \\
b_{11,5} &= \frac{3260}{9}\zeta_{11}^9 + \frac{4816}{9}\zeta_{11}^8 - \frac{15724}{3}\zeta_{11}^7 + \frac{17776}{9}\zeta_{11}^6 + \frac{17776}{9}\zeta_{11}^5 - \frac{15724}{3}\zeta_{11}^4 + \frac{4816}{9}\zeta_{11}^3 + \frac{3260}{9}\zeta_{11}^2 - \frac{15088}{9}, \\
b_{11,7} &= \frac{116156}{9}\zeta_{11}^9 + \frac{46756}{9}\zeta_{11}^8 + \frac{24860}{3}\zeta_{11}^7 + \frac{79048}{9}\zeta_{11}^6 + \frac{79048}{9}\zeta_{11}^5 + \frac{24860}{3}\zeta_{11}^4 + \frac{46756}{9}\zeta_{11}^3 + \frac{116156}{9}\zeta_{11}^2 + \frac{135644}{9}, \\
b_{11,9} &= -\frac{76120}{9}\zeta_{11}^9 - \frac{27476}{9}\zeta_{11}^8 - \frac{13120}{3}\zeta_{11}^7 - \frac{31580}{9}\zeta_{11}^6 - \frac{31580}{9}\zeta_{11}^5 - \frac{13120}{3}\zeta_{11}^4 - \frac{27476}{9}\zeta_{11}^3 - \frac{76120}{9}\zeta_{11}^2 - \frac{43672}{9}, \\
c_{11,1} &= -\frac{96176}{3}\zeta_{11}^9 - \frac{26152}{3}\zeta_{11}^8 - 19032\zeta_{11}^7 - \frac{54766}{3}\zeta_{11}^6 - \frac{54766}{3}\zeta_{11}^5 - 19032\zeta_{11}^4 - \frac{26152}{3}\zeta_{11}^3 - \frac{96176}{3}\zeta_{11}^2 - \frac{102740}{3}, \\
c_{11,3} &= \frac{60344}{3}\zeta_{11}^9 + \frac{7480}{3}\zeta_{11}^8 + 16658\zeta_{11}^7 + \frac{20836}{3}\zeta_{11}^6 + \frac{20836}{3}\zeta_{11}^5 + 16658\zeta_{11}^4 + \frac{7480}{3}\zeta_{11}^3 + \frac{60344}{3}\zeta_{11}^2 + \frac{48332}{3},
\end{aligned}$$

$$\begin{aligned}
c_{11,5} &= \frac{78520}{3}\zeta_{11}^9 + \frac{15818}{3}\zeta_{11}^8 + 17412\zeta_{11}^7 + \frac{39212}{3}\zeta_{11}^6 + \frac{39212}{3}\zeta_{11}^5 + 17412\zeta_{11}^4 + \frac{15818}{3}\zeta_{11}^3 + \frac{78520}{3}\zeta_{11}^2 + \frac{71860}{3}, \\
c_{11,7} &= \frac{2830}{3}\zeta_{11}^9 - \frac{12796}{3}\zeta_{11}^8 + 6204\zeta_{11}^7 - \frac{12892}{3}\zeta_{11}^6 - \frac{12892}{3}\zeta_{11}^5 + 6204\zeta_{11}^4 - \frac{12796}{3}\zeta_{11}^3 + \frac{2830}{3}\zeta_{11}^2 - \frac{13772}{3}, \\
c_{11,9} &= \frac{45778}{3}\zeta_{11}^9 + \frac{15914}{3}\zeta_{11}^8 + 4726\zeta_{11}^7 + \frac{34586}{3}\zeta_{11}^6 + \frac{34586}{3}\zeta_{11}^5 + 4726\zeta_{11}^4 + \frac{15914}{3}\zeta_{11}^3 + \frac{45778}{3}\zeta_{11}^2 + \frac{52570}{3}, \\
d_{11,1} &= 30976\zeta_{11}^9 + 7744\zeta_{11}^8 + 15488\zeta_{11}^7 + 15488\zeta_{11}^6 + 15488\zeta_{11}^5 + 15488\zeta_{11}^4 + 7744\zeta_{11}^3 + 30976\zeta_{11}^2 + 30976, \\
d_{11,3} &= -23232\zeta_{11}^9 - 15488\zeta_{11}^8 - 15488\zeta_{11}^7 - 15488\zeta_{11}^6 - 15488\zeta_{11}^5 - 15488\zeta_{11}^4 - 15488\zeta_{11}^3 - 23232\zeta_{11}^2 - 30976, \\
d_{11,5} &= -30976\zeta_{11}^9 - 23232\zeta_{11}^7 - 15488\zeta_{11}^6 - 15488\zeta_{11}^5 - 23232\zeta_{11}^4 - 30976\zeta_{11}^2 - 30976, \\
d_{11,7} &= 7744\zeta_{11}^9 + 7744\zeta_{11}^8 + 7744\zeta_{11}^7 + 7744\zeta_{11}^6 + 7744\zeta_{11}^5 + 7744\zeta_{11}^4 + 7744\zeta_{11}^3 + 7744\zeta_{11}^2 + 23232, \\
d_{11,9} &= 30976\zeta_{11}^9 + 30976\zeta_{11}^7 + 7744\zeta_{11}^6 + 7744\zeta_{11}^5 + 30976\zeta_{11}^4 + 30976\zeta_{11}^2 + 30976, \\
e_{11,1} &= -\frac{198032}{9}\zeta_{11}^9 - \frac{54412}{9}\zeta_{11}^8 - \frac{34940}{3}\zeta_{11}^7 - \frac{128788}{9}\zeta_{11}^6 - \frac{128788}{9}\zeta_{11}^5 - \frac{34940}{3}\zeta_{11}^4 - \frac{54412}{9}\zeta_{11}^3 - \frac{198032}{9}\zeta_{11}^2 - \frac{223640}{9}, \\
e_{11,3} &= -\frac{187232}{9}\zeta_{11}^9 - \frac{13732}{9}\zeta_{11}^8 - \frac{44264}{3}\zeta_{11}^7 - \frac{73276}{9}\zeta_{11}^6 - \frac{73276}{9}\zeta_{11}^5 - \frac{44264}{3}\zeta_{11}^4 - \frac{13732}{9}\zeta_{11}^3 - \frac{187232}{9}\zeta_{11}^2 - \frac{136556}{9}, \\
e_{11,5} &= \frac{238684}{9}\zeta_{11}^9 + \frac{79508}{9}\zeta_{11}^8 + \frac{46780}{3}\zeta_{11}^7 + \frac{137144}{9}\zeta_{11}^6 + \frac{137144}{9}\zeta_{11}^5 + \frac{46780}{3}\zeta_{11}^4 + \frac{79508}{9}\zeta_{11}^3 + \frac{238684}{9}\zeta_{11}^2 + \frac{248680}{9}, \\
e_{11,7} &= \frac{155848}{9}\zeta_{11}^9 + \frac{5132}{9}\zeta_{11}^8 + \frac{51940}{3}\zeta_{11}^7 + \frac{40736}{9}\zeta_{11}^6 + \frac{40736}{9}\zeta_{11}^5 + \frac{51940}{3}\zeta_{11}^4 + \frac{5132}{9}\zeta_{11}^3 + \frac{155848}{9}\zeta_{11}^2 + \frac{132148}{9}, \\
e_{11,9} &= \frac{111532}{9}\zeta_{11}^9 + \frac{43904}{9}\zeta_{11}^8 + \frac{10684}{3}\zeta_{11}^7 + \frac{84584}{9}\zeta_{11}^6 + \frac{84584}{9}\zeta_{11}^5 + \frac{10684}{3}\zeta_{11}^4 + \frac{43904}{9}\zeta_{11}^3 + \frac{111532}{9}\zeta_{11}^2 + \frac{145468}{9}, \\
g_{11,1} &= -\frac{36608}{9}\zeta_{11}^9 - \frac{81664}{9}\zeta_{11}^8 + \frac{5632}{3}\zeta_{11}^7 - \frac{81664}{9}\zeta_{11}^6 - \frac{81664}{9}\zeta_{11}^5 + \frac{5632}{3}\zeta_{11}^4 - \frac{81664}{9}\zeta_{11}^3 - \frac{36608}{9}\zeta_{11}^2 - \frac{91520}{9}, \\
g_{11,3} &= -\frac{315392}{9}\zeta_{11}^9 + \frac{19712}{9}\zeta_{11}^8 - \frac{87296}{3}\zeta_{11}^7 - \frac{107008}{9}\zeta_{11}^6 - \frac{107008}{9}\zeta_{11}^5 - \frac{87296}{3}\zeta_{11}^4 + \frac{19712}{9}\zeta_{11}^3 - \frac{315392}{9}\zeta_{11}^2 - \frac{268928}{9}, \\
g_{11,5} &= \frac{140800}{9}\zeta_{11}^9 + \frac{45056}{9}\zeta_{11}^8 + \frac{14080}{3}\zeta_{11}^7 + \frac{70400}{9}\zeta_{11}^6 + \frac{70400}{9}\zeta_{11}^5 + \frac{14080}{3}\zeta_{11}^4 + \frac{45056}{9}\zeta_{11}^3 + \frac{140800}{9}\zeta_{11}^2 + \frac{85888}{9}, \\
g_{11,7} &= \frac{90112}{9}\zeta_{11}^9 + \frac{45056}{9}\zeta_{11}^8 + \frac{39424}{3}\zeta_{11}^7 + \frac{45056}{9}\zeta_{11}^6 + \frac{45056}{9}\zeta_{11}^5 + \frac{39424}{3}\zeta_{11}^4 + \frac{45056}{9}\zeta_{11}^3 + \frac{90112}{9}\zeta_{11}^2 + \frac{136576}{9}, \\
g_{11,9} &= \frac{267520}{9}\zeta_{11}^9 + \frac{45056}{9}\zeta_{11}^8 + \frac{64768}{3}\zeta_{11}^7 + \frac{146432}{9}\zeta_{11}^6 + \frac{146432}{9}\zeta_{11}^5 + \frac{64768}{3}\zeta_{11}^4 + \frac{45056}{9}\zeta_{11}^3 + \frac{267520}{9}\zeta_{11}^2 + \frac{339328}{9}, \\
h_{11,1} &= -\frac{2816}{3}\zeta_{11}^9 - \frac{26752}{3}\zeta_{11}^8 + 1408\zeta_{11}^7 - \frac{22528}{3}\zeta_{11}^6 - \frac{22528}{3}\zeta_{11}^5 + 1408\zeta_{11}^4 - \frac{26752}{3}\zeta_{11}^3 - \frac{2816}{3}\zeta_{11}^2 - \frac{28160}{3}, \\
h_{11,3} &= -\frac{95744}{3}\zeta_{11}^9 + \frac{15488}{3}\zeta_{11}^8 - 26752\zeta_{11}^7 - \frac{26752}{3}\zeta_{11}^6 - \frac{26752}{3}\zeta_{11}^5 - 26752\zeta_{11}^4 + \frac{15488}{3}\zeta_{11}^3 - \frac{95744}{3}\zeta_{11}^2 - \frac{74624}{3}, \\
h_{11,5} &= \frac{1408}{3}\zeta_{11}^9 + \frac{11264}{3}\zeta_{11}^8 - 7040\zeta_{11}^7 + \frac{2816}{3}\zeta_{11}^6 + \frac{2816}{3}\zeta_{11}^5 - 7040\zeta_{11}^4 + \frac{11264}{3}\zeta_{11}^3 + \frac{1408}{3}\zeta_{11}^2 - \frac{19712}{3}, \\
h_{11,7} &= \frac{35200}{3}\zeta_{11}^9 + \frac{15488}{3}\zeta_{11}^8 + 15488\zeta_{11}^7 + \frac{19712}{3}\zeta_{11}^6 + \frac{19712}{3}\zeta_{11}^5 + 15488\zeta_{11}^4 + \frac{15488}{3}\zeta_{11}^3 + \frac{35200}{3}\zeta_{11}^2 + \frac{60544}{3}, \\
h_{11,9} &= \frac{107008}{3}\zeta_{11}^9 + \frac{7040}{3}\zeta_{11}^8 + 28160\zeta_{11}^7 + \frac{49280}{3}\zeta_{11}^6 + \frac{49280}{3}\zeta_{11}^5 + 28160\zeta_{11}^4 + \frac{7040}{3}\zeta_{11}^3 + \frac{107008}{3}\zeta_{11}^2 + \frac{123904}{3}, \\
m_{11,1} &= -\frac{55946}{9}\zeta_{11}^9 - \frac{8866}{9}\zeta_{11}^8 - \frac{13904}{3}\zeta_{11}^7 - \frac{24112}{9}\zeta_{11}^6 - \frac{24112}{9}\zeta_{11}^5 - \frac{13904}{3}\zeta_{11}^4 - \frac{8866}{9}\zeta_{11}^3 - \frac{55946}{9}\zeta_{11}^2 - \frac{50468}{9}, \\
m_{11,3} &= -\frac{15334}{9}\zeta_{11}^9 - \frac{8756}{9}\zeta_{11}^8 + \frac{44}{3}\zeta_{11}^7 - \frac{16478}{9}\zeta_{11}^6 - \frac{16478}{9}\zeta_{11}^5 + \frac{44}{3}\zeta_{11}^4 - \frac{8756}{9}\zeta_{11}^3 - \frac{15334}{9}\zeta_{11}^2 - \frac{23980}{9}, \\
m_{11,5} &= -\frac{20482}{9}\zeta_{11}^9 + \frac{1342}{9}\zeta_{11}^8 - \frac{7612}{3}\zeta_{11}^7 - \frac{3014}{9}\zeta_{11}^6 - \frac{3014}{9}\zeta_{11}^5 - \frac{7612}{3}\zeta_{11}^4 + \frac{1342}{9}\zeta_{11}^3 - \frac{20482}{9}\zeta_{11}^2 - \frac{11506}{9}, \\
m_{11,7} &= \frac{46838}{9}\zeta_{11}^9 + \frac{16588}{9}\zeta_{11}^8 + \frac{7502}{3}\zeta_{11}^7 + \frac{31042}{9}\zeta_{11}^6 + \frac{31042}{9}\zeta_{11}^5 + \frac{7502}{3}\zeta_{11}^4 + \frac{16588}{9}\zeta_{11}^3 + \frac{46838}{9}\zeta_{11}^2 + \frac{53438}{9}, \\
m_{11,9} &= -\frac{57112}{9}\zeta_{11}^9 - \frac{13112}{9}\zeta_{11}^8 - \frac{11374}{3}\zeta_{11}^7 - \frac{30734}{9}\zeta_{11}^6 - \frac{30734}{9}\zeta_{11}^5 - \frac{11374}{3}\zeta_{11}^4 - \frac{13112}{9}\zeta_{11}^3 - \frac{57112}{9}\zeta_{11}^2 - \frac{54868}{9}, \\
n_{11,1} &= -\frac{15224}{3}\zeta_{11}^9 - \frac{3520}{3}\zeta_{11}^8 - 3432\zeta_{11}^7 - \frac{7744}{3}\zeta_{11}^6 - \frac{7744}{3}\zeta_{11}^5 - 3432\zeta_{11}^4 - \frac{3520}{3}\zeta_{11}^3 - \frac{15224}{3}\zeta_{11}^2 - \frac{15488}{3},
\end{aligned}$$

$$\begin{aligned}
n_{11,3} &= \frac{3608}{3}\zeta_{11}^9 - \frac{440}{3}\zeta_{11}^8 + 1496\zeta_{11}^7 - \frac{176}{3}\zeta_{11}^6 - \frac{176}{3}\zeta_{11}^5 + 1496\zeta_{11}^4 - \frac{440}{3}\zeta_{11}^3 + \frac{3608}{3}\zeta_{11}^2 + \frac{1496}{3}, \\
n_{11,5} &= -\frac{7744}{3}\zeta_{11}^9 - \frac{968}{3}\zeta_{11}^8 - 2024\zeta_{11}^7 - \frac{3080}{3}\zeta_{11}^6 - \frac{3080}{3}\zeta_{11}^5 - 2024\zeta_{11}^4 - \frac{968}{3}\zeta_{11}^3 - \frac{7744}{3}\zeta_{11}^2 - \frac{5896}{3}, \\
n_{11,7} &= \frac{5456}{3}\zeta_{11}^9 + \frac{3256}{3}\zeta_{11}^8 + 352\zeta_{11}^7 + \frac{4840}{3}\zeta_{11}^6 + \frac{4840}{3}\zeta_{11}^5 + 352\zeta_{11}^4 + \frac{3256}{3}\zeta_{11}^3 + \frac{5456}{3}\zeta_{11}^2 + \frac{7568}{3}, \\
n_{11,9} &= -\frac{10912}{3}\zeta_{11}^9 - \frac{2552}{3}\zeta_{11}^8 - 1848\zeta_{11}^7 - \frac{6512}{3}\zeta_{11}^6 - \frac{6512}{3}\zeta_{11}^5 - 1848\zeta_{11}^4 - \frac{2552}{3}\zeta_{11}^3 - \frac{10912}{3}\zeta_{11}^2 - \frac{10912}{3}.
\end{aligned}$$

8.1.3. **Identity for $\mathcal{OK}_{11,1}^{(C)}(\zeta_{11}, z)$ - a quadratic non-residue case ($m = 1$).**

$$\begin{aligned}
(8.3) \quad \mathcal{OK}_{11,1}^{(C)}(\zeta_{11}, z) &= q^{\frac{1}{11}}(q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{M}(k, 11, 11n+1) \zeta_{11}^k \right) q^n \\
&= \frac{f_{22,4}(z)f_{22,7}(z)}{f_{22,5}(z)f_{22,6}(z)} \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
&\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
&\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
&\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
\end{aligned}$$

where the coefficients are :

$$\begin{aligned}
a_{11,1} &= -\frac{361}{9}\zeta_{11}^9 - \frac{464}{9}\zeta_{11}^8 - 219\zeta_{11}^7 + \frac{52}{3}\zeta_{11}^6 + \frac{52}{3}\zeta_{11}^5 - 219\zeta_{11}^4 - \frac{464}{9}\zeta_{11}^3 - \frac{361}{9}\zeta_{11}^2 - \frac{1960}{9}, \\
a_{11,3} &= \frac{514}{9}\zeta_{11}^9 + \frac{185}{9}\zeta_{11}^8 + 130\zeta_{11}^7 + \frac{8}{3}\zeta_{11}^6 + \frac{8}{3}\zeta_{11}^5 + 130\zeta_{11}^4 + \frac{185}{9}\zeta_{11}^3 + \frac{514}{9}\zeta_{11}^2 + \frac{1870}{9}, \\
a_{11,5} &= -\frac{1}{9}\zeta_{11}^9 + \frac{472}{9}\zeta_{11}^8 + 179\zeta_{11}^7 - \frac{164}{3}\zeta_{11}^6 - \frac{164}{3}\zeta_{11}^5 + 179\zeta_{11}^4 + \frac{472}{9}\zeta_{11}^3 - \frac{1}{9}\zeta_{11}^2 + \frac{2108}{9}, \\
a_{11,7} &= -\frac{350}{9}\zeta_{11}^9 - \frac{346}{9}\zeta_{11}^8 - 36\zeta_{11}^7 - \frac{58}{3}\zeta_{11}^6 - \frac{58}{3}\zeta_{11}^5 - 36\zeta_{11}^4 - \frac{346}{9}\zeta_{11}^3 - \frac{350}{9}\zeta_{11}^2 - \frac{1343}{9}, \\
a_{11,9} &= \frac{1180}{9}\zeta_{11}^9 - \frac{499}{9}\zeta_{11}^8 - 131\zeta_{11}^7 + \frac{281}{3}\zeta_{11}^6 + \frac{281}{3}\zeta_{11}^5 - 131\zeta_{11}^4 - \frac{499}{9}\zeta_{11}^3 + \frac{1180}{9}\zeta_{11}^2 - \frac{668}{9}, \\
b_{11,1} &= -\frac{3920}{9}\zeta_{11}^9 - \frac{2860}{9}\zeta_{11}^8 - 314\zeta_{11}^7 - \frac{1132}{3}\zeta_{11}^6 - \frac{1132}{3}\zeta_{11}^5 - 314\zeta_{11}^4 - \frac{2860}{9}\zeta_{11}^3 - \frac{3920}{9}\zeta_{11}^2 - \frac{6560}{9}, \\
b_{11,3} &= -\frac{5926}{9}\zeta_{11}^9 - \frac{2540}{9}\zeta_{11}^8 - 844\zeta_{11}^7 - \frac{704}{3}\zeta_{11}^6 - \frac{704}{3}\zeta_{11}^5 - 844\zeta_{11}^4 - \frac{2540}{9}\zeta_{11}^3 - \frac{5926}{9}\zeta_{11}^2 - \frac{9892}{9}, \\
b_{11,5} &= \frac{6016}{9}\zeta_{11}^9 - \frac{3688}{9}\zeta_{11}^8 - 866\zeta_{11}^7 + \frac{1142}{3}\zeta_{11}^6 + \frac{1142}{3}\zeta_{11}^5 - 866\zeta_{11}^4 - \frac{3688}{9}\zeta_{11}^3 + \frac{6016}{9}\zeta_{11}^2 - \frac{7298}{9}, \\
b_{11,7} &= -\frac{4046}{9}\zeta_{11}^9 + \frac{3602}{9}\zeta_{11}^8 + 364\zeta_{11}^7 - \frac{148}{3}\zeta_{11}^6 - \frac{148}{3}\zeta_{11}^5 + 364\zeta_{11}^4 + \frac{3602}{9}\zeta_{11}^3 - \frac{4046}{9}\zeta_{11}^2 + \frac{3052}{9}, \\
b_{11,9} &= -\frac{5846}{9}\zeta_{11}^9 + \frac{848}{9}\zeta_{11}^8 + 134\zeta_{11}^7 - \frac{988}{3}\zeta_{11}^6 - \frac{988}{3}\zeta_{11}^5 + 134\zeta_{11}^4 + \frac{848}{9}\zeta_{11}^3 - \frac{5846}{9}\zeta_{11}^2 + \frac{2332}{9}, \\
c_{11,1} &= \frac{638}{3}\zeta_{11}^9 + \frac{1048}{3}\zeta_{11}^8 + 1529\zeta_{11}^7 - 235\zeta_{11}^6 - 235\zeta_{11}^5 + 1529\zeta_{11}^4 + \frac{1048}{3}\zeta_{11}^3 + \frac{638}{3}\zeta_{11}^2 + \frac{5192}{3}, \\
c_{11,3} &= -\frac{1667}{3}\zeta_{11}^9 - \frac{361}{3}\zeta_{11}^8 - 494\zeta_{11}^7 - 205\zeta_{11}^6 - 205\zeta_{11}^5 - 494\zeta_{11}^4 - \frac{361}{3}\zeta_{11}^3 - \frac{1667}{3}\zeta_{11}^2 - \frac{3560}{3}, \\
c_{11,5} &= -\frac{469}{3}\zeta_{11}^9 - \frac{1004}{3}\zeta_{11}^8 - 1143\zeta_{11}^7 + 214\zeta_{11}^6 + 214\zeta_{11}^5 - 1143\zeta_{11}^4 - \frac{1004}{3}\zeta_{11}^3 - \frac{469}{3}\zeta_{11}^2 - \frac{5131}{3},
\end{aligned}$$

$$\begin{aligned}
c_{11,7} &= -\frac{1333}{3}\zeta_{11}^9 + \frac{205}{3}\zeta_{11}^8 + 491\zeta_{11}^7 - 330\zeta_{11}^6 - 330\zeta_{11}^5 + 491\zeta_{11}^4 + \frac{205}{3}\zeta_{11}^3 - \frac{1333}{3}\zeta_{11}^2 - \frac{364}{3}, \\
c_{11,9} &= \frac{1247}{3}\zeta_{11}^9 - \frac{1304}{3}\zeta_{11}^8 - 1327\zeta_{11}^7 + 494\zeta_{11}^6 + 494\zeta_{11}^5 - 1327\zeta_{11}^4 - \frac{1304}{3}\zeta_{11}^3 + \frac{1247}{3}\zeta_{11}^2 - \frac{3469}{3}, \\
d_{11,1} &= d_{11,3} = d_{11,5} = d_{11,7} = d_{11,9} = 0, \\
e_{11,1} &= \frac{2582}{9}\zeta_{11}^9 + \frac{904}{9}\zeta_{11}^8 + 1010\zeta_{11}^7 - \frac{908}{3}\zeta_{11}^6 - \frac{908}{3}\zeta_{11}^5 + 1010\zeta_{11}^4 + \frac{904}{9}\zeta_{11}^3 + \frac{2582}{9}\zeta_{11}^2 + \frac{9632}{9}, \\
e_{11,3} &= \frac{2690}{9}\zeta_{11}^9 + \frac{5548}{9}\zeta_{11}^8 + 1180\zeta_{11}^7 + \frac{304}{3}\zeta_{11}^6 + \frac{304}{3}\zeta_{11}^5 + 1180\zeta_{11}^4 + \frac{5548}{9}\zeta_{11}^3 + \frac{2690}{9}\zeta_{11}^2 + \frac{18308}{9}, \\
e_{11,5} &= \frac{2906}{9}\zeta_{11}^9 - \frac{1598}{9}\zeta_{11}^8 - 1098\zeta_{11}^7 + \frac{1408}{3}\zeta_{11}^6 + \frac{1408}{3}\zeta_{11}^5 - 1098\zeta_{11}^4 - \frac{1598}{9}\zeta_{11}^3 + \frac{2906}{9}\zeta_{11}^2 - \frac{12256}{9}, \\
e_{11,7} &= -\frac{12016}{9}\zeta_{11}^9 + \frac{2632}{9}\zeta_{11}^8 + 468\zeta_{11}^7 - \frac{2084}{3}\zeta_{11}^6 - \frac{2084}{3}\zeta_{11}^5 + 468\zeta_{11}^4 + \frac{2632}{9}\zeta_{11}^3 - \frac{12016}{9}\zeta_{11}^2 - \frac{1276}{9}, \\
e_{11,9} &= -\frac{1684}{9}\zeta_{11}^9 - \frac{5324}{9}\zeta_{11}^8 - 1624\zeta_{11}^7 + \frac{190}{3}\zeta_{11}^6 + \frac{190}{3}\zeta_{11}^5 - 1624\zeta_{11}^4 - \frac{5324}{9}\zeta_{11}^3 - \frac{1684}{9}\zeta_{11}^2 - \frac{13894}{9}, \\
g_{11,1} &= -\frac{3520}{9}\zeta_{11}^9 - \frac{3344}{9}\zeta_{11}^8 + 176\zeta_{11}^7 - \frac{2288}{3}\zeta_{11}^6 - \frac{2288}{3}\zeta_{11}^5 + 176\zeta_{11}^4 - \frac{3344}{9}\zeta_{11}^3 - \frac{3520}{9}\zeta_{11}^2 - \frac{880}{9}, \\
g_{11,3} &= \frac{10736}{9}\zeta_{11}^9 + \frac{7744}{9}\zeta_{11}^8 + 1408\zeta_{11}^7 + \frac{1936}{3}\zeta_{11}^6 + \frac{1936}{3}\zeta_{11}^5 + 1408\zeta_{11}^4 + \frac{7744}{9}\zeta_{11}^3 + \frac{10736}{9}\zeta_{11}^2 + \frac{22880}{9}, \\
g_{11,5} &= \frac{9152}{9}\zeta_{11}^9 - \frac{4928}{9}\zeta_{11}^8 - 1584\zeta_{11}^7 + \frac{1936}{3}\zeta_{11}^6 + \frac{1936}{3}\zeta_{11}^5 - 1584\zeta_{11}^4 - \frac{4928}{9}\zeta_{11}^3 + \frac{9152}{9}\zeta_{11}^2 - \frac{16720}{9}, \\
g_{11,7} &= -\frac{13024}{9}\zeta_{11}^9 + \frac{7744}{9}\zeta_{11}^8 + 1232\zeta_{11}^7 - \frac{1760}{3}\zeta_{11}^6 - \frac{1760}{3}\zeta_{11}^5 + 1232\zeta_{11}^4 + \frac{7744}{9}\zeta_{11}^3 - \frac{13024}{9}\zeta_{11}^2 + \frac{8624}{9}, \\
g_{11,9} &= -\frac{13024}{9}\zeta_{11}^9 - \frac{3344}{9}\zeta_{11}^8 - 1232\zeta_{11}^7 - \frac{1760}{3}\zeta_{11}^6 - \frac{1760}{3}\zeta_{11}^5 - 1232\zeta_{11}^4 - \frac{3344}{9}\zeta_{11}^3 - \frac{13024}{9}\zeta_{11}^2 - \frac{11968}{9}, \\
h_{11,1} &= -\frac{352}{3}\zeta_{11}^9 - \frac{704}{3}\zeta_{11}^8 - 352\zeta_{11}^7 - 352\zeta_{11}^6 - \frac{704}{3}\zeta_{11}^5 - \frac{352}{3}\zeta_{11}^4 - \frac{352}{3}, \\
h_{11,3} &= \frac{1760}{3}\zeta_{11}^9 + \frac{1408}{3}\zeta_{11}^8 + 704\zeta_{11}^7 + 352\zeta_{11}^6 + 352\zeta_{11}^5 + 704\zeta_{11}^4 + \frac{1408}{3}\zeta_{11}^3 + \frac{1760}{3}\zeta_{11}^2 + \frac{3872}{3}, \\
h_{11,5} &= \frac{1760}{3}\zeta_{11}^9 - \frac{704}{3}\zeta_{11}^8 - 704\zeta_{11}^7 + 352\zeta_{11}^6 + 352\zeta_{11}^5 - 704\zeta_{11}^4 - \frac{704}{3}\zeta_{11}^3 + \frac{1760}{3}\zeta_{11}^2 - \frac{2464}{3}, \\
h_{11,7} &= -\frac{2464}{3}\zeta_{11}^9 + \frac{1408}{3}\zeta_{11}^8 + 704\zeta_{11}^7 - 352\zeta_{11}^6 - 352\zeta_{11}^5 + 704\zeta_{11}^4 + \frac{1408}{3}\zeta_{11}^3 - \frac{2464}{3}\zeta_{11}^2 + \frac{1760}{3}, \\
h_{11,9} &= -\frac{2464}{3}\zeta_{11}^9 - \frac{704}{3}\zeta_{11}^8 - 704\zeta_{11}^7 - 352\zeta_{11}^6 - 352\zeta_{11}^5 - 704\zeta_{11}^4 - \frac{704}{3}\zeta_{11}^3 - \frac{2464}{3}\zeta_{11}^2 - \frac{2464}{3}, \\
m_{11,1} &= \frac{1175}{9}\zeta_{11}^9 + \frac{511}{9}\zeta_{11}^8 + 223\zeta_{11}^7 + \frac{58}{3}\zeta_{11}^6 + \frac{58}{3}\zeta_{11}^5 + 223\zeta_{11}^4 + \frac{511}{9}\zeta_{11}^3 + \frac{1175}{9}\zeta_{11}^2 + \frac{3374}{9}, \\
m_{11,3} &= -\frac{905}{9}\zeta_{11}^9 + \frac{479}{9}\zeta_{11}^8 + 185\zeta_{11}^7 - \frac{292}{3}\zeta_{11}^6 - \frac{292}{3}\zeta_{11}^5 + 185\zeta_{11}^4 + \frac{479}{9}\zeta_{11}^3 - \frac{905}{9}\zeta_{11}^2 + \frac{793}{9}, \\
m_{11,5} &= \frac{814}{9}\zeta_{11}^9 + \frac{47}{9}\zeta_{11}^8 + 4\zeta_{11}^7 + \frac{110}{3}\zeta_{11}^6 + \frac{110}{3}\zeta_{11}^5 + 4\zeta_{11}^4 + \frac{47}{9}\zeta_{11}^3 + \frac{814}{9}\zeta_{11}^2 + \frac{1414}{9}, \\
m_{11,7} &= \frac{346}{9}\zeta_{11}^9 - \frac{673}{9}\zeta_{11}^8 - 319\zeta_{11}^7 + \frac{278}{3}\zeta_{11}^6 + \frac{278}{3}\zeta_{11}^5 - 319\zeta_{11}^4 - \frac{673}{9}\zeta_{11}^3 + \frac{346}{9}\zeta_{11}^2 - \frac{2708}{9}, \\
m_{11,9} &= -\frac{5}{9}\zeta_{11}^9 + \frac{1010}{9}\zeta_{11}^8 + 354\zeta_{11}^7 - \frac{223}{3}\zeta_{11}^6 - \frac{223}{3}\zeta_{11}^5 + 354\zeta_{11}^4 + \frac{1010}{9}\zeta_{11}^3 - \frac{5}{9}\zeta_{11}^2 + \frac{4042}{9}, \\
n_{11,1} &= \frac{176}{3}\zeta_{11}^9 + \frac{88}{3}\zeta_{11}^8 + 132\zeta_{11}^7 + 132\zeta_{11}^6 + \frac{88}{3}\zeta_{11}^5 + \frac{176}{3}\zeta_{11}^4 + \frac{572}{3}, \\
n_{11,3} &= -\frac{176}{3}\zeta_{11}^9 + \frac{32}{3}\zeta_{11}^8 + 40\zeta_{11}^7 - 44\zeta_{11}^6 - 44\zeta_{11}^5 + 40\zeta_{11}^4 + \frac{32}{3}\zeta_{11}^3 - \frac{176}{3}\zeta_{11}^2 - \frac{68}{3}, \\
n_{11,5} &= \frac{88}{3}\zeta_{11}^9 + \frac{44}{3}\zeta_{11}^8 + 44\zeta_{11}^7 + 44\zeta_{11}^6 + \frac{44}{3}\zeta_{11}^5 + \frac{88}{3}\zeta_{11}^4 + \frac{352}{3}, \\
n_{11,7} &= \frac{88}{3}\zeta_{11}^9 - \frac{76}{3}\zeta_{11}^8 - 128\zeta_{11}^7 + 44\zeta_{11}^6 + 44\zeta_{11}^5 - 128\zeta_{11}^4 - \frac{76}{3}\zeta_{11}^3 + \frac{88}{3}\zeta_{11}^2 - \frac{284}{3}, \\
n_{11,9} &= -\frac{44}{3}\zeta_{11}^9 + \frac{164}{3}\zeta_{11}^8 + 172\zeta_{11}^7 - 44\zeta_{11}^6 - 44\zeta_{11}^5 + 172\zeta_{11}^4 + \frac{164}{3}\zeta_{11}^3 - \frac{44}{3}\zeta_{11}^2 + \frac{592}{3}.
\end{aligned}$$

8.1.4. *Identity for $\mathcal{OK}_{11,1}^{(R)}(\zeta_{11}, z)$ - a quadratic non-residue case ($m = 1$).*

$$\begin{aligned}
 \mathcal{OK}_{11,1}^{(R)}(\zeta_{11}, z) &= q^{\frac{1}{11}}(q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{N}(k, 11, 11n+1) \zeta_{11}^k \right) q^n \\
 (8.4) \quad &= \frac{f_{22,4}(z)f_{22,7}(z)}{f_{22,5}(z)f_{22,6}(z)} \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
 &\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
 &\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
 &\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
 \end{aligned}$$

where the coefficients are :

$$\begin{aligned}
 a_{11,1} &= -\frac{37346}{9}\zeta_{11}^9 - \frac{16636}{9}\zeta_{11}^8 - \frac{14338}{3}\zeta_{11}^7 - \frac{17894}{9}\zeta_{11}^6 - \frac{17894}{9}\zeta_{11}^5 - \frac{14338}{3}\zeta_{11}^4 - \frac{16636}{9}\zeta_{11}^3 - \frac{37346}{9}\zeta_{11}^2 - \frac{64010}{9}, \\
 a_{11,3} &= \frac{37886}{9}\zeta_{11}^9 + \frac{9040}{9}\zeta_{11}^8 + \frac{11164}{3}\zeta_{11}^7 + \frac{14942}{9}\zeta_{11}^6 + \frac{14942}{9}\zeta_{11}^5 + \frac{11164}{3}\zeta_{11}^4 + \frac{9040}{9}\zeta_{11}^3 + \frac{37886}{9}\zeta_{11}^2 + \frac{48224}{9}, \\
 a_{11,5} &= \frac{38920}{9}\zeta_{11}^9 + \frac{13244}{9}\zeta_{11}^8 + \frac{14138}{3}\zeta_{11}^7 + \frac{15892}{9}\zeta_{11}^6 + \frac{15892}{9}\zeta_{11}^5 + \frac{14138}{3}\zeta_{11}^4 + \frac{13244}{9}\zeta_{11}^3 + \frac{38920}{9}\zeta_{11}^2 + \frac{61900}{9}, \\
 a_{11,7} &= -\frac{13846}{9}\zeta_{11}^9 + \frac{814}{9}\zeta_{11}^8 - \frac{4376}{3}\zeta_{11}^7 - \frac{2068}{9}\zeta_{11}^6 - \frac{2068}{9}\zeta_{11}^5 - \frac{4376}{3}\zeta_{11}^4 + \frac{814}{9}\zeta_{11}^3 - \frac{13846}{9}\zeta_{11}^2 - \frac{13462}{9}, \\
 a_{11,9} &= \frac{10796}{9}\zeta_{11}^9 - \frac{8816}{9}\zeta_{11}^8 - \frac{1964}{3}\zeta_{11}^7 + \frac{2882}{9}\zeta_{11}^6 + \frac{2882}{9}\zeta_{11}^5 - \frac{1964}{3}\zeta_{11}^4 - \frac{8816}{9}\zeta_{11}^3 + \frac{10796}{9}\zeta_{11}^2 - \frac{12130}{9}, \\
 b_{11,1} &= -\frac{138796}{9}\zeta_{11}^9 + \frac{952}{9}\zeta_{11}^8 - \frac{43592}{3}\zeta_{11}^7 - \frac{40024}{9}\zeta_{11}^6 - \frac{40024}{9}\zeta_{11}^5 - \frac{43592}{3}\zeta_{11}^4 + \frac{952}{9}\zeta_{11}^3 - \frac{138796}{9}\zeta_{11}^2 - \frac{138988}{9}, \\
 b_{11,3} &= -\frac{313256}{9}\zeta_{11}^9 - \frac{81916}{9}\zeta_{11}^8 - \frac{94312}{3}\zeta_{11}^7 - \frac{146636}{9}\zeta_{11}^6 - \frac{146636}{9}\zeta_{11}^5 - \frac{94312}{3}\zeta_{11}^4 - \frac{81916}{9}\zeta_{11}^3 - \frac{313256}{9}\zeta_{11}^2 - \frac{412172}{9}, \\
 b_{11,5} &= \frac{84404}{9}\zeta_{11}^9 - \frac{51572}{9}\zeta_{11}^8 - \frac{16904}{3}\zeta_{11}^7 + \frac{34856}{9}\zeta_{11}^6 + \frac{34856}{9}\zeta_{11}^5 - \frac{16904}{3}\zeta_{11}^4 - \frac{51572}{9}\zeta_{11}^3 + \frac{84404}{9}\zeta_{11}^2 - \frac{69112}{9}, \\
 b_{11,7} &= -\frac{139084}{9}\zeta_{11}^9 - \frac{34796}{9}\zeta_{11}^8 - \frac{33644}{3}\zeta_{11}^7 - \frac{60940}{9}\zeta_{11}^6 - \frac{60940}{9}\zeta_{11}^5 - \frac{33644}{3}\zeta_{11}^4 - \frac{34796}{9}\zeta_{11}^3 - \frac{139084}{9}\zeta_{11}^2 - \frac{177040}{9}, \\
 b_{11,9} &= \frac{32060}{9}\zeta_{11}^9 + \frac{34828}{9}\zeta_{11}^8 + \frac{24820}{3}\zeta_{11}^7 + \frac{2996}{9}\zeta_{11}^6 + \frac{2996}{9}\zeta_{11}^5 + \frac{24820}{3}\zeta_{11}^4 + \frac{34828}{9}\zeta_{11}^3 + \frac{32060}{9}\zeta_{11}^2 + \frac{100664}{9}, \\
 c_{11,1} &= \frac{106006}{3}\zeta_{11}^9 + \frac{39476}{3}\zeta_{11}^8 + 37276\zeta_{11}^7 + \frac{47896}{3}\zeta_{11}^6 + \frac{47896}{3}\zeta_{11}^5 + 37276\zeta_{11}^4 + \frac{39476}{3}\zeta_{11}^3 + \frac{106006}{3}\zeta_{11}^2 + \frac{166654}{3}, \\
 c_{11,3} &= -\frac{78616}{3}\zeta_{11}^9 - \frac{10964}{3}\zeta_{11}^8 - 21578\zeta_{11}^7 - \frac{28066}{3}\zeta_{11}^6 - \frac{28066}{3}\zeta_{11}^5 - 21578\zeta_{11}^4 - \frac{10964}{3}\zeta_{11}^3 - \frac{78616}{3}\zeta_{11}^2 - \frac{86434}{3}, \\
 c_{11,5} &= -\frac{86120}{3}\zeta_{11}^9 - \frac{26818}{3}\zeta_{11}^8 - 30446\zeta_{11}^7 - \frac{32930}{3}\zeta_{11}^6 - \frac{32930}{3}\zeta_{11}^5 - 30446\zeta_{11}^4 - \frac{26818}{3}\zeta_{11}^3 - \frac{86120}{3}\zeta_{11}^2 - \frac{131072}{3}, \\
 c_{11,7} &= -\frac{18896}{3}\zeta_{11}^9 + \frac{15296}{3}\zeta_{11}^8 + 34\zeta_{11}^7 - \frac{68}{3}\zeta_{11}^6 - \frac{68}{3}\zeta_{11}^5 + 34\zeta_{11}^4 + \frac{15296}{3}\zeta_{11}^3 - \frac{18896}{3}\zeta_{11}^2 + \frac{11902}{3}, \\
 c_{11,9} &= -\frac{31730}{3}\zeta_{11}^9 - \frac{31762}{3}\zeta_{11}^8 - 21082\zeta_{11}^7 - \frac{17228}{3}\zeta_{11}^6 - \frac{17228}{3}\zeta_{11}^5 - 21082\zeta_{11}^4 - \frac{31762}{3}\zeta_{11}^3 - \frac{31730}{3}\zeta_{11}^2 - \frac{97322}{3}, \\
 d_{11,1} &= -30976\zeta_{11}^9 - 15488\zeta_{11}^8 - 38720\zeta_{11}^7 - 15488\zeta_{11}^6 - 15488\zeta_{11}^5 - 38720\zeta_{11}^4 - 15488\zeta_{11}^3 - 30976\zeta_{11}^2 - 54208, \\
 d_{11,3} &= 30976\zeta_{11}^9 + 7744\zeta_{11}^8 + 15488\zeta_{11}^7 + 15488\zeta_{11}^6 + 15488\zeta_{11}^5 + 15488\zeta_{11}^4 + 7744\zeta_{11}^3 + 30976\zeta_{11}^2 + 30976, \\
 d_{11,5} &= 38720\zeta_{11}^9 + 7744\zeta_{11}^8 + 38720\zeta_{11}^7 + 15488\zeta_{11}^6 + 15488\zeta_{11}^5 + 38720\zeta_{11}^4 + 7744\zeta_{11}^3 + 38720\zeta_{11}^2 + 54208,
 \end{aligned}$$

$$\begin{aligned}
d_{11,7} &= -23232\zeta_{11}^9 - 15488\zeta_{11}^6 - 15488\zeta_{11}^5 - 23232\zeta_{11}^2 - 7744, \\
d_{11,9} &= -54208\zeta_{11}^9 - 30976\zeta_{11}^7 - 23232\zeta_{11}^6 - 23232\zeta_{11}^5 - 30976\zeta_{11}^4 - 54208\zeta_{11}^2 - 38720, \\
e_{11,1} &= \frac{208948}{9}\zeta_{11}^9 + \frac{92300}{9}\zeta_{11}^8 + \frac{81464}{3}\zeta_{11}^7 + \frac{97396}{9}\zeta_{11}^6 + \frac{97396}{9}\zeta_{11}^5 + \frac{81464}{3}\zeta_{11}^4 + \frac{92300}{9}\zeta_{11}^3 + \frac{208948}{9}\zeta_{11}^2 + \frac{373156}{9}, \\
e_{11,3} &= \frac{200992}{9}\zeta_{11}^9 + \frac{60260}{9}\zeta_{11}^8 + \frac{78500}{3}\zeta_{11}^7 + \frac{67012}{9}\zeta_{11}^6 + \frac{67012}{9}\zeta_{11}^5 + \frac{78500}{3}\zeta_{11}^4 + \frac{60260}{9}\zeta_{11}^3 + \frac{200992}{9}\zeta_{11}^2 + \frac{315880}{9}, \\
e_{11,5} &= -\frac{248144}{9}\zeta_{11}^9 - \frac{98068}{9}\zeta_{11}^8 - \frac{86884}{3}\zeta_{11}^7 - \frac{108596}{9}\zeta_{11}^6 - \frac{108596}{9}\zeta_{11}^5 - \frac{86884}{3}\zeta_{11}^4 - \frac{98068}{9}\zeta_{11}^3 - \frac{248144}{9}\zeta_{11}^2 - \frac{396632}{9}, \\
e_{11,7} &= -\frac{257360}{9}\zeta_{11}^9 + \frac{248}{9}\zeta_{11}^8 - \frac{49312}{3}\zeta_{11}^7 - \frac{97976}{9}\zeta_{11}^6 - \frac{97976}{9}\zeta_{11}^5 - \frac{49312}{3}\zeta_{11}^4 + \frac{248}{9}\zeta_{11}^3 - \frac{257360}{9}\zeta_{11}^2 - \frac{192260}{9}, \\
e_{11,9} &= -\frac{99104}{9}\zeta_{11}^9 - \frac{84748}{9}\zeta_{11}^8 - \frac{53776}{3}\zeta_{11}^7 - \frac{62696}{9}\zeta_{11}^6 - \frac{62696}{9}\zeta_{11}^5 - \frac{53776}{3}\zeta_{11}^4 - \frac{84748}{9}\zeta_{11}^3 - \frac{99104}{9}\zeta_{11}^2 - \frac{249896}{9}, \\
g_{11,1} &= -\frac{8096}{9}\zeta_{11}^9 + \frac{50864}{9}\zeta_{11}^8 + \frac{3344}{3}\zeta_{11}^7 + \frac{7216}{9}\zeta_{11}^6 + \frac{7216}{9}\zeta_{11}^5 + \frac{3344}{3}\zeta_{11}^4 + \frac{50864}{9}\zeta_{11}^3 - \frac{8096}{9}\zeta_{11}^2 + \frac{88000}{9}, \\
g_{11,3} &= \frac{435424}{9}\zeta_{11}^9 + \frac{60368}{9}\zeta_{11}^8 + \frac{133232}{3}\zeta_{11}^7 + \frac{168784}{9}\zeta_{11}^6 + \frac{168784}{9}\zeta_{11}^5 + \frac{133232}{3}\zeta_{11}^4 + \frac{60368}{9}\zeta_{11}^3 + \frac{435424}{9}\zeta_{11}^2 + \frac{515680}{9}, \\
g_{11,5} &= -\frac{52448}{9}\zeta_{11}^9 - \frac{101200}{9}\zeta_{11}^8 - \frac{67408}{3}\zeta_{11}^7 - \frac{11792}{9}\zeta_{11}^6 - \frac{11792}{9}\zeta_{11}^5 - \frac{67408}{3}\zeta_{11}^4 - \frac{101200}{9}\zeta_{11}^3 - \frac{52448}{9}\zeta_{11}^2 - \frac{273152}{9}, \\
g_{11,7} &= -\frac{220352}{9}\zeta_{11}^9 + \frac{31856}{9}\zeta_{11}^8 - \frac{4048}{3}\zeta_{11}^7 - \frac{97328}{9}\zeta_{11}^6 - \frac{97328}{9}\zeta_{11}^5 - \frac{4048}{3}\zeta_{11}^4 + \frac{31856}{9}\zeta_{11}^3 - \frac{220352}{9}\zeta_{11}^2 - \frac{57728}{9}, \\
g_{11,9} &= -\frac{400928}{9}\zeta_{11}^9 - \frac{75856}{9}\zeta_{11}^8 - \frac{102256}{3}\zeta_{11}^7 - \frac{198704}{9}\zeta_{11}^6 - \frac{198704}{9}\zeta_{11}^5 - \frac{102256}{3}\zeta_{11}^4 - \frac{75856}{9}\zeta_{11}^3 - \frac{400928}{9}\zeta_{11}^2 - \frac{456896}{9}, \\
h_{11,1} &= \frac{1408}{3}\zeta_{11}^9 + \frac{7040}{3}\zeta_{11}^8 - 5632\zeta_{11}^7 + \frac{5632}{3}\zeta_{11}^6 + \frac{5632}{3}\zeta_{11}^5 - 5632\zeta_{11}^4 + \frac{7040}{3}\zeta_{11}^3 + \frac{1408}{3}\zeta_{11}^2 + \frac{5632}{3}, \\
h_{11,3} &= \frac{128128}{3}\zeta_{11}^9 + \frac{15488}{3}\zeta_{11}^8 + 42240\zeta_{11}^7 + \frac{47872}{3}\zeta_{11}^6 + \frac{47872}{3}\zeta_{11}^5 + 42240\zeta_{11}^4 + \frac{15488}{3}\zeta_{11}^3 + \frac{128128}{3}\zeta_{11}^2 + \frac{153472}{3}, \\
h_{11,5} &= \frac{43648}{3}\zeta_{11}^9 - \frac{22528}{3}\zeta_{11}^8 - 4224\zeta_{11}^7 + \frac{22528}{3}\zeta_{11}^6 + \frac{22528}{3}\zeta_{11}^5 - 4224\zeta_{11}^4 - \frac{22528}{3}\zeta_{11}^3 + \frac{43648}{3}\zeta_{11}^2 - \frac{15488}{3}, \\
h_{11,7} &= -\frac{87296}{3}\zeta_{11}^9 + \frac{11264}{3}\zeta_{11}^8 - 2816\zeta_{11}^7 - \frac{40832}{3}\zeta_{11}^6 - \frac{40832}{3}\zeta_{11}^5 - 2816\zeta_{11}^4 + \frac{11264}{3}\zeta_{11}^3 - \frac{87296}{3}\zeta_{11}^2 - \frac{28160}{3}, \\
h_{11,9} &= -\frac{159104}{3}\zeta_{11}^9 - \frac{22528}{3}\zeta_{11}^8 - 40832\zeta_{11}^7 - \frac{74624}{3}\zeta_{11}^6 - \frac{74624}{3}\zeta_{11}^5 - 40832\zeta_{11}^4 - \frac{22528}{3}\zeta_{11}^3 - \frac{159104}{3}\zeta_{11}^2 - \frac{171776}{3}, \\
m_{11,1} &= \frac{69460}{9}\zeta_{11}^9 + \frac{14744}{9}\zeta_{11}^8 + \frac{21110}{3}\zeta_{11}^7 + \frac{26848}{9}\zeta_{11}^6 + \frac{26848}{9}\zeta_{11}^5 + \frac{21110}{3}\zeta_{11}^4 + \frac{14744}{9}\zeta_{11}^3 + \frac{69460}{9}\zeta_{11}^2 + \frac{88066}{9}, \\
m_{11,3} &= \frac{6536}{9}\zeta_{11}^9 + \frac{15262}{9}\zeta_{11}^8 + \frac{7432}{3}\zeta_{11}^7 + \frac{6776}{9}\zeta_{11}^6 + \frac{6776}{9}\zeta_{11}^5 + \frac{7432}{3}\zeta_{11}^4 + \frac{15262}{9}\zeta_{11}^3 + \frac{6536}{9}\zeta_{11}^2 + \frac{38060}{9}, \\
m_{11,5} &= \frac{32114}{9}\zeta_{11}^9 - \frac{1892}{9}\zeta_{11}^8 + \frac{6772}{3}\zeta_{11}^7 + \frac{8954}{9}\zeta_{11}^6 + \frac{8954}{9}\zeta_{11}^5 + \frac{6772}{3}\zeta_{11}^4 - \frac{1892}{9}\zeta_{11}^3 + \frac{32114}{9}\zeta_{11}^2 + \frac{24056}{9}, \\
m_{11,7} &= -\frac{44818}{9}\zeta_{11}^9 - \frac{24374}{9}\zeta_{11}^8 - \frac{18698}{3}\zeta_{11}^7 - \frac{21898}{9}\zeta_{11}^6 - \frac{21898}{9}\zeta_{11}^5 - \frac{18698}{3}\zeta_{11}^4 - \frac{24374}{9}\zeta_{11}^3 - \frac{44818}{9}\zeta_{11}^2 - \frac{86716}{9}, \\
m_{11,9} &= \frac{58664}{9}\zeta_{11}^9 + \frac{23560}{9}\zeta_{11}^8 + \frac{23074}{3}\zeta_{11}^7 + \frac{23966}{9}\zeta_{11}^6 + \frac{23966}{9}\zeta_{11}^5 + \frac{23074}{3}\zeta_{11}^4 + \frac{23560}{9}\zeta_{11}^3 + \frac{58664}{9}\zeta_{11}^2 + \frac{100196}{9}, \\
n_{11,1} &= \frac{18040}{3}\zeta_{11}^9 + \frac{4928}{3}\zeta_{11}^8 + 5544\zeta_{11}^7 + \frac{7744}{3}\zeta_{11}^6 + \frac{7744}{3}\zeta_{11}^5 + 5544\zeta_{11}^4 + \frac{4928}{3}\zeta_{11}^3 + \frac{18040}{3}\zeta_{11}^2 + \frac{24376}{3}, \\
n_{11,3} &= -\frac{6136}{3}\zeta_{11}^9 + \frac{880}{3}\zeta_{11}^8 - 1040\zeta_{11}^7 - \frac{1672}{3}\zeta_{11}^6 - \frac{1672}{3}\zeta_{11}^5 - 1040\zeta_{11}^4 + \frac{880}{3}\zeta_{11}^3 - \frac{6136}{3}\zeta_{11}^2 - \frac{2944}{3}, \\
n_{11,5} &= \frac{8888}{3}\zeta_{11}^9 + \frac{1672}{3}\zeta_{11}^8 + 2816\zeta_{11}^7 + \frac{2816}{3}\zeta_{11}^6 + \frac{2816}{3}\zeta_{11}^5 + 2816\zeta_{11}^4 + \frac{1672}{3}\zeta_{11}^3 + \frac{8888}{3}\zeta_{11}^2 + \frac{11528}{3}, \\
n_{11,7} &= -\frac{3808}{3}\zeta_{11}^9 - \frac{4400}{3}\zeta_{11}^8 - 2216\zeta_{11}^7 - \frac{2728}{3}\zeta_{11}^6 - \frac{2728}{3}\zeta_{11}^5 - 2216\zeta_{11}^4 - \frac{4400}{3}\zeta_{11}^3 - \frac{3808}{3}\zeta_{11}^2 - \frac{11488}{3}, \\
n_{11,9} &= \frac{9968}{3}\zeta_{11}^9 + \frac{5368}{3}\zeta_{11}^8 + 4592\zeta_{11}^7 + \frac{4400}{3}\zeta_{11}^6 + \frac{4400}{3}\zeta_{11}^5 + 4592\zeta_{11}^4 + \frac{5368}{3}\zeta_{11}^3 + \frac{9968}{3}\zeta_{11}^2 + \frac{20288}{3}.
\end{aligned}$$

8.1.5. **Identity for $\mathcal{OK}_{11,2}^{(C)}(\zeta_{11}, z)$ - a quadratic residue case ($m = 2$).**

$$\begin{aligned}
 \mathcal{OK}_{11,2}^{(C)}(\zeta_{11}, z) &= q^{\frac{2}{11}}(q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{M}(k, 11, 11n+2) \zeta_{11}^k \right) q^n \\
 (8.5) \quad &= \frac{f_{22,2}(z)f_{22,4}(z)f_{22,6}(z)f_{22,8}(z)}{f_{22,1}(z)f_{22,5}(z)f_{22,7}(z)f_{22,9}(z)} \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
 &\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
 &\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
 &\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
 \end{aligned}$$

where the coefficients are :

$$\begin{aligned}
 a_{11,1} &= 92 + \frac{607}{9}\zeta_{11}^8 - \frac{40}{3}\zeta_{11}^7 + \frac{530}{9}\zeta_{11}^6 + \frac{530}{9}\zeta_{11}^5 - \frac{40}{3}\zeta_{11}^4 + \frac{607}{9}\zeta_{11}^3 + \frac{422}{9}\zeta_{11}^9 + \frac{422}{9}\zeta_{11}^2, \\
 a_{11,3} &= -23 - \frac{220}{9}\zeta_{11}^8 - \frac{68}{3}\zeta_{11}^7 - \frac{296}{9}\zeta_{11}^6 - \frac{296}{9}\zeta_{11}^5 - \frac{68}{3}\zeta_{11}^4 - \frac{220}{9}\zeta_{11}^3 - \frac{260}{9}\zeta_{11}^9 - \frac{260}{9}\zeta_{11}^2, \\
 a_{11,5} &= -43 - \frac{383}{9}\zeta_{11}^8 - \frac{52}{3}\zeta_{11}^7 - \frac{505}{9}\zeta_{11}^6 - \frac{505}{9}\zeta_{11}^5 - \frac{52}{3}\zeta_{11}^4 - \frac{383}{9}\zeta_{11}^3 - \frac{352}{9}\zeta_{11}^9 - \frac{352}{9}\zeta_{11}^2, \\
 a_{11,7} &= -82 - \frac{553}{9}\zeta_{11}^8 + \frac{118}{3}\zeta_{11}^7 - \frac{278}{9}\zeta_{11}^6 - \frac{278}{9}\zeta_{11}^5 + \frac{118}{3}\zeta_{11}^4 - \frac{553}{9}\zeta_{11}^3 - \frac{98}{9}\zeta_{11}^9 - \frac{98}{9}\zeta_{11}^2, \\
 a_{11,9} &= 61 + \frac{446}{9}\zeta_{11}^8 - \frac{95}{3}\zeta_{11}^7 + \frac{352}{9}\zeta_{11}^6 + \frac{352}{9}\zeta_{11}^5 - \frac{95}{3}\zeta_{11}^4 + \frac{446}{9}\zeta_{11}^3 + \frac{208}{9}\zeta_{11}^9 + \frac{208}{9}\zeta_{11}^2, \\
 b_{11,1} &= -452 - \frac{4354}{9}\zeta_{11}^8 + \frac{244}{3}\zeta_{11}^7 - \frac{3524}{9}\zeta_{11}^6 - \frac{3524}{9}\zeta_{11}^5 + \frac{244}{3}\zeta_{11}^4 - \frac{4354}{9}\zeta_{11}^3 - \frac{1094}{9}\zeta_{11}^9 - \frac{1094}{9}\zeta_{11}^2, \\
 b_{11,3} &= 96 - \frac{182}{9}\zeta_{11}^8 + \frac{212}{3}\zeta_{11}^7 + \frac{500}{9}\zeta_{11}^6 + \frac{500}{9}\zeta_{11}^5 + \frac{212}{3}\zeta_{11}^4 - \frac{182}{9}\zeta_{11}^3 + \frac{1166}{9}\zeta_{11}^9 + \frac{1166}{9}\zeta_{11}^2, \\
 b_{11,5} &= 118 + \frac{596}{9}\zeta_{11}^8 - \frac{410}{3}\zeta_{11}^7 + \frac{688}{9}\zeta_{11}^6 + \frac{688}{9}\zeta_{11}^5 - \frac{410}{3}\zeta_{11}^4 + \frac{596}{9}\zeta_{11}^3 + \frac{724}{9}\zeta_{11}^9 + \frac{724}{9}\zeta_{11}^2, \\
 b_{11,7} &= 248 + \frac{2486}{9}\zeta_{11}^8 + \frac{244}{3}\zeta_{11}^7 + \frac{2830}{9}\zeta_{11}^6 + \frac{2830}{9}\zeta_{11}^5 + \frac{244}{3}\zeta_{11}^4 + \frac{2486}{9}\zeta_{11}^3 + \frac{1570}{9}\zeta_{11}^9 + \frac{1570}{9}\zeta_{11}^2, \\
 b_{11,9} &= 2 - \frac{142}{9}\zeta_{11}^8 - \frac{170}{3}\zeta_{11}^7 - \frac{392}{9}\zeta_{11}^6 - \frac{392}{9}\zeta_{11}^5 - \frac{170}{3}\zeta_{11}^4 - \frac{142}{9}\zeta_{11}^3 - \frac{824}{9}\zeta_{11}^9 - \frac{824}{9}\zeta_{11}^2, \\
 c_{11,1} &= -566 - \frac{1370}{3}\zeta_{11}^8 - 16\zeta_{11}^7 - \frac{1375}{3}\zeta_{11}^6 - \frac{1375}{3}\zeta_{11}^5 - 16\zeta_{11}^4 - \frac{1370}{3}\zeta_{11}^3 - \frac{1036}{3}\zeta_{11}^9 - \frac{1036}{3}\zeta_{11}^2, \\
 c_{11,3} &= -173 - \frac{346}{3}\zeta_{11}^8 + 246\zeta_{11}^7 + \frac{46}{3}\zeta_{11}^6 + \frac{46}{3}\zeta_{11}^5 + 246\zeta_{11}^4 - \frac{346}{3}\zeta_{11}^3 + \frac{244}{3}\zeta_{11}^9 + \frac{244}{3}\zeta_{11}^2, \\
 c_{11,5} &= 124 + \frac{466}{3}\zeta_{11}^8 + 168\zeta_{11}^7 + \frac{836}{3}\zeta_{11}^6 + \frac{836}{3}\zeta_{11}^5 + 168\zeta_{11}^4 + \frac{466}{3}\zeta_{11}^3 + \frac{680}{3}\zeta_{11}^9 + \frac{680}{3}\zeta_{11}^2, \\
 c_{11,7} &= -600 - \frac{1382}{3}\zeta_{11}^8 + 267\zeta_{11}^7 - \frac{889}{3}\zeta_{11}^6 - \frac{889}{3}\zeta_{11}^5 + 267\zeta_{11}^4 - \frac{1382}{3}\zeta_{11}^3 - \frac{478}{3}\zeta_{11}^9 - \frac{478}{3}\zeta_{11}^2, \\
 c_{11,9} &= 534 + \frac{1267}{3}\zeta_{11}^8 - 143\zeta_{11}^7 + \frac{1148}{3}\zeta_{11}^6 + \frac{1148}{3}\zeta_{11}^5 - 143\zeta_{11}^4 + \frac{1267}{3}\zeta_{11}^3 + \frac{782}{3}\zeta_{11}^9 + \frac{782}{3}\zeta_{11}^2, \\
 d_{11,1} &= d_{11,3} = d_{11,5} = d_{11,7} = d_{11,9} = 0, \\
 e_{11,1} &= -588 - \frac{4946}{9}\zeta_{11}^8 - \frac{112}{3}\zeta_{11}^7 - \frac{5158}{9}\zeta_{11}^6 - \frac{5158}{9}\zeta_{11}^5 - \frac{112}{3}\zeta_{11}^4 - \frac{4946}{9}\zeta_{11}^3 - \frac{2908}{9}\zeta_{11}^9 - \frac{2908}{9}\zeta_{11}^2,
 \end{aligned}$$

$$\begin{aligned}
e_{11,3} &= 402 + \frac{3262}{9}\zeta_{11}^8 - \frac{598}{3}\zeta_{11}^7 + \frac{1358}{9}\zeta_{11}^6 + \frac{1358}{9}\zeta_{11}^5 - \frac{598}{3}\zeta_{11}^4 + \frac{3262}{9}\zeta_{11}^3 - \frac{334}{9}\zeta_{11}^9 - \frac{334}{9}\zeta_{11}^2, \\
e_{11,5} &= 472 + \frac{4126}{9}\zeta_{11}^8 + \frac{188}{3}\zeta_{11}^7 + \frac{4400}{9}\zeta_{11}^6 + \frac{4400}{9}\zeta_{11}^5 + \frac{188}{3}\zeta_{11}^4 + \frac{4126}{9}\zeta_{11}^3 + \frac{2888}{9}\zeta_{11}^9 + \frac{2888}{9}\zeta_{11}^2, \\
e_{11,7} &= -328 - \frac{2426}{9}\zeta_{11}^8 + \frac{968}{3}\zeta_{11}^7 - \frac{1126}{9}\zeta_{11}^6 - \frac{1126}{9}\zeta_{11}^5 + \frac{968}{3}\zeta_{11}^4 - \frac{2426}{9}\zeta_{11}^3 - \frac{208}{9}\zeta_{11}^9 - \frac{208}{9}\zeta_{11}^2, \\
e_{11,9} &= 426 + \frac{2272}{9}\zeta_{11}^8 - \frac{478}{3}\zeta_{11}^7 + \frac{1682}{9}\zeta_{11}^6 + \frac{1682}{9}\zeta_{11}^5 - \frac{478}{3}\zeta_{11}^4 + \frac{2272}{9}\zeta_{11}^3 + \frac{1646}{9}\zeta_{11}^9 + \frac{1646}{9}\zeta_{11}^2, \\
g_{11,1} &= -848 - \frac{8168}{9}\zeta_{11}^8 - \frac{16}{3}\zeta_{11}^7 - \frac{7408}{9}\zeta_{11}^6 - \frac{7408}{9}\zeta_{11}^5 - \frac{16}{3}\zeta_{11}^4 - \frac{8168}{9}\zeta_{11}^3 - \frac{3088}{9}\zeta_{11}^9 - \frac{3088}{9}\zeta_{11}^2, \\
g_{11,3} &= 640 + \frac{6232}{9}\zeta_{11}^8 - \frac{928}{3}\zeta_{11}^7 + \frac{3392}{9}\zeta_{11}^6 + \frac{3392}{9}\zeta_{11}^5 - \frac{928}{3}\zeta_{11}^4 + \frac{6232}{9}\zeta_{11}^3 + \frac{224}{9}\zeta_{11}^9 + \frac{224}{9}\zeta_{11}^2, \\
g_{11,5} &= 224 + \frac{2200}{9}\zeta_{11}^8 - \frac{256}{3}\zeta_{11}^7 + \frac{2240}{9}\zeta_{11}^6 + \frac{2240}{9}\zeta_{11}^5 - \frac{256}{3}\zeta_{11}^4 + \frac{2200}{9}\zeta_{11}^3 + \frac{2096}{9}\zeta_{11}^9 + \frac{2096}{9}\zeta_{11}^2, \\
g_{11,7} &= 112 + \frac{1624}{9}\zeta_{11}^8 + \frac{992}{3}\zeta_{11}^7 + \frac{1952}{9}\zeta_{11}^6 + \frac{1952}{9}\zeta_{11}^5 + \frac{992}{3}\zeta_{11}^4 + \frac{1624}{9}\zeta_{11}^3 + \frac{512}{9}\zeta_{11}^9 + \frac{512}{9}\zeta_{11}^2, \\
g_{11,9} &= 400 + \frac{1192}{9}\zeta_{11}^8 + \frac{32}{3}\zeta_{11}^7 + \frac{1232}{9}\zeta_{11}^6 + \frac{1232}{9}\zeta_{11}^5 + \frac{32}{3}\zeta_{11}^4 + \frac{1192}{9}\zeta_{11}^3 + \frac{1664}{9}\zeta_{11}^9 + \frac{1664}{9}\zeta_{11}^2, \\
h_{11,1} &= -480 - \frac{1472}{3}\zeta_{11}^8 - \frac{1312}{3}\zeta_{11}^6 - \frac{1312}{3}\zeta_{11}^5 - \frac{1472}{3}\zeta_{11}^3 - \frac{544}{3}\zeta_{11}^9 - \frac{544}{3}\zeta_{11}^2, \\
h_{11,3} &= 416 + \frac{1408}{3}\zeta_{11}^8 - 192\zeta_{11}^7 + \frac{800}{3}\zeta_{11}^6 + \frac{800}{3}\zeta_{11}^5 - 192\zeta_{11}^4 + \frac{1408}{3}\zeta_{11}^3 + \frac{32}{3}\zeta_{11}^9 + \frac{32}{3}\zeta_{11}^2, \\
h_{11,5} &= 96 + \frac{448}{3}\zeta_{11}^8 - 64\zeta_{11}^7 + \frac{416}{3}\zeta_{11}^6 + \frac{416}{3}\zeta_{11}^5 - 64\zeta_{11}^4 + \frac{448}{3}\zeta_{11}^3 + \frac{416}{3}\zeta_{11}^9 + \frac{416}{3}\zeta_{11}^2, \\
h_{11,7} &= 96 + \frac{256}{3}\zeta_{11}^8 + 192\zeta_{11}^7 + \frac{224}{3}\zeta_{11}^6 + \frac{224}{3}\zeta_{11}^5 + 192\zeta_{11}^4 + \frac{256}{3}\zeta_{11}^3 + \frac{32}{3}\zeta_{11}^9 + \frac{32}{3}\zeta_{11}^2, \\
h_{11,9} &= 224 + \frac{64}{3}\zeta_{11}^8 + 64\zeta_{11}^7 + \frac{224}{3}\zeta_{11}^6 + \frac{224}{3}\zeta_{11}^5 + 64\zeta_{11}^4 + \frac{64}{3}\zeta_{11}^3 + \frac{416}{3}\zeta_{11}^9 + \frac{416}{3}\zeta_{11}^2, \\
m_{11,1} &= -8 - \frac{110}{9}\zeta_{11}^8 - \frac{148}{3}\zeta_{11}^7 - \frac{355}{9}\zeta_{11}^6 - \frac{355}{9}\zeta_{11}^5 - \frac{148}{3}\zeta_{11}^4 - \frac{110}{9}\zeta_{11}^3 - \frac{364}{9}\zeta_{11}^9 - \frac{364}{9}\zeta_{11}^2, \\
m_{11,3} &= -127 - \frac{880}{9}\zeta_{11}^8 + \frac{136}{3}\zeta_{11}^7 - \frac{680}{9}\zeta_{11}^6 - \frac{680}{9}\zeta_{11}^5 + \frac{136}{3}\zeta_{11}^4 - \frac{880}{9}\zeta_{11}^3 - \frac{410}{9}\zeta_{11}^9 - \frac{410}{9}\zeta_{11}^2, \\
m_{11,5} &= 84 + \frac{497}{9}\zeta_{11}^8 - \frac{188}{3}\zeta_{11}^7 + \frac{175}{9}\zeta_{11}^6 + \frac{175}{9}\zeta_{11}^5 - \frac{188}{3}\zeta_{11}^4 + \frac{497}{9}\zeta_{11}^3 + \frac{58}{9}\zeta_{11}^9 + \frac{58}{9}\zeta_{11}^2, \\
m_{11,7} &= 150 + \frac{1100}{9}\zeta_{11}^8 - \frac{68}{3}\zeta_{11}^7 + \frac{976}{9}\zeta_{11}^6 + \frac{976}{9}\zeta_{11}^5 - \frac{68}{3}\zeta_{11}^4 + \frac{1100}{9}\zeta_{11}^3 + \frac{670}{9}\zeta_{11}^9 + \frac{670}{9}\zeta_{11}^2, \\
m_{11,9} &= -69 - \frac{556}{9}\zeta_{11}^8 - \frac{53}{3}\zeta_{11}^7 - \frac{707}{9}\zeta_{11}^6 - \frac{707}{9}\zeta_{11}^5 - \frac{53}{3}\zeta_{11}^4 - \frac{556}{9}\zeta_{11}^3 - \frac{572}{9}\zeta_{11}^9 - \frac{572}{9}\zeta_{11}^2, \\
n_{11,1} &= -36 - \frac{80}{3}\zeta_{11}^8 - 20\zeta_{11}^7 - \frac{112}{3}\zeta_{11}^6 - \frac{112}{3}\zeta_{11}^5 - 20\zeta_{11}^4 - \frac{80}{3}\zeta_{11}^3 - \frac{112}{3}\zeta_{11}^9 - \frac{112}{3}\zeta_{11}^2, \\
n_{11,3} &= -52 - \frac{124}{3}\zeta_{11}^8 + 28\zeta_{11}^7 - \frac{92}{3}\zeta_{11}^6 - \frac{92}{3}\zeta_{11}^5 + 28\zeta_{11}^4 - \frac{124}{3}\zeta_{11}^3 - \frac{32}{3}\zeta_{11}^9 - \frac{32}{3}\zeta_{11}^2, \\
n_{11,5} &= -\frac{32}{3}\zeta_{11}^9 + \frac{44}{3}\zeta_{11}^8 - 28\zeta_{11}^7 - \frac{8}{3}\zeta_{11}^6 - \frac{8}{3}\zeta_{11}^5 - 28\zeta_{11}^4 + \frac{44}{3}\zeta_{11}^3 - \frac{32}{3}\zeta_{11}^2 + 24, \\
n_{11,7} &= \frac{112}{3}\zeta_{11}^9 + \frac{200}{3}\zeta_{11}^8 - 20\zeta_{11}^7 + \frac{172}{3}\zeta_{11}^6 + \frac{172}{3}\zeta_{11}^5 - 20\zeta_{11}^4 + \frac{200}{3}\zeta_{11}^3 + \frac{112}{3}\zeta_{11}^2 + 84, \\
n_{11,9} &= -48 - \frac{112}{3}\zeta_{11}^8 - \frac{140}{3}\zeta_{11}^6 - \frac{140}{3}\zeta_{11}^5 - \frac{112}{3}\zeta_{11}^3 - \frac{116}{3}\zeta_{11}^9 - \frac{116}{3}\zeta_{11}^2.
\end{aligned}$$

8.1.6. **Identity for $\mathcal{OK}_{11,2}^{(R)}(\zeta_{11}, z)$ - a quadratic residue case ($m = 2$).**

$$\mathcal{OK}_{11,2}^{(R)}(\zeta_{11}, z) = q^{\frac{2}{11}}(q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \bar{N}(k, 11, 11n+2) \zeta_{11}^k \right) q^n$$

(8.6)

$$\begin{aligned}
& + \frac{2q^2(1 - \zeta_{11})}{1 + \zeta_{11}}(\zeta_{11}^{-6} - \zeta_{11}^6)\Phi_{11,3}(q) \\
& = \frac{f_{22,2}(z)f_{22,4}(z)f_{22,6}(z)f_{22,8}(z)}{f_{22,1}(z)f_{22,5}(z)f_{22,7}(z)f_{22,9}(z)} \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
& \quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
& \quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
& \quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
\end{aligned}$$

where the coefficients are :

$$\begin{aligned}
a_{11,1} &= -\frac{3904}{9} + \frac{2224}{9}\zeta_{11}^9 - \frac{2242}{3}\zeta_{11}^8 + \frac{3974}{9}\zeta_{11}^7 - \frac{994}{3}\zeta_{11}^6 - \frac{994}{3}\zeta_{11}^5 + \frac{3974}{9}\zeta_{11}^4 - \frac{2242}{3}\zeta_{11}^3 + \frac{2224}{9}\zeta_{11}^2, \\
a_{11,3} &= -\frac{3602}{9} - \frac{4168}{9}\zeta_{11}^9 - \frac{200}{3}\zeta_{11}^8 - \frac{4046}{9}\zeta_{11}^7 - \frac{902}{3}\zeta_{11}^6 - \frac{902}{3}\zeta_{11}^5 - \frac{4046}{9}\zeta_{11}^4 - \frac{200}{3}\zeta_{11}^3 - \frac{4168}{9}\zeta_{11}^2, \\
a_{11,5} &= -\frac{2770}{9} - \frac{5462}{9}\zeta_{11}^9 + \frac{428}{3}\zeta_{11}^8 - \frac{5566}{9}\zeta_{11}^7 - \frac{694}{3}\zeta_{11}^6 - \frac{694}{3}\zeta_{11}^5 - \frac{5566}{9}\zeta_{11}^4 + \frac{428}{3}\zeta_{11}^3 - \frac{5462}{9}\zeta_{11}^2, \\
a_{11,7} &= \frac{10348}{9} + \frac{4454}{9}\zeta_{11}^9 + \frac{2818}{3}\zeta_{11}^8 + \frac{598}{9}\zeta_{11}^7 + \frac{2464}{3}\zeta_{11}^6 + \frac{2464}{3}\zeta_{11}^5 + \frac{598}{9}\zeta_{11}^4 + \frac{2818}{3}\zeta_{11}^3 + \frac{4454}{9}\zeta_{11}^2, \\
a_{11,9} &= -\frac{9938}{9} - \frac{2566}{9}\zeta_{11}^9 - \frac{2978}{3}\zeta_{11}^8 + \frac{958}{9}\zeta_{11}^7 - \frac{2108}{3}\zeta_{11}^6 - \frac{2108}{3}\zeta_{11}^5 + \frac{958}{9}\zeta_{11}^4 - \frac{2978}{3}\zeta_{11}^3 - \frac{2566}{9}\zeta_{11}^2, \\
b_{11,1} &= \frac{89884}{9} + \frac{41792}{9}\zeta_{11}^9 + \frac{21508}{3}\zeta_{11}^8 + \frac{3292}{9}\zeta_{11}^7 + \frac{18172}{3}\zeta_{11}^6 + \frac{18172}{3}\zeta_{11}^5 + \frac{3292}{9}\zeta_{11}^4 + \frac{21508}{3}\zeta_{11}^3 + \frac{41792}{9}\zeta_{11}^2, \\
b_{11,3} &= \frac{22292}{9} + \frac{35068}{9}\zeta_{11}^9 - \frac{3232}{3}\zeta_{11}^8 + \frac{29792}{9}\zeta_{11}^7 + \frac{2324}{3}\zeta_{11}^6 + \frac{2324}{3}\zeta_{11}^5 + \frac{29792}{9}\zeta_{11}^4 - \frac{3232}{3}\zeta_{11}^3 + \frac{35068}{9}\zeta_{11}^2, \\
b_{11,5} &= -\frac{41516}{9} - \frac{12748}{9}\zeta_{11}^9 - \frac{10844}{3}\zeta_{11}^8 - \frac{6104}{9}\zeta_{11}^7 - \frac{8168}{3}\zeta_{11}^6 - \frac{8168}{3}\zeta_{11}^5 - \frac{6104}{9}\zeta_{11}^4 - \frac{10844}{3}\zeta_{11}^3 - \frac{12748}{9}\zeta_{11}^2, \\
b_{11,7} &= -\frac{11492}{9} + \frac{14576}{9}\zeta_{11}^9 - \frac{6932}{3}\zeta_{11}^8 + \frac{27340}{9}\zeta_{11}^7 - \frac{512}{3}\zeta_{11}^6 - \frac{512}{3}\zeta_{11}^5 + \frac{27340}{9}\zeta_{11}^4 - \frac{6932}{3}\zeta_{11}^3 + \frac{14576}{9}\zeta_{11}^2, \\
b_{11,9} &= -\frac{10016}{9} - \frac{16852}{9}\zeta_{11}^9 + \frac{1300}{3}\zeta_{11}^8 - \frac{15572}{9}\zeta_{11}^7 - \frac{2624}{3}\zeta_{11}^6 - \frac{2624}{3}\zeta_{11}^5 - \frac{15572}{9}\zeta_{11}^4 + \frac{1300}{3}\zeta_{11}^3 - \frac{16852}{9}\zeta_{11}^2, \\
c_{11,1} &= \frac{2870}{3} - \frac{9602}{3}\zeta_{11}^9 + 4012\zeta_{11}^8 - \frac{13102}{3}\zeta_{11}^7 + 796\zeta_{11}^6 + 796\zeta_{11}^5 - \frac{13102}{3}\zeta_{11}^4 + 4012\zeta_{11}^3 - \frac{9602}{3}\zeta_{11}^2, \\
c_{11,3} &= \frac{19270}{3} + \frac{12110}{3}\zeta_{11}^9 + 3908\zeta_{11}^8 + \frac{6622}{3}\zeta_{11}^7 + 4412\zeta_{11}^6 + 4412\zeta_{11}^5 + \frac{6622}{3}\zeta_{11}^4 + 3908\zeta_{11}^3 + \frac{12110}{3}\zeta_{11}^2, \\
c_{11,5} &= \frac{8732}{3} + \frac{11974}{3}\zeta_{11}^9 + 228\zeta_{11}^8 + \frac{11120}{3}\zeta_{11}^7 + 2370\zeta_{11}^6 + 2370\zeta_{11}^5 + \frac{11120}{3}\zeta_{11}^4 + 228\zeta_{11}^3 + \frac{11974}{3}\zeta_{11}^2, \\
c_{11,7} &= \frac{26972}{3} + \frac{9250}{3}\zeta_{11}^9 + 7832\zeta_{11}^8 - \frac{562}{3}\zeta_{11}^7 + 6150\zeta_{11}^6 + 6150\zeta_{11}^5 - \frac{562}{3}\zeta_{11}^4 + 7832\zeta_{11}^3 + \frac{9250}{3}\zeta_{11}^2, \\
c_{11,9} &= -\frac{15268}{3} + \frac{790}{3}\zeta_{11}^9 - 5948\zeta_{11}^8 + \frac{6494}{3}\zeta_{11}^7 - 3324\zeta_{11}^6 - 3324\zeta_{11}^5 + \frac{6494}{3}\zeta_{11}^4 - 5948\zeta_{11}^3 + \frac{790}{3}\zeta_{11}^2, \\
d_{11,1} &= 5632\zeta_{11}^9 - 5632\zeta_{11}^8 + 4928\zeta_{11}^7 - 1408\zeta_{11}^6 - 1408\zeta_{11}^5 + 4928\zeta_{11}^4 - 5632\zeta_{11}^3 + 5632\zeta_{11}^2 + 704, \\
d_{11,3} &= -1408\zeta_{11}^9 + 9152\zeta_{11}^8 - 7040\zeta_{11}^7 + 4224\zeta_{11}^6 + 4224\zeta_{11}^5 - 7040\zeta_{11}^4 + 9152\zeta_{11}^3 - 1408\zeta_{11}^2 + 5632, \\
d_{11,5} &= -10560\zeta_{11}^9 - 4928\zeta_{11}^8 - 6336\zeta_{11}^7 - 7040\zeta_{11}^6 - 7040\zeta_{11}^5 - 6336\zeta_{11}^4 - 4928\zeta_{11}^3 - 10560\zeta_{11}^2 - 11968, \\
d_{11,7} &= -704\zeta_{11}^9 - 7040\zeta_{11}^8 + 4224\zeta_{11}^7 - 5632\zeta_{11}^6 - 5632\zeta_{11}^5 + 4224\zeta_{11}^4 - 7040\zeta_{11}^3 - 704\zeta_{11}^2 - 4928,
\end{aligned}$$

$$\begin{aligned}
d_{11,9} &= 11968\zeta_{11}^9 + 11264\zeta_{11}^8 + 5632\zeta_{11}^7 + 10560\zeta_{11}^6 + 10560\zeta_{11}^5 + 5632\zeta_{11}^4 + 11264\zeta_{11}^3 + 11968\zeta_{11}^2 + 21824, \\
e_{11,1} &= \frac{23240}{9} - \frac{15848}{9}\zeta_{11}^9 + \frac{12116}{3}\zeta_{11}^8 - \frac{27868}{9}\zeta_{11}^7 + \frac{3980}{3}\zeta_{11}^6 + \frac{3980}{3}\zeta_{11}^5 - \frac{27868}{9}\zeta_{11}^4 + \frac{12116}{3}\zeta_{11}^3 - \frac{15848}{9}\zeta_{11}^2, \\
e_{11,3} &= -\frac{41812}{9} - \frac{31688}{9}\zeta_{11}^9 - \frac{8128}{3}\zeta_{11}^8 - \frac{16528}{9}\zeta_{11}^7 - \frac{10624}{3}\zeta_{11}^6 - \frac{10624}{3}\zeta_{11}^5 - \frac{16528}{9}\zeta_{11}^4 - \frac{8128}{3}\zeta_{11}^3 - \frac{31688}{9}\zeta_{11}^2, \\
e_{11,5} &= -\frac{9088}{9} + \frac{26092}{9}\zeta_{11}^9 - \frac{10960}{3}\zeta_{11}^8 + \frac{40352}{9}\zeta_{11}^7 - \frac{1768}{3}\zeta_{11}^6 - \frac{1768}{3}\zeta_{11}^5 + \frac{40352}{9}\zeta_{11}^4 - \frac{10960}{3}\zeta_{11}^3 + \frac{26092}{9}\zeta_{11}^2, \\
e_{11,7} &= \frac{77024}{9} + \frac{41140}{9}\zeta_{11}^9 + \frac{17240}{3}\zeta_{11}^8 + \frac{15872}{9}\zeta_{11}^7 + \frac{16772}{3}\zeta_{11}^6 + \frac{16772}{3}\zeta_{11}^5 + \frac{15872}{9}\zeta_{11}^4 + \frac{17240}{3}\zeta_{11}^3 + \frac{41140}{9}\zeta_{11}^2, \\
e_{11,9} &= -\frac{61576}{9} - \frac{16568}{9}\zeta_{11}^9 - \frac{19576}{3}\zeta_{11}^8 + \frac{4028}{9}\zeta_{11}^7 - \frac{14440}{3}\zeta_{11}^6 - \frac{14440}{3}\zeta_{11}^5 + \frac{4028}{9}\zeta_{11}^4 - \frac{19576}{3}\zeta_{11}^3 - \frac{16568}{9}\zeta_{11}^2, \\
g_{11,1} &= \frac{95744}{9} + \frac{24688}{9}\zeta_{11}^9 + \frac{26144}{3}\zeta_{11}^8 - \frac{24736}{9}\zeta_{11}^7 + \frac{16112}{3}\zeta_{11}^6 + \frac{16112}{3}\zeta_{11}^5 - \frac{24736}{9}\zeta_{11}^4 + \frac{26144}{3}\zeta_{11}^3 + \frac{24688}{9}\zeta_{11}^2, \\
g_{11,3} &= -\frac{125728}{9} - \frac{72656}{9}\zeta_{11}^9 - \frac{26080}{3}\zeta_{11}^8 - \frac{23008}{9}\zeta_{11}^7 - \frac{24976}{3}\zeta_{11}^6 - \frac{24976}{3}\zeta_{11}^5 - \frac{23008}{9}\zeta_{11}^4 - \frac{26080}{3}\zeta_{11}^3 - \frac{72656}{9}\zeta_{11}^2, \\
g_{11,5} &= -\frac{17152}{9} + \frac{10000}{9}\zeta_{11}^9 - \frac{8896}{3}\zeta_{11}^8 + \frac{12128}{9}\zeta_{11}^7 - \frac{2896}{3}\zeta_{11}^6 - \frac{2896}{3}\zeta_{11}^5 + \frac{12128}{9}\zeta_{11}^4 - \frac{8896}{3}\zeta_{11}^3 + \frac{10000}{9}\zeta_{11}^2, \\
g_{11,7} &= \frac{19424}{9} + \frac{14032}{9}\zeta_{11}^9 + \frac{3872}{3}\zeta_{11}^8 + \frac{24800}{9}\zeta_{11}^7 + \frac{5936}{3}\zeta_{11}^6 + \frac{5936}{3}\zeta_{11}^5 + \frac{24800}{9}\zeta_{11}^4 + \frac{3872}{3}\zeta_{11}^3 + \frac{14032}{9}\zeta_{11}^2, \\
g_{11,9} &= \frac{12224}{9} + \frac{27280}{9}\zeta_{11}^9 - \frac{6304}{3}\zeta_{11}^8 + \frac{29120}{9}\zeta_{11}^7 - \frac{1744}{3}\zeta_{11}^6 - \frac{1744}{3}\zeta_{11}^5 + \frac{29120}{9}\zeta_{11}^4 - \frac{6304}{3}\zeta_{11}^3 + \frac{27280}{9}\zeta_{11}^2, \\
h_{11,1} &= \frac{32384}{3} + \frac{10240}{3}\zeta_{11}^9 + 8512\zeta_{11}^8 - \frac{8512}{3}\zeta_{11}^7 + 5696\zeta_{11}^6 + 5696\zeta_{11}^5 - \frac{8512}{3}\zeta_{11}^4 + 8512\zeta_{11}^3 + \frac{10240}{3}\zeta_{11}^2, \\
h_{11,3} &= -\frac{55936}{3} - \frac{32000}{3}\zeta_{11}^9 - 11328\zeta_{11}^8 - \frac{8128}{3}\zeta_{11}^7 - 10560\zeta_{11}^6 - 10560\zeta_{11}^5 - \frac{8128}{3}\zeta_{11}^4 - 11328\zeta_{11}^3 - \frac{32000}{3}\zeta_{11}^2, \\
h_{11,5} &= -\frac{20992}{3} - \frac{8192}{3}\zeta_{11}^9 - 4928\zeta_{11}^8 - \frac{3904}{3}\zeta_{11}^7 - 3264\zeta_{11}^6 - 3264\zeta_{11}^5 - \frac{3904}{3}\zeta_{11}^4 - 4928\zeta_{11}^3 - \frac{8192}{3}\zeta_{11}^2, \\
h_{11,7} &= \frac{10880}{3} + \frac{7936}{3}\zeta_{11}^9 + 1216\zeta_{11}^8 + \frac{12608}{3}\zeta_{11}^7 + 1856\zeta_{11}^6 + 1856\zeta_{11}^5 + \frac{12608}{3}\zeta_{11}^4 + 1216\zeta_{11}^3 + \frac{7936}{3}\zeta_{11}^2, \\
h_{11,9} &= \frac{30848}{3} + \frac{24832}{3}\zeta_{11}^9 + 3008\zeta_{11}^8 + \frac{15680}{3}\zeta_{11}^7 + 4160\zeta_{11}^6 + 4160\zeta_{11}^5 + \frac{15680}{3}\zeta_{11}^4 + 3008\zeta_{11}^3 + \frac{24832}{3}\zeta_{11}^2, \\
m_{11,1} &= -\frac{11386}{9} - \frac{9446}{9}\zeta_{11}^9 - \frac{1666}{3}\zeta_{11}^8 - \frac{6622}{9}\zeta_{11}^7 - \frac{2590}{3}\zeta_{11}^6 - \frac{2590}{3}\zeta_{11}^5 - \frac{6622}{9}\zeta_{11}^4 - \frac{1666}{3}\zeta_{11}^3 - \frac{9446}{9}\zeta_{11}^2, \\
m_{11,3} &= \frac{12520}{9} + \frac{1760}{9}\zeta_{11}^9 + \frac{4336}{3}\zeta_{11}^8 - \frac{2918}{9}\zeta_{11}^7 + \frac{2890}{3}\zeta_{11}^6 + \frac{2890}{3}\zeta_{11}^5 - \frac{2918}{9}\zeta_{11}^4 + \frac{4336}{3}\zeta_{11}^3 + \frac{1760}{9}\zeta_{11}^2, \\
m_{11,5} &= -\frac{15290}{9} - \frac{7222}{9}\zeta_{11}^9 - \frac{3908}{3}\zeta_{11}^8 - \frac{2648}{9}\zeta_{11}^7 - \frac{3584}{3}\zeta_{11}^6 - \frac{3584}{3}\zeta_{11}^5 - \frac{2648}{9}\zeta_{11}^4 - \frac{3908}{3}\zeta_{11}^3 - \frac{7222}{9}\zeta_{11}^2, \\
m_{11,7} &= -\frac{8918}{9} + \frac{2408}{9}\zeta_{11}^9 - \frac{4136}{3}\zeta_{11}^8 + \frac{6964}{9}\zeta_{11}^7 - \frac{1988}{3}\zeta_{11}^6 - \frac{1988}{3}\zeta_{11}^5 + \frac{6964}{9}\zeta_{11}^4 - \frac{4136}{3}\zeta_{11}^3 + \frac{2408}{9}\zeta_{11}^2, \\
m_{11,9} &= -\frac{1448}{9} - \frac{6880}{9}\zeta_{11}^9 + \frac{1312}{3}\zeta_{11}^8 - \frac{7580}{9}\zeta_{11}^7 - \frac{482}{3}\zeta_{11}^6 - \frac{482}{3}\zeta_{11}^5 - \frac{7580}{9}\zeta_{11}^4 + \frac{1312}{3}\zeta_{11}^3 - \frac{6880}{9}\zeta_{11}^2, \\
n_{11,1} &= -\frac{1456}{3} - \frac{2384}{3}\zeta_{11}^9 + 176\zeta_{11}^8 - \frac{2320}{3}\zeta_{11}^7 - 272\zeta_{11}^6 - 272\zeta_{11}^5 - \frac{2320}{3}\zeta_{11}^4 + 176\zeta_{11}^3 - \frac{2384}{3}\zeta_{11}^2, \\
n_{11,3} &= \frac{3736}{3} + \frac{1664}{3}\zeta_{11}^9 + 896\zeta_{11}^8 + \frac{424}{3}\zeta_{11}^7 + 792\zeta_{11}^6 + 792\zeta_{11}^5 + \frac{424}{3}\zeta_{11}^4 + 896\zeta_{11}^3 + \frac{1664}{3}\zeta_{11}^2, \\
n_{11,5} &= -\frac{1888}{3}\zeta_{11}^9 - 464\zeta_{11}^8 - \frac{1208}{3}\zeta_{11}^7 - 624\zeta_{11}^6 - 624\zeta_{11}^5 - \frac{1208}{3}\zeta_{11}^4 - 464\zeta_{11}^3 - \frac{1888}{3}\zeta_{11}^2 - \frac{2504}{3}, \\
n_{11,7} &= -\frac{640}{3}\zeta_{11}^9 - 1248\zeta_{11}^8 + \frac{856}{3}\zeta_{11}^7 - 824\zeta_{11}^6 - 824\zeta_{11}^5 + \frac{856}{3}\zeta_{11}^4 - 1248\zeta_{11}^3 - \frac{640}{3}\zeta_{11}^2 - \frac{3752}{3}, \\
n_{11,9} &= \frac{616}{3} - \frac{1264}{3}\zeta_{11}^9 + 552\zeta_{11}^8 - \frac{1520}{3}\zeta_{11}^7 + 112\zeta_{11}^6 + 112\zeta_{11}^5 - \frac{1520}{3}\zeta_{11}^4 + 552\zeta_{11}^3 - \frac{1264}{3}\zeta_{11}^2.
\end{aligned}$$

8.1.7. *Another identity for $\mathcal{OK}_{11,0}^{(C)}(\zeta_{11}, z)$ with coefficients in $\mathbb{Z}[\zeta_{11}]$.*

$$\begin{aligned}
 \mathcal{OK}_{11,0}^{(C)}(\zeta_{11}, z) &= (q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{M}(k, 11, 11n) \zeta_{11}^k \right) q^n \\
 (8.7) \quad &+ 8(2 - 2(\zeta_{11}^3 + \zeta_{11}^8) + 2(\zeta_{11}^4 + \zeta_{11}^7) - (\zeta_{11}^5 + \zeta_{11}^6)) \frac{\eta(11z)^7 \eta(2z)^{12}}{\eta(22z)^4 \eta(z)^{13}} \\
 &= \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
 &\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
 &\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) + \\
 &\quad \left. \left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z) \right),
 \end{aligned}$$

where the coefficients are :

$$\begin{aligned}
 a_{11,1} &= 111\zeta_{11}^9 - 324\zeta_{11}^8 + 325\zeta_{11}^7 - 134\zeta_{11}^6 - 134\zeta_{11}^5 + 325\zeta_{11}^4 - 324\zeta_{11}^3 + 111\zeta_{11}^2 + 249, \\
 a_{11,3} &= -76\zeta_{11}^9 + 303\zeta_{11}^8 - 324\zeta_{11}^7 + 58\zeta_{11}^6 + 58\zeta_{11}^5 - 324\zeta_{11}^4 + 303\zeta_{11}^3 - 76\zeta_{11}^2 - 346, \\
 a_{11,5} &= -110\zeta_{11}^9 - 248\zeta_{11}^8 + 214\zeta_{11}^7 - 190\zeta_{11}^6 - 190\zeta_{11}^5 + 214\zeta_{11}^4 - 248\zeta_{11}^3 - 110\zeta_{11}^2 + 269, \\
 a_{11,7} &= 55\zeta_{11}^9 + 325\zeta_{11}^8 - 248\zeta_{11}^7 + 245\zeta_{11}^6 + 245\zeta_{11}^5 - 248\zeta_{11}^4 + 325\zeta_{11}^3 + 55\zeta_{11}^2 - 137, \\
 a_{11,9} &= 22\zeta_{11}^9 + 214\zeta_{11}^8 - 303\zeta_{11}^7 + 80\zeta_{11}^6 + 80\zeta_{11}^5 - 303\zeta_{11}^4 + 214\zeta_{11}^3 + 22\zeta_{11}^2 - 413, \\
 b_{11,1} &= 346\zeta_{11}^9 + 738\zeta_{11}^8 - 278\zeta_{11}^7 + 582\zeta_{11}^6 + 582\zeta_{11}^5 - 278\zeta_{11}^4 + 738\zeta_{11}^3 + 346\zeta_{11}^2 + 458, \\
 b_{11,3} &= 736\zeta_{11}^9 - 82\zeta_{11}^8 + 738\zeta_{11}^7 + 154\zeta_{11}^6 + 154\zeta_{11}^5 + 738\zeta_{11}^4 - 82\zeta_{11}^3 + 736\zeta_{11}^2 + 934, \\
 b_{11,5} &= 114\zeta_{11}^9 + 2\zeta_{11}^8 - 624\zeta_{11}^7 + 156\zeta_{11}^6 + 156\zeta_{11}^5 - 624\zeta_{11}^4 + 2\zeta_{11}^3 + 114\zeta_{11}^2 - 704, \\
 b_{11,7} &= 80\zeta_{11}^9 + 278\zeta_{11}^8 - 2\zeta_{11}^7 + 236\zeta_{11}^6 + 236\zeta_{11}^5 - 2\zeta_{11}^4 + 278\zeta_{11}^3 + 80\zeta_{11}^2 - 348, \\
 b_{11,9} &= 196\zeta_{11}^9 + 624\zeta_{11}^8 - 82\zeta_{11}^7 + 42\zeta_{11}^6 + 42\zeta_{11}^5 - 82\zeta_{11}^4 + 624\zeta_{11}^3 + 196\zeta_{11}^2 - 196, \\
 c_{11,1} &= -812\zeta_{11}^9 + 1323\zeta_{11}^8 - 1413\zeta_{11}^7 + 270\zeta_{11}^6 + 270\zeta_{11}^5 - 1413\zeta_{11}^4 + 1323\zeta_{11}^3 - 812\zeta_{11}^2 - 947, \\
 c_{11,3} &= 466\zeta_{11}^9 - 886\zeta_{11}^8 + 1323\zeta_{11}^7 + 196\zeta_{11}^6 + 196\zeta_{11}^5 + 1323\zeta_{11}^4 - 886\zeta_{11}^3 + 466\zeta_{11}^2 + 1850, \\
 c_{11,5} &= 722\zeta_{11}^9 + 857\zeta_{11}^8 - 601\zeta_{11}^7 + 1053\zeta_{11}^6 + 1053\zeta_{11}^5 - 601\zeta_{11}^4 + 857\zeta_{11}^3 + 722\zeta_{11}^2 - 630, \\
 c_{11,7} &= 29\zeta_{11}^9 + 1413\zeta_{11}^8 - 857\zeta_{11}^7 + 1082\zeta_{11}^6 + 1082\zeta_{11}^5 - 857\zeta_{11}^4 + 1413\zeta_{11}^3 + 29\zeta_{11}^2 - 45, \\
 c_{11,9} &= 527\zeta_{11}^9 + 601\zeta_{11}^8 - 886\zeta_{11}^7 + 331\zeta_{11}^6 + 331\zeta_{11}^5 - 886\zeta_{11}^4 + 601\zeta_{11}^3 + 527\zeta_{11}^2 - 1608, \\
 d_{11,1} &= -3872\zeta_{11}^8 + 3872\zeta_{11}^7 - 1936\zeta_{11}^6 - 1936\zeta_{11}^5 + 3872\zeta_{11}^4 - 3872\zeta_{11}^3 + 3872, \\
 d_{11,3} &= 3872\zeta_{11}^8 - 3872\zeta_{11}^7 + 1936\zeta_{11}^6 + 1936\zeta_{11}^5 - 3872\zeta_{11}^4 + 3872\zeta_{11}^3 - 3872, \\
 d_{11,5} &= -3872\zeta_{11}^8 + 3872\zeta_{11}^7 - 1936\zeta_{11}^6 - 1936\zeta_{11}^5 + 3872\zeta_{11}^4 - 3872\zeta_{11}^3 + 3872, \\
 d_{11,7} &= -3872\zeta_{11}^8 + 3872\zeta_{11}^7 - 1936\zeta_{11}^6 - 1936\zeta_{11}^5 + 3872\zeta_{11}^4 - 3872\zeta_{11}^3 + 3872, \\
 d_{11,9} &= -3872\zeta_{11}^8 + 3872\zeta_{11}^7 - 1936\zeta_{11}^6 - 1936\zeta_{11}^5 + 3872\zeta_{11}^4 - 3872\zeta_{11}^3 + 3872, \\
 e_{11,1} &= -692\zeta_{11}^9 - 984\zeta_{11}^8 + 636\zeta_{11}^7 - 665\zeta_{11}^6 - 665\zeta_{11}^5 + 636\zeta_{11}^4 - 984\zeta_{11}^3 - 692\zeta_{11}^2 + 1452,
 \end{aligned}$$

$$\begin{aligned}
e_{11,3} &= -816\zeta_{11}^9 - 1454\zeta_{11}^8 + 984\zeta_{11}^7 - 1481\zeta_{11}^6 - 1481\zeta_{11}^5 + 984\zeta_{11}^4 - 1454\zeta_{11}^3 - 816\zeta_{11}^2 + 166, \\
e_{11,5} &= 344\zeta_{11}^9 - 1800\zeta_{11}^8 + 1328\zeta_{11}^7 - 319\zeta_{11}^6 - 319\zeta_{11}^5 + 1328\zeta_{11}^4 - 1800\zeta_{11}^3 + 344\zeta_{11}^2 + 982, \\
e_{11,7} &= 346\zeta_{11}^9 - 636\zeta_{11}^8 + 1800\zeta_{11}^7 + 27\zeta_{11}^6 + 27\zeta_{11}^5 + 1800\zeta_{11}^4 - 636\zeta_{11}^3 + 346\zeta_{11}^2 + 2492, \\
e_{11,9} &= 818\zeta_{11}^9 - 1328\zeta_{11}^8 + 1454\zeta_{11}^7 - 663\zeta_{11}^6 - 663\zeta_{11}^5 + 1454\zeta_{11}^4 - 1328\zeta_{11}^3 + 818\zeta_{11}^2 + 1110, \\
g_{11,1} &= -1232\zeta_{11}^8 + 1408\zeta_{11}^7 - 616\zeta_{11}^6 - 616\zeta_{11}^5 + 1408\zeta_{11}^4 - 1232\zeta_{11}^3 + 2816, \\
g_{11,3} &= -1408\zeta_{11}^9 - 2640\zeta_{11}^8 + 1232\zeta_{11}^7 - 2024\zeta_{11}^6 - 2024\zeta_{11}^5 + 1232\zeta_{11}^4 - 2640\zeta_{11}^3 - 1408\zeta_{11}^2, \\
g_{11,5} &= 176\zeta_{11}^9 - 2640\zeta_{11}^8 + 1408\zeta_{11}^7 - 616\zeta_{11}^6 - 616\zeta_{11}^5 + 1408\zeta_{11}^4 - 2640\zeta_{11}^3 + 176\zeta_{11}^2 + 1408, \\
g_{11,7} &= -1408\zeta_{11}^8 + 2640\zeta_{11}^7 - 616\zeta_{11}^6 - 616\zeta_{11}^5 + 2640\zeta_{11}^4 - 1408\zeta_{11}^3 + 2640, \\
g_{11,9} &= 1232\zeta_{11}^9 - 1408\zeta_{11}^8 + 2640\zeta_{11}^7 - 792\zeta_{11}^6 - 792\zeta_{11}^5 + 2640\zeta_{11}^4 - 1408\zeta_{11}^3 + 1232\zeta_{11}^2 + 2464, \\
h_{11,1} &= -1056\zeta_{11}^8 + 1056\zeta_{11}^7 - 528\zeta_{11}^6 - 528\zeta_{11}^5 + 1056\zeta_{11}^4 - 1056\zeta_{11}^3 + 1760, \\
h_{11,3} &= -704\zeta_{11}^9 - 1760\zeta_{11}^8 + 1056\zeta_{11}^7 - 1232\zeta_{11}^6 - 1232\zeta_{11}^5 + 1056\zeta_{11}^4 - 1760\zeta_{11}^3 - 704\zeta_{11}^2 + 352, \\
h_{11,5} &= -1760\zeta_{11}^8 + 1056\zeta_{11}^7 - 528\zeta_{11}^6 - 528\zeta_{11}^5 + 1056\zeta_{11}^4 - 1760\zeta_{11}^3 + 1056, \\
h_{11,7} &= -1056\zeta_{11}^8 + 1760\zeta_{11}^7 - 528\zeta_{11}^6 - 528\zeta_{11}^5 + 1760\zeta_{11}^4 - 1056\zeta_{11}^3 + 1760, \\
h_{11,9} &= 704\zeta_{11}^9 - 1056\zeta_{11}^8 + 1760\zeta_{11}^7 - 528\zeta_{11}^6 - 528\zeta_{11}^5 + 1760\zeta_{11}^4 - 1056\zeta_{11}^3 + 704\zeta_{11}^2 + 1760, \\
m_{11,1} &= -165\zeta_{11}^9 + 143\zeta_{11}^8 - 253\zeta_{11}^7 - 77\zeta_{11}^6 - 77\zeta_{11}^5 - 253\zeta_{11}^4 + 143\zeta_{11}^3 - 165\zeta_{11}^2 - 308, \\
m_{11,3} &= -55\zeta_{11}^9 - 66\zeta_{11}^8 + 143\zeta_{11}^7 + 22\zeta_{11}^6 + 22\zeta_{11}^5 + 143\zeta_{11}^4 - 66\zeta_{11}^3 - 55\zeta_{11}^2 + 330, \\
m_{11,5} &= -55\zeta_{11}^9 - 198\zeta_{11}^8 + 88\zeta_{11}^7 - 220\zeta_{11}^6 - 220\zeta_{11}^5 + 88\zeta_{11}^4 - 198\zeta_{11}^3 - 55\zeta_{11}^2 - 44, \\
m_{11,7} &= 132\zeta_{11}^9 - 253\zeta_{11}^8 + 198\zeta_{11}^7 - 88\zeta_{11}^6 - 88\zeta_{11}^5 + 198\zeta_{11}^4 - 253\zeta_{11}^3 + 132\zeta_{11}^2 + 33, \\
m_{11,9} &= -187\zeta_{11}^9 - 88\zeta_{11}^8 + 66\zeta_{11}^7 - 165\zeta_{11}^6 - 165\zeta_{11}^5 + 66\zeta_{11}^4 - 88\zeta_{11}^3 - 187\zeta_{11}^2 + 121, \\
n_{11,1} &= -88\zeta_{11}^9 + 176\zeta_{11}^8 - 220\zeta_{11}^7 + 22\zeta_{11}^6 + 22\zeta_{11}^5 - 220\zeta_{11}^4 + 176\zeta_{11}^3 - 88\zeta_{11}^2 - 220, \\
n_{11,3} &= -132\zeta_{11}^8 + 176\zeta_{11}^7 - 22\zeta_{11}^6 - 22\zeta_{11}^5 + 176\zeta_{11}^4 - 132\zeta_{11}^3 + 264, \\
n_{11,5} &= -44\zeta_{11}^9 - 176\zeta_{11}^8 + 132\zeta_{11}^7 - 154\zeta_{11}^6 - 154\zeta_{11}^5 + 132\zeta_{11}^4 - 176\zeta_{11}^3 - 44\zeta_{11}^2 + 88, \\
n_{11,7} &= 44\zeta_{11}^9 - 220\zeta_{11}^8 + 176\zeta_{11}^7 - 110\zeta_{11}^6 - 110\zeta_{11}^5 + 176\zeta_{11}^4 - 220\zeta_{11}^3 + 44\zeta_{11}^2 + 88, \\
n_{11,9} &= -88\zeta_{11}^9 - 132\zeta_{11}^8 + 132\zeta_{11}^7 - 110\zeta_{11}^6 - 110\zeta_{11}^5 + 132\zeta_{11}^4 - 132\zeta_{11}^3 - 88\zeta_{11}^2 + 176.
\end{aligned}$$

8.1.8. **Another identity for $\mathcal{OK}_{11,0}^{(R)}(\zeta_{11}, z)$ with coefficients in $\mathbb{Z}[\zeta_{11}]$.**

$$\begin{aligned}
\mathcal{OK}_{11,0}^{(R)}(\zeta_{11}, z) &= (q^{11}; q^{11})_{\infty} \sum_{n=0}^{\infty} \left(\sum_{k=0}^{10} \overline{N}(k, 11, 11n) \zeta_{11}^k \right) q^n \\
(8.8) \quad &- 8(4 - 2(\zeta_{11}^2 + \zeta_{11}^9) + (\zeta_{11}^3 + \zeta_{11}^8) - 3(\zeta_{11}^4 + \zeta_{11}^7) + (\zeta_{11}^5 + \zeta_{11}^6)) \frac{\eta(11z)^7 \eta(2z)^{12}}{\eta(22z)^4 \eta(z)^{13}} \\
&= \sum_{\substack{r=1 \\ r \text{ odd}}}^9 \left(a_{11,r} j(22, \pi_r(\vec{n}_1), z) + b_{11,r} j(22, \pi_r(\vec{n}_2), z) + \right. \\
&\quad \left(c_{11,r} + d_{11,r} \frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} \right) j(22, \pi_r(\vec{n}_3), z) + \\
&\quad \left(f_{11,r} + g_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^2 + h_{11,r} \left(\frac{\eta(22z) \eta(2z)}{\eta(11z) \eta(z)} \right)^4 \right) j(22, \pi_r(\vec{n}_4), z) +
\end{aligned}$$

$$\left(m_{11,r} + n_{11,r} \left(\frac{\eta(22z)\eta(2z)}{\eta(11z)\eta(z)} \right)^2 \right) j(22, \pi_r(\vec{n}_5), z),$$

where the coefficients are :

$$\begin{aligned} a_{11,1} &= 4206\zeta_{11}^9 + 1266\zeta_{11}^8 + 1696\zeta_{11}^7 + 2476\zeta_{11}^6 + 2476\zeta_{11}^5 + 1696\zeta_{11}^4 + 1266\zeta_{11}^3 + 4206\zeta_{11}^2 + 3791, \\ a_{11,3} &= -3760\zeta_{11}^9 - 1002\zeta_{11}^8 - 2108\zeta_{11}^7 - 1748\zeta_{11}^6 - 1748\zeta_{11}^5 - 2108\zeta_{11}^4 - 1002\zeta_{11}^3 - 3760\zeta_{11}^2 - 2729, \\ a_{11,5} &= -3708\zeta_{11}^9 - 690\zeta_{11}^8 - 2922\zeta_{11}^7 - 2030\zeta_{11}^6 - 2030\zeta_{11}^5 - 2922\zeta_{11}^4 - 690\zeta_{11}^3 - 3708\zeta_{11}^2 - 4475, \\ a_{11,7} &= 874\zeta_{11}^9 - 520\zeta_{11}^8 + 1692\zeta_{11}^7 - 168\zeta_{11}^6 - 168\zeta_{11}^5 + 1692\zeta_{11}^4 - 520\zeta_{11}^3 + 874\zeta_{11}^2 + 695, \\ a_{11,9} &= -136\zeta_{11}^9 + 338\zeta_{11}^8 - 440\zeta_{11}^7 + 602\zeta_{11}^6 + 602\zeta_{11}^5 - 440\zeta_{11}^4 + 338\zeta_{11}^3 - 136\zeta_{11}^2 + 1025, \\ b_{11,1} &= 10136\zeta_{11}^9 - 4208\zeta_{11}^8 + 11400\zeta_{11}^7 + 452\zeta_{11}^6 + 452\zeta_{11}^5 + 11400\zeta_{11}^4 - 4208\zeta_{11}^3 + 10136\zeta_{11}^2 + 5536, \\ b_{11,3} &= 27336\zeta_{11}^9 + 4664\zeta_{11}^8 + 17824\zeta_{11}^7 + 13896\zeta_{11}^6 + 13896\zeta_{11}^5 + 17824\zeta_{11}^4 + 4664\zeta_{11}^3 + 27336\zeta_{11}^2 + 28804, \\ b_{11,5} &= 20\zeta_{11}^9 + 364\zeta_{11}^8 - 4728\zeta_{11}^7 + 1804\zeta_{11}^6 + 1804\zeta_{11}^5 - 4728\zeta_{11}^4 + 364\zeta_{11}^3 + 20\zeta_{11}^2 - 992, \\ b_{11,7} &= 12564\zeta_{11}^9 + 5024\zeta_{11}^8 + 8800\zeta_{11}^7 + 8612\zeta_{11}^6 + 8612\zeta_{11}^5 + 8800\zeta_{11}^4 + 5024\zeta_{11}^3 + 12564\zeta_{11}^2 + 15756, \\ b_{11,9} &= -8800\zeta_{11}^9 - 3224\zeta_{11}^8 - 3860\zeta_{11}^7 - 3680\zeta_{11}^6 - 3680\zeta_{11}^5 - 3860\zeta_{11}^4 - 3224\zeta_{11}^3 - 8800\zeta_{11}^2 - 4168, \\ c_{11,1} &= -33024\zeta_{11}^9 - 9200\zeta_{11}^8 - 17584\zeta_{11}^7 - 18738\zeta_{11}^6 - 18738\zeta_{11}^5 - 17584\zeta_{11}^4 - 9200\zeta_{11}^3 - 33024\zeta_{11}^2 - 32316, \\ c_{11,3} &= 21080\zeta_{11}^9 + 2976\zeta_{11}^8 + 15210\zeta_{11}^7 + 7428\zeta_{11}^6 + 7428\zeta_{11}^5 + 15210\zeta_{11}^4 + 2976\zeta_{11}^3 + 21080\zeta_{11}^2 + 14180, \\ c_{11,5} &= 25208\zeta_{11}^9 + 4790\zeta_{11}^8 + 18860\zeta_{11}^7 + 12588\zeta_{11}^6 + 12588\zeta_{11}^5 + 18860\zeta_{11}^4 + 4790\zeta_{11}^3 + 25208\zeta_{11}^2 + 25884, \\ c_{11,7} &= -22\zeta_{11}^9 - 4748\zeta_{11}^8 + 7652\zeta_{11}^7 - 4780\zeta_{11}^6 - 4780\zeta_{11}^5 + 7652\zeta_{11}^4 - 4748\zeta_{11}^3 - 22\zeta_{11}^2 - 2660, \\ c_{11,9} &= 14294\zeta_{11}^9 + 4822\zeta_{11}^8 + 6174\zeta_{11}^7 + 11046\zeta_{11}^6 + 11046\zeta_{11}^5 + 6174\zeta_{11}^4 + 4822\zeta_{11}^3 + 14294\zeta_{11}^2 + 19454, \\ d_{11,1} &= 34848\zeta_{11}^9 + 9680\zeta_{11}^8 + 9680\zeta_{11}^7 + 17424\zeta_{11}^6 + 17424\zeta_{11}^5 + 9680\zeta_{11}^4 + 9680\zeta_{11}^3 + 34848\zeta_{11}^2 + 23232, \\ d_{11,3} &= -27104\zeta_{11}^9 - 17424\zeta_{11}^8 - 9680\zeta_{11}^7 - 17424\zeta_{11}^6 - 17424\zeta_{11}^5 - 9680\zeta_{11}^4 - 17424\zeta_{11}^3 - 27104\zeta_{11}^2 - 23232, \\ d_{11,5} &= -27104\zeta_{11}^9 + 1936\zeta_{11}^8 - 29040\zeta_{11}^7 - 13552\zeta_{11}^6 - 13552\zeta_{11}^5 - 29040\zeta_{11}^4 + 1936\zeta_{11}^3 - 27104\zeta_{11}^2 - 38720, \\ d_{11,7} &= 11616\zeta_{11}^9 + 9680\zeta_{11}^8 + 1936\zeta_{11}^7 + 9680\zeta_{11}^6 + 9680\zeta_{11}^5 + 1936\zeta_{11}^4 + 9680\zeta_{11}^3 + 11616\zeta_{11}^2 + 15488, \\ d_{11,9} &= 34848\zeta_{11}^9 + 1936\zeta_{11}^8 + 25168\zeta_{11}^7 + 9680\zeta_{11}^6 + 9680\zeta_{11}^5 + 25168\zeta_{11}^4 + 1936\zeta_{11}^3 + 34848\zeta_{11}^2 + 23232, \\ e_{11,1} &= -20818\zeta_{11}^9 - 5453\zeta_{11}^8 - 13425\zeta_{11}^7 - 13717\zeta_{11}^6 - 13717\zeta_{11}^5 - 13425\zeta_{11}^4 - 5453\zeta_{11}^3 - 20818\zeta_{11}^2 - 27220, \\ e_{11,3} &= -19618\zeta_{11}^9 - 933\zeta_{11}^8 - 16533\zeta_{11}^7 - 7549\zeta_{11}^6 - 7549\zeta_{11}^5 - 16533\zeta_{11}^4 - 933\zeta_{11}^3 - 19618\zeta_{11}^2 - 17544, \\ e_{11,5} &= 27706\zeta_{11}^9 + 9427\zeta_{11}^8 + 13815\zeta_{11}^7 + 15831\zeta_{11}^6 + 15831\zeta_{11}^5 + 13815\zeta_{11}^4 + 9427\zeta_{11}^3 + 27706\zeta_{11}^2 + 25260, \\ e_{11,7} &= 18502\zeta_{11}^9 + 1163\zeta_{11}^8 + 15535\zeta_{11}^7 + 5119\zeta_{11}^6 + 5119\zeta_{11}^5 + 15535\zeta_{11}^4 + 1163\zeta_{11}^3 + 18502\zeta_{11}^2 + 12312, \\ e_{11,9} &= 13578\zeta_{11}^9 + 5471\zeta_{11}^8 + 1783\zeta_{11}^7 + 9991\zeta_{11}^6 + 9991\zeta_{11}^5 + 1783\zeta_{11}^4 + 5471\zeta_{11}^3 + 13578\zeta_{11}^2 + 13792, \\ g_{11,1} &= -2288\zeta_{11}^9 - 8184\zeta_{11}^8 - 792\zeta_{11}^7 - 8184\zeta_{11}^6 - 8184\zeta_{11}^5 - 792\zeta_{11}^4 - 8184\zeta_{11}^3 - 2288\zeta_{11}^2 - 13728, \\ g_{11,3} &= -33264\zeta_{11}^9 + 3080\zeta_{11}^8 - 31768\zeta_{11}^7 - 11000\zeta_{11}^6 - 11000\zeta_{11}^5 - 31768\zeta_{11}^4 + 3080\zeta_{11}^3 - 33264\zeta_{11}^2 - 33440, \\ g_{11,5} &= 17424\zeta_{11}^9 + 5896\zeta_{11}^8 + 2024\zeta_{11}^7 + 8712\zeta_{11}^6 + 8712\zeta_{11}^5 + 2024\zeta_{11}^4 + 5896\zeta_{11}^3 + 17424\zeta_{11}^2 + 5984, \\ g_{11,7} &= 11792\zeta_{11}^9 + 5896\zeta_{11}^8 + 10472\zeta_{11}^7 + 5896\zeta_{11}^6 + 5896\zeta_{11}^5 + 10472\zeta_{11}^4 + 5896\zeta_{11}^3 + 11792\zeta_{11}^2 + 11616, \\ g_{11,9} &= 31504\zeta_{11}^9 + 5896\zeta_{11}^8 + 18920\zeta_{11}^7 + 17160\zeta_{11}^6 + 17160\zeta_{11}^5 + 18920\zeta_{11}^4 + 5896\zeta_{11}^3 + 31504\zeta_{11}^2 + 34144, \\ h_{11,1} &= 352\zeta_{11}^9 - 8272\zeta_{11}^8 - 528\zeta_{11}^7 - 6864\zeta_{11}^6 - 6864\zeta_{11}^5 - 528\zeta_{11}^4 - 8272\zeta_{11}^3 + 352\zeta_{11}^2 - 11968, \\ h_{11,3} &= -30624\zeta_{11}^9 + 5808\zeta_{11}^8 - 28688\zeta_{11}^7 - 8272\zeta_{11}^6 - 8272\zeta_{11}^5 - 28688\zeta_{11}^4 + 5808\zeta_{11}^3 - 30624\zeta_{11}^2 - 27456, \\ h_{11,5} &= 1760\zeta_{11}^9 + 4400\zeta_{11}^8 - 8976\zeta_{11}^7 + 1584\zeta_{11}^6 + 1584\zeta_{11}^5 - 8976\zeta_{11}^4 + 4400\zeta_{11}^3 + 1760\zeta_{11}^2 - 9152, \\ h_{11,7} &= 13024\zeta_{11}^9 + 5808\zeta_{11}^8 + 13552\zeta_{11}^7 + 7216\zeta_{11}^6 + 7216\zeta_{11}^5 + 13552\zeta_{11}^4 + 5808\zeta_{11}^3 + 13024\zeta_{11}^2 + 17600, \end{aligned}$$

$$\begin{aligned}
h_{11,9} &= 36960\zeta_{11}^9 + 2992\zeta_{11}^8 + 26224\zeta_{11}^7 + 17072\zeta_{11}^6 + 17072\zeta_{11}^5 + 26224\zeta_{11}^4 + 2992\zeta_{11}^3 + 36960\zeta_{11}^2 + 38720, \\
m_{11,1} &= -6358\zeta_{11}^9 - 1056\zeta_{11}^8 - 4422\zeta_{11}^7 - 2750\zeta_{11}^6 - 2750\zeta_{11}^5 - 4422\zeta_{11}^4 - 1056\zeta_{11}^3 - 6358\zeta_{11}^2 - 5324, \\
m_{11,3} &= -1562\zeta_{11}^9 - 902\zeta_{11}^8 - 198\zeta_{11}^7 - 1760\zeta_{11}^6 - 1760\zeta_{11}^5 - 198\zeta_{11}^4 - 902\zeta_{11}^3 - 1562\zeta_{11}^2 - 2948, \\
m_{11,5} &= -2134\zeta_{11}^9 + 220\zeta_{11}^8 - 2750\zeta_{11}^7 - 264\zeta_{11}^6 - 264\zeta_{11}^5 - 2750\zeta_{11}^4 + 220\zeta_{11}^3 - 2134\zeta_{11}^2 - 1562, \\
m_{11,7} &= 5346\zeta_{11}^9 + 1914\zeta_{11}^8 + 2288\zeta_{11}^7 + 3520\zeta_{11}^6 + 3520\zeta_{11}^5 + 2288\zeta_{11}^4 + 1914\zeta_{11}^3 + 5346\zeta_{11}^2 + 5654, \\
m_{11,9} &= -6204\zeta_{11}^9 - 1386\zeta_{11}^8 - 4004\zeta_{11}^7 - 3344\zeta_{11}^6 - 3344\zeta_{11}^5 - 4004\zeta_{11}^4 - 1386\zeta_{11}^3 - 6204\zeta_{11}^2 - 6380, \\
n_{11,1} &= -5236\zeta_{11}^9 - 1254\zeta_{11}^8 - 3190\zeta_{11}^7 - 2662\zeta_{11}^6 - 2662\zeta_{11}^5 - 3190\zeta_{11}^4 - 1254\zeta_{11}^3 - 5236\zeta_{11}^2 - 4840, \\
n_{11,3} &= 1364\zeta_{11}^9 - 66\zeta_{11}^8 + 1254\zeta_{11}^7 + 22\zeta_{11}^6 + 22\zeta_{11}^5 + 1254\zeta_{11}^4 - 66\zeta_{11}^3 + 1364\zeta_{11}^2 + 176, \\
n_{11,5} &= -2420\zeta_{11}^9 - 242\zeta_{11}^8 - 2266\zeta_{11}^7 - 946\zeta_{11}^6 - 946\zeta_{11}^5 - 2266\zeta_{11}^4 - 242\zeta_{11}^3 - 2420\zeta_{11}^2 - 2288, \\
n_{11,7} &= 1980\zeta_{11}^9 + 1166\zeta_{11}^8 + 110\zeta_{11}^7 + 1694\zeta_{11}^6 + 1694\zeta_{11}^5 + 110\zeta_{11}^4 + 1166\zeta_{11}^3 + 1980\zeta_{11}^2 + 2200, \\
n_{11,9} &= -3476\zeta_{11}^9 - 770\zeta_{11}^8 - 2090\zeta_{11}^7 - 2090\zeta_{11}^6 - 2090\zeta_{11}^5 - 2090\zeta_{11}^4 - 770\zeta_{11}^3 - 3476\zeta_{11}^2 - 3960.
\end{aligned}$$

8.2. Proof of the identities. Since, in Theorems 4.4 and 5.8, we have established that $\mathcal{OK}_{11,m}^{(C)}(\zeta_{11}, z)$ and $\mathcal{OK}_{11,m}^{(R)}(\zeta_{11}, z)$ are weakly holomorphic modular forms of weight 1 on $\Gamma(2p)$, the proofs of identities (8.1)-(8.8) primarily involve showing that the other side is also a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$ and establishing the equality using the Valence formula. This in turn involves studying the transformation of the theta functions $f_{N,\rho}(z)$ due to Biagioli, recorded below. We also need the set of inequivalent cusps $\mathcal{S}_p$ for $\Gamma_1(2p)$ given in Proposition 7.3 and the lower bounds for the orders of $\mathcal{OK}_{11,m}^{(C)}(\zeta_{11}, z)$ and $\mathcal{OK}_{11,m}^{(R)}(\zeta_{11}, z)$ calculated in Theorems 7.4 and 7.5.

Let $\text{ord}(f; \zeta)$ denote the invariant order of meromorphic function f at cusp ζ .

Theorem 8.1 (The Valence Formula [17](p.98)). *Let $f \neq 0$ be a modular form of weight k with respect to a subgroup Γ of finite index in $\Gamma(1) = SL_2(\mathbb{Z})$. Then*

$$(8.9) \quad \text{ORD}(f, \Gamma) = \frac{1}{12} \mu k,$$

where μ is index $\widehat{\Gamma}$ in $\widehat{\Gamma(1)}$,

$$\text{ORD}(f, \Gamma) := \sum_{\zeta \in R^*} \text{ORD}(f, \zeta, \Gamma),$$

R^* is a fundamental region for Γ , and

$$(8.10) \quad \text{ORD}(f; \zeta; \Gamma) = n(\Gamma; \zeta) \text{ord}(f; \zeta),$$

for a cusp ζ and $n(\Gamma; \zeta)$ denotes the fan width of the cusp $\zeta \pmod{\Gamma}$.

Proposition 8.2 (Lemma 3.2 [2] (p.285)). *Let $1 \leq N \nmid \rho$, and $(a, c) = 1$. Then*

$$\text{ord}\left(f_{N,\rho}(z), \frac{a}{c}\right) = \frac{g^2}{2N} \left(\frac{a\rho}{g} - \left\lfloor \frac{a\rho}{g} \right\rfloor - \frac{1}{2} \right)^2,$$

and

$$\text{ord}\left(\eta(Nz), \frac{a}{c}\right) = \frac{g^2}{24N},$$

where $g = (N, c)$.

The following proposition follows easily from (8.10) and Propositions 7.3 and 8.2.

Proposition 8.3. *Let $p \geq 3$ be prime and suppose $s = \frac{a}{c}$ is one of the cusps listed in Proposition 7.3 with $i\infty$ represented by $\frac{1}{p}$. Then*

(i) *If $(c, 2p) = 1$ then*

$$ORD(j(2p, \vec{n}, z), s, \Gamma_1(2p)) = \frac{1}{24} \left(n_0 + 3 \sum_{j=1}^p n_j \right)$$

(ii) *If $(c, 2p) = 2$ then*

$$ORD(j(2p, \vec{n}, z), s, \Gamma_1(2p)) = \frac{1}{48} \left(n_0 + 3 \sum_{j=1}^p n_j \right).$$

(iii) *If $c = p$ then*

$$ORD(j(2p, \vec{n}, z), s, \Gamma_1(2p)) = \frac{p}{24} \left(n_0 + 12 \sum_{j=1}^p n_j \left(\frac{aj}{p} - \left\lfloor \frac{aj}{p} \right\rfloor - \frac{1}{2} \right)^2 \right).$$

(iv) *If $c = 2p$ then*

$$ORD(j(2p, \vec{n}, z), s, \Gamma_1(2p)) = \frac{p}{12} \left(n_0 + 12 \sum_{j=1}^p n_j \left(\frac{aj}{p} - \left\lfloor \frac{aj}{p} \right\rfloor - \frac{1}{2} \right)^2 \right).$$

An Algorithm for Proving Overpartition Crank and Rank Mod p Identities. We describe an algorithm for proving crank and rank mod p identities that utilizes the Valence Formula. This algorithm is analogous to the one we developed in our previous paper [Section 7 [11] (p.1922)]. We apply this algorithm with the aid of the MAPLE packages THETAIDS and ETA developed by the first author.

We note that, just like in [11], in the actual identities proved in the subsequent subsection, we write them in terms of the permutation π_r and the generalized eta-functions $j(2p, \pi_r(\vec{n}), z)$ defined as in Definition 2.4 and 2.1. In our algorithm, we write $\mathcal{OK}_{p,m}^*(\zeta_p, z)$, where $*$ = (R) or (C) in terms of generalized eta functions $j_{p,m,k}$, written in a general sense as in

$$(8.11) \quad \mathcal{OK}_{p,m}^*(\zeta_p, z) = \sum_{k=1}^r c_{2p,m,k} j_{2p,m,k}(z)$$

where $j_{2p,m,k}(z)$ are quotients of generalized eta-functions, and the $c_{2p,m,k}$ are non-zero cyclotomic integers in $\mathbb{Q}[\zeta_{11}]$ or $\mathbb{Z}[\zeta_{11}]$.

Step 1. Use Theorem 3.1 to check the three modularity conditions for each $j_{2p,m,k}(z)$, $1 \leq k \leq r$ in the RHS of the expression (8.11). This shows that the RHS of (8.11) is a weakly holomorphic modular form of weight 1 on $\Gamma(2p)$ satisfying the same modularity conditions as $\mathcal{OK}_{p,m}^{(C)}(\zeta_p, z)$ and $\mathcal{OK}_{p,m}^{(R)}(\zeta_p, z)$ in Theorems 4.4 and 5.8 respectively.

Calculate orders at $i\infty$ of each generalized eta quotient $j_{2p,m,k}$.

Step 2. For the cases when $m \neq 0$, convert the eta quotients to weight 0 by dividing each by the eta quotient having the lowest order at $i\infty$ i.e. choose $k = k_0$ such that

$$\text{ORD}(j_{2p,m,k_0}(z), i\infty, \Gamma_1(2p)) = \min_{1 \leq k \leq r} \text{ORD}(j_{2p,m,k}(z), i\infty, \Gamma_1(2p)).$$

Let $j_0 = j_{2p,m,k_0}(z)$. Then (8.11) has the following equivalent form :

$$(8.12) \quad \frac{\mathcal{OK}_{p,m}^*(\zeta_p, z)}{j_0} = \sum_{k=1}^r c_{2p,m,k} \frac{j_{2p,m,k}(z)}{j_0}$$

We note that the LHS and each term on the RHS is a modular function on $\Gamma_1(2p)$. When $m = 0$, we skip this step.

Step 3. At each cusp $s \in \mathcal{S}_p$ given in Proposition 7.3, calculate

$$\begin{aligned} & \text{ORD}(j_{2p,m,k}(z), s, \Gamma_1(2p)) \quad \text{when } m = 0, \\ & \text{ORD}\left(\frac{j_{2p,m,k}(z)}{j_0}, s, \Gamma_1(2p)\right) \quad \text{when } m \neq 0. \end{aligned}$$

for $1 \leq k \leq r$, using Proposition 8.3.

Step 4. At each cusp s , calculate the lower bound $\lambda(p, m, s)$ of

$$\begin{aligned} & \text{ORD}(\mathcal{OK}_{p,m}^*(\zeta_p, z), s, \Gamma_1(2p)) \quad \text{when } m = 0, \\ & \text{ORD}\left(\frac{\mathcal{OK}_{p,m}^*(\zeta_p, z)}{j_0}, s, \Gamma_1(2p)\right) \quad \text{when } m \neq 0. \end{aligned}$$

using Theorems 7.4 and 7.5. We note that value is an integer.

Step 5. Calculate

$$(8.13) \quad B = \begin{cases} \sum_{s \in \mathcal{S}_p, s \neq i\infty} \min(\{\text{ORD}(j_{2p,m,k}(z), s, \Gamma_1(2p)) : 1 \leq k \leq r\} \cup \{\lambda(p, m, s)\}), & \text{if } m = 0 \\ \sum_{s \in \mathcal{S}_p, s \neq i\infty} \min\left(\left\{\text{ORD}\left(\frac{j_{2p,m,k}(z)}{j_0}, s, \Gamma_1(2p)\right) : 1 \leq k \leq r\right\} \cup \{\lambda(p, m, s)\}\right), & \text{if } m \neq 0 \end{cases}$$

Step 6. Show

$$\text{ORD}(\text{LHS} - \text{RHS of (8.11)}, i\infty, \Gamma_1(2p)) \geq -B + 1 + \frac{\mu}{12} \text{ if } m = 0,$$

$$\text{ORD}(\text{LHS} - \text{RHS of (8.12)}, i\infty, \Gamma_1(2p)) \geq -B + 1 \text{ if } m \neq 0.$$

Here

$$(8.14) \quad \mu = \frac{3}{2}(p^2 - 1),$$

which is the index of $\widehat{\Gamma_1(2p)}$ in $\widehat{\Gamma(1)}$ obtained by combining Theorems 4.2.5 (2) and (3) [18, p.106]. Then the Valence formula in Theorem 8.1 implies LHS=RHS and (8.11) is proved.

We show how to prove one identity namely Equation (8.1) using the algorithm. The proof of the others follow suit. First we make a few remarks.

- (1) All the identities for the elements $\mathcal{OK}_{11,m}^{(C)}(\zeta_{11}, z)$ and $\mathcal{OK}_{11,m}^{(R)}(\zeta_{11}, z)$ where $0 \leq m \leq 10$ give a complete 11-dissection of the overpartition crank and rank generating functions $\mathcal{O}_C(z, \tau)$ and $\mathcal{O}_R(z, \tau)$ respectively. As remarked earlier, these identities in terms of generalized eta functions are completely determined by three particular ones, namely

$$\mathcal{OK}_{11,0}^{(C)}(\zeta_p, z), \mathcal{OK}_{11,m^+}^{(C)}(\zeta_p, z), \mathcal{OK}_{11,m^-}^{(C)}(\zeta_p, z),$$

where $\left(\frac{-m^+}{p}\right) = 1$, $\left(\frac{-m^-}{p}\right) = -1$ for the crank and similarly for the rank function.

- (2) The eta quotients appearing in the RHS of the identities can be incorporated in the definition of the $j(22, \vec{n}, z)$ functions. This will give rise to a new $\vec{n}$ vector. For instance, the function

$$\frac{\eta(22z)^7 \eta(2z)^3}{\eta(11z)^3 \eta(z)^7} j(22, \vec{n}_3, z)$$

where

$$\vec{n}_3 = (15, -2, -1, 0, -3, 0, -2, -2, 0, -1, -3, 1)$$

appearing in the identity for $\mathcal{OK}_{11,0}^{(C)}(\zeta_{11}, z)$ can be equivalently written as $j(22, \vec{n}_3^t, z)$ where

$$\vec{n}_3^t = (75, -9, -5, -7, -7, -7, -6, -9, -4, -8, -7, -4).$$

This implies that we can also interpret the identities to have 8 new vectors in case of the crank identities and 9 new ones for the rank identities. We adopt this in the MAPLE program described in the next point.

- (3) A detailed calculation done in MAPLE using packages THETAIDS and ETA developed by the first author is available here. The program contains proof of each of the identities (8.1)- (8.8).

Proof of (8.1).

Step 1 We check the conditions for modularity as in Theorem 3.1 for $j(22, \pi_r(\vec{n}_1^t), z)$, $1 \leq r \leq 9$, r odd, involved in (8.1). Here, $p = 11, m = 0, n_0 = 3$. By Theorem 3.3, we only need to check modularity for $r = 1$. With $\vec{n}_1^t = (3, 0, 0, -1, -1, 1, 0, 1, -2, 1, -2, 2)$, we easily see that

$$n_0 + \sum_{k=1}^{11} n_k = 3 + (-1) = 2,$$

$$n_0 + 3 \sum_{k=1}^{11} n_k = 0 \equiv 0 \pmod{24},$$

$$\sum_{k=1}^{11} k^2 n_k = 44 \equiv 0 \pmod{11},$$

as required.

Since $m = 0$, we skip Step 2.

Step 3 Using Proposition 8.3, we calculate the orders of each of the forty functions f at each cusp s of $\Gamma_1(22)$. We omit producing this lengthy table here, but the lower bounds for the entire combination of these functions on the RHS is calculated in the table in Step 5.

Step 4 Considering the LHS of equation (8.1) we now calculate lower bounds $\lambda(11, 0, s)$ of the orders $\text{ORD}\left(\mathcal{K}_{11,0}^{(C)}(\zeta_{11}, z), s, \Gamma_1(22)\right)$ for the cusps s of $\Gamma_1(22)$.

cusp s	$\lambda(11, 0, s)$
$i\infty$	
0	0
$1/2n$	0
$1/(2n' - 1)$	-5
$n/11$	2
$n/22$	0

where $1 \leq n \leq 5$ and $2 \leq n' \leq 5$, using Theorem 7.4. Again we note that each value is an integer.

Step 5 We summarize the calculations in Steps 4 and 5 in a table. This gives lower bounds for the LHS and RHS of equation (8.1) at the cusps s .

cusp ζ	$\text{ORD}(LHS; \zeta)$	$\text{ORD}(RHS; \zeta)$	$\text{ORD}(LHS - RHS; \zeta)$
$i\infty$			
0	≥ 0	≥ -5	≥ -5
$1/2n$	≥ 0	≥ -1	≥ -1
$1/(2n' - 1)$	≥ -5	≥ -5	≥ -5
$n/11$	≥ 2	≥ 0	≥ 0
$n/22$	≥ 0	≥ 0	≥ 0

where $1 \leq n \leq 5$ and $2 \leq n' \leq 5$. The constant B (in equation (8.13)) is the sum of the lower bounds in the last column, so that $B = -30$.

Step 6 The LHS and RHS are weakly holomorphic modular forms of weight 1 on $\Gamma_1(22)$. So in the Valence Formula, $\frac{\mu k}{12} = 180$ where μ is calculated using (8.14). The result follows provided we can show that $\text{ORD}(LHS - RHS, i\infty, \Gamma_1(22)) \geq 46$. This is easily verified using MAPLE.

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